No Space Like Home?
Selection and Efficiency in Hospital-at-Home

ANDREW CAPRON †
DEPARTMENT OF ECONOMICS, UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

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ABSTRACT: Hospital capacity is costly to expand, and the value of an available inpatient bed varies with congestion. Hospital-at-Home (HaH) provides inpatient-level care in a patient's residence and can release a brick-and-mortar bed without requiring permanent capacity expansion. However, this release is only realized when the hospital offers HaH and the patient accepts. I study how these decisions determine the use and value of flexible hospital capacity. Using clinical, accounting, census, and staffing data from four hospitals, supplemented by a novel patient survey, I jointly estimate patient acceptance and hospital offers, accounting for persistent patient preferences and endogenous offer selection. Patients differ systematically in their willingness to receive home care, and hospitals target likely adopters while balancing financial returns, clinical needs, and capacity relief. At median occupancy levels, the estimated capacity-relief value is equivalent to between \$1,150 and \$1,570 per discounted bed-day at the two largest hospitals. A counterfactual 10% increase in HaH reimbursement generates approximately 12,000 potential B&M additional bed-days released and raises aggregate expected patient option value by 91%. Although induced transfers raise predicted treatment costs by more than the additional reimbursement, reducing contribution from assessed patients by \$4.1 million, filling 50–100% of released bed-days yields a net increase in combined hospital cash contribution of \$3.7–\$17.0 million. These findings show how consumer adoption shapes flexible capacity deployment and how the value of released capacity can make expansion privately attractive even when induced transfers raise treatment costs.

KEYWORDS: Hospital-at-home, healthcare innovation, capacity management, demand-mediated capacity-constrained production
JEL CLASSIFICATION: D12, D24, I11, I18, L21, L23

† Correspondence: acapron@unc.edu.

1 Introduction

The United States healthcare system faces an overcrowding problem. This challenge is particularly acute in inpatient hospital care, with many facilities operating at or near capacity. Available bed capacity has declined since the Covid-19 public health emergency, even as patient demand has remained strong. Population aging is expected to increase annual hospitalizations by approximately 11 percent between 2025 and 2035, pushing national average hospital occupancy to 85 percent as early as 2032 if bed supply does not expand (Leuchter et al., 2025).[1] This presents both a clinical and economic problem. When capacity binds, admitting one patient can delay care for others, raise treatment costs, and worsen health outcomes. A ten-minute increase in emergency department wait time raises hospital costs for seriously ill patients by approximately 6 percent, emergency department crowding increases mortality, and ICU congestion during the Covid-19 pandemic increased inpatient mortality (Woodworth and Holmes, 2020; Woodworth, 2020; Gowrisankaran et al., 2026). Hospitals cannot costlessly eliminate these losses by simply building more beds. Physical expansion requires substantial upfront investment that is slow to deliver and difficult to reverse (Pindyck, 1988; Janke and Venkatesh, 2025). Capacity built to accommodate high demand may remain underused when demand recedes, making permanent expansion risky even when hospitals are currently crowded (Joskow, 1980; Gaynor and Anderson, 1995). Financing constraints further affect hospital investment (Adelino et al., 2015), while certificate-of-need requirements and related regulation can further constrain expansion (Courtemanche and Garuccio, 2025; Chiu, 2021). Against this backdrop, the recent expansion of Hospital-at-Home (HaH), an innovative care delivery model that provides inpatient-level care in patients' homes, offers a flexible way to relieve capacity pressure without requiring permanent capacity expansion.

HaH turns this capacity problem into a two-sided allocation problem. The hospital determines which clinically appropriate patients receive an offer, while patients determine whether that offer translates into treatment at home and released B&M capacity. HaH can improve allocation by allowing clinically suitable patients who prefer home treatment to receive it, while making hospital beds available for patients who require scarce facility-based care. These interacting decisions motivate the paper's central question: how do firm selection and consumer preferences shape the use and value of flexible capacity when conventional production is capacity constrained? I combine evidence on treatment-location effects with structural models of patient acceptance and hospital offers to characterize these decisions and recover the private value of released capacity. The

[1] An occupancy rate of 85 percent is a commonly used healthcare standard for capacity pressure because hospitals require slack to accommodate stochastic demand and patient flow. Occupancy at this level is associated with regular bed shortages and periodic capacity crises (Bagust et al., 1999; National Guideline Centre, 2018). However, recent evidence suggests that 85 percent is a conservative upper bound for the onset of capacity strain, which may occur at lower occupancy rates depending on the ward type and care setting (Pratt and Wood, 2021).

framework then evaluates how reimbursement, eligibility rules, information, and hospital capacity shocks affect HaH use, capacity relief, and associated patient and hospital outcomes.

The empirical analysis links four complementary sources of information from four hospitals in a large southeastern U.S. health system between September 2021 and February 2026. Detailed electronic health records (EHR) identify clinical eligibility, health outcomes, and each stage of the HaH screening, offer, and acceptance process. Granular census, capacity, and staffing records measure the hospital state in which these decisions are made and trace how admissions, HaH transfers, and discharges affect subsequent congestion. Payor negotiated rates and patient-level revenues and costs reveal the pecuniary incentives associated with treating a patient at home rather than in the hospital. Finally, a novel survey linked to the EHR provides supplementary evidence on patients' expectations about care quality, comfort, safety, and convenience, together with household non-medical spending and caregiver circumstances that administrative records do not capture.

I jointly estimate a panel model of patient acceptance and a structural model of hospital offers using 31,944 inpatient stays from 21,354 unique patients. Repeated encounters distinguish time-invariant differences in willingness to receive care at home from changes associated with prior interactions, while correlated encounter shocks account for endogenous offer selection. Hospitals anticipate acceptance when weighing financial returns, patient surplus, clinical needs, HaH program strain, and B&M bed-capacity released by a transfer. Expected remaining treatment duration determines the extent of the release, and B&M occupancy and staffed capacity determine the measured pressure it relieves. Complementary fuzzy spatial regression discontinuity evidence exploits the program's geographic eligibility boundary to provide context on health and utilization effects.

The estimates show that flexible capacity depends jointly on patient willingness and hospital targeting. Hospitals value expected patient surplus alongside the financial and capacity consequences of accepted transfers. The estimated gross private value of released B&M capacity rises with occupancy and, at hospital-specific median levels, is equivalent to approximately $1,570 and $1,150 per discounted bed-day at the two largest hospitals. A reimbursement policy counterfactual demonstrates how this capacity value can make expansion financially attractive even when induced transfers raise treatment costs. A 10 percent reimbursement increase generates 11,763 additional potential B&M bed-days and raises aggregate expected patient option value by 90.8 percent. Higher predicted treatment costs exceed additional reimbursement, reducing focal contribution by $4.1 million. Filling 50–100 percent of released bed-days nevertheless yields a net hospital cash gain of $3.7–$17.0 million, illustrating the importance of valuing the capacity made available alongside the returns to treating patients at home. Additional simulations examine how a public health emergency capacity shock, improved information that helps hospitals target offers toward likely adopters, and expanded HaH staffing capacity affect allocation and associated outcomes.

This paper builds on a clinical health literature showing that HaH can safely substitute for conventional B&M hospitalization for selected patients and may improve satisfaction, reduce resource utilization, or lower treatment costs in some settings (Leff et al., 2005; Levine et al., 2020; Shepperd et al., 2009; Caplan et al., 2012; Arsenault-Lapierre et al., 2021). More recent evidence under the federal Acute Hospital Care at Home initiative documents the feasibility of operating HaH at broader scale but continues to emphasize the importance of patient and program selection (Adams et al., 2023; Centers for Medicare & Medicaid Services, 2024). Existing studies primarily evaluate clinical outcomes among patients who enter HaH programs. This paper instead treats entry itself as an economic outcome, modeling both the hospital's decision to make HaH available and the patient's decision to accept it.

The economic value of HaH begins with the limits of conventional B&M expansion. Hospitals balance the costs of reserve capacity against its benefits for accommodating demand fluctuations and maintaining patient flow (Joskow, 1980; Gaynor and Anderson, 1995; Carey, 1998; Bagust et al., 1999; Pratt and Wood, 2021). Permanent expansion commits hospitals to long-lived, partly irreversible investments under uncertain future demand (Pindyck, 1988). Investment also depends on reimbursement and financing conditions (Wedig et al., 1989; Adelino et al., 2015; Gao et al., 2022), while certificate-of-need requirements can further constrain adjustment (Chiu, 2021; Courtemanche and Garuccio, 2025). Population aging increases near-term capacity pressure, but the longer-run volume, location, and treatment requirements of inpatient demand remain uncertain. HaH offers a flexible alternative, allowing hospitals to release capacity in response to operating conditions when suitable patients accept treatment at home.

The paper's first central contribution is to recover the model-implied shadow value of released-bed capacity, which I refer to as simply the private capacity value. Existing work shows that empty-bed availability affects hospital assignment and admission decisions (Kleiner, 2019; Freedman, 2016), while congestion changes care and generates health losses across patients (Woodworth, 2020; Francetic et al., 2024). Most closely related, Gowrisankaran et al. (2026) model how patient selection and congestion interact across hospitals during the Covid-19 pandemic. I study a within-hospital margin, in which transferring a patient to HaH makes B&M capacity available for operating slack or another admission. This connects the paper to research on production under capacity constraints and the opportunity cost of reserving capacity for future demand (Jofre-Bonet and Pesendorfer, 2003; Ryan, 2012; Kalouptsidi, 2014; Jeziorski and Krasnokutskaya, 2016). Patient-level offer decisions reveal how the hospital values measured capacity relief relative to accounting returns, conditional on expected patient acceptance.

The paper's second central contribution is to model patient acceptance as an input into the firm production process. More broadly, patient demand shapes both the allocation of healthcare production and the development and diffusion of medical technologies (Chandra et al., 2016; Ace-

moglu and Linn, 2004; Hamilton et al., 2021). Gandhi (2023), which studies selective admissions at skilled nursing facilities, is closest in spirit because firm admission decisions vary with census and patient type, but the firm in that setting controls the relevant capacity margin directly. On the supply side, I allow offers to reflect financial returns, clinical needs, offering frictions, and congestion relief, consistent with research showing that provider objectives need not reduce to contemporaneous profit maximization (Ellis and McGuire, 1986; Jack, 2005; Brekke et al., 2011). On the demand side, existing work studies hospital choice, the value of access, and intermediary incentives (Capps et al., 2003; Ho and Pakes, 2014). In this setting, the hospital first determines whether HaH enters the patient's choice set, making patient demand a problem of endogenous consideration (Goeree, 2008). I account for dependence between hospital offers and patient acceptance by modeling the correlation in unobserved encounter-specific shocks (Van de Ven and Van Praag, 1981), while using repeated encounters to distinguish time-invariant preference heterogeneity from history dependence. Supplementary survey evidence connects choice heterogeneity to beliefs about care and household circumstances (Ben-Akiva et al., 1994; Allcott and Wozny, 2014; Fuster and Zafar, 2016; Stantcheva, 2023; Ferrario and Stantcheva, 2025; Haaland et al., 2025). Related allocation problems arise when airlines compensate passengers to relinquish seats or utilities induce consumers to shift electricity use away from peak periods. In each case, relief of a capacity constraint depends on consumers accepting an alternative.

These findings make the allocation of offers central to evaluating HaH policy. Changes in reimbursement, eligibility, and information can expand opportunities for home treatment, but realized capacity relief depends on which patients receive offers, whether they accept, and when and how released beds are used. Hospital financial gains need not coincide with social resource savings, because reimbursement changes provider incentives while treatment costs and benefits extend across hospitals, patients, and households. More broadly, the paper provides a framework for understanding flexible supply technologies whose deployment requires consumer adoption. Their value depends on the joint distribution of gains from reallocating production, consumers' willingness to adopt, and the shadow value of production capacity.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting. Section 3 develops a framework which distinguishes the private and social value of HaH. Section 4 describes the novel data and reduced-form evidence that supports the model. Section 5 develops a joint model of patient acceptance and hospital offers, and Section 6 discusses estimation and identification. Section 7 interprets the results and Section 8 evaluates the counterfactual analyses. Section 9 concludes.

2 Institutional Background: Hospital-at-Home

Hospital-at-Home (HaH) provides acute, hospital-level care in a patient's residence as a substitute for treatment that would otherwise require an inpatient admission. It is distinct from conventional home health, which primarily provides intermittent post-acute or chronic care, and from telemedicine alone. HaH episodes may include physician oversight, in-person nursing or paramedic visits, remote monitoring, medication and intravenous therapy, laboratory and diagnostic services, oxygen delivery, and transfer back to a physical hospital when necessary. Programs are commonly classified as either admission-avoidance models, in which the patient enters HaH directly, or early-transfer models, in which part of an inpatient episode is completed at home (Shepperd et al., 2009; Caplan et al., 2012; Arsenault-Lapierre et al., 2021; Leong et al., 2021). The program studied in this paper contains elements of both: patients generally enter through a hospital emergency department and may transfer to HaH immediately or after receiving inpatient care in the B&M hospital.

Clinical experimentation with home-based acute care dates at least to randomized trials in the United Kingdom in the 1970s (Hill et al., 1978). In the United States, Johns Hopkins developed an early physician-led HaH program in the 1990s. Its pilot and subsequent multi-site evaluation demonstrated the feasibility of treating selected older patients with conditions such as pneumonia, heart failure, chronic obstructive pulmonary disease, and cellulitis outside a conventional hospital (Leff et al., 1999, 2005). Nevertheless, U.S. adoption remained limited for the next two decades. Operating a program required hospitals to develop patient-selection protocols, mobile clinical teams, pharmacy and diagnostic delivery, remote monitoring, and emergency escalation systems, while reimbursement for home-based inpatient care remained uncertain (Applebaum and Rosemary, 2023; Brody et al., 2019). HaH consequently remained concentrated in demonstration programs, integrated delivery systems, and alternative-payment arrangements.

The Covid-19 pandemic produced the regulatory change underlying the present U.S. HaH market. CMS introduced its Hospitals Without Walls flexibilities in March 2020 and launched the Acute Hospital Care at Home (AHCAH) initiative in November 2020 (Centers for Medicare & Medicaid Services, 2020). The initiative permits approved Medicare-certified hospitals to treat qualifying patients as hospital inpatients in their homes through waivers of selected hospital Conditions of Participation. Participating hospitals must perform an in-person evaluation before home treatment, apply medical and non-medical screening protocols, provide daily evaluation by a physician or advanced-practice provider and registered nurse, arrange at least two in-person visits per day, maintain immediate remote communication, and provide a response to clinical deterioration within 30 minutes. The initiative has been extended beyond the public-health emergency and is currently authorized through September 30, 2030 (Centers for Medicare & Medicaid Services,

2026). Regulatory flexibility, inpatient reimbursement, and pandemic-era capacity pressure generated rapid adoption, with several hundred hospitals entering the initiative during its first years (Applebaum and Rosemary, 2023).

The waiver changes both the production technology and the financial incentives for acute care. It creates a capacity margin that does not require new B&M construction, but capacity is released only when the hospital offers HaH and the patient accepts it.[2] HaH patients retain inpatient status, so hospital reimbursement is generally determined by the inpatient episode and diagnosis-related group rather than by the physical location of each service. A hospital may therefore receive similar revenue while substituting mobile labor, monitoring, transportation, and household inputs for hospital rooms and facility-based services. HaH relaxes the physical-space constraint, but mobile staffing, logistics, and emergency-response requirements limit the capacity it can release. The resulting accounting return need not equal the change in social resource use. HaH may be privately profitable without generating equivalent resource savings, or may conserve resources while remaining unattractive under a particular payer contract. Hospital accounting data also omit inputs supplied by patients and households, including time, utilities, housing space, and caregiving. These distinctions make reimbursement, hospital costs, and social costs separate empirical objects.

Access to HaH is determined through a sequential and highly selective process. Patients must first satisfy condition, payer, and geographic eligibility rules. In the program studied here, geographic eligibility is based on residence within an operational service area constructed around CMS's 30-minute emergency response requirement. Hospital staff then assess clinical stability, anticipated procedures and monitoring needs, and whether the patient's condition can be managed safely outside the hospital. A subsequent social-stability assessment may consider the safety of the residence, access to utilities and communication, caregiver or household support, and the patient's ability to participate in home treatment. Operational constraints, including mobile staffing and current program capacity, can further affect whether an otherwise suitable patient receives an offer. Only after the hospital makes HaH available does the patient or caregiver decide whether to accept. Acceptance may depend on perceived safety, trust in home-based care, prior experience, caregiver burden, housing conditions, and the convenience or comfort of remaining at home (Saenger et al., 2020; Centers for Medicare & Medicaid Services, 2024). A review of patient and caregiver perceptions likewise highlights the importance of the experience of receiving acute care at home, motivating attention to preferences beyond clinical outcomes (Wang et al., 2024).

Clinical evidence generally finds that HaH can provide safe care for appropriately selected patients. Trials and systematic reviews report outcomes that are often comparable to, and in some

[2] In North Carolina, additions to existing hospital bed capacity generally require certificate-of-need approval. The State Medical Facilities Plan determines the number of additional beds that may be approved, and competing applications undergo comparative review (North Carolina Department of Health and Human Services, 2024).

settings better than, conventional hospitalization, including possible reductions in mortality, readmissions, institutional care, and measured treatment expenditures (Leff et al., 2005; Shepperd et al., 2009; Caplan et al., 2012; Arsenault-Lapierre et al., 2021). A prominent randomized trial found comparable or improved clinical outcomes and substantially lower measured expenditures under HaH (Levine et al., 2020).[3] The findings are not uniform across admission-avoidance and early-transfer programs, clinical conditions, outcome windows, or cost perspectives. Many trials are small and use tightly controlled selection protocols, while observational comparisons face selection by both providers and patients. Evidence from the AHCAH period similarly suggests favorable or comparable short-run outcomes among treated patients but documents substantial differences between HaH and conventional inpatient populations (Adams et al., 2023; Centers for Medicare & Medicaid Services, 2024; Vakkalanka et al., 2026). Existing evidence therefore establishes the clinical feasibility of HaH more clearly than its causal effects in routine implementation or its value as a system-wide capacity instrument.

These institutions link the allocation of treatment across settings to the capacity available for other patients. An accepted HaH offer can improve alignment with patient preferences and release a hospital bed, while changing treatment costs and shifting some inputs to households. Hospitals need not value these consequences in the same way as a social planner. Section 3 formalizes these channels and characterizes the resulting differences between the private and social value of an HaH offer.

3 Economic Framework and Welfare

The central economic object is the shadow value of B&M capacity released by an accepted HaH offer. The hospital determines whether home treatment is available, and the patient determines whether the release occurs. This section describes how patient demand, treatment returns, and the value of freed capacity shape hospital and social planner incentives. Three wedges govern their potential disagreement: patient and health benefits, resource costs and profitability, and the value assigned to released capacity.

Patient choice and the consequences of HaH. Consider an HaH-eligible patient i admitted when hospital congestion is c. If offered HaH, the patient chooses between home treatment, h, and B&M care, b. Let

$$\Delta u_i = u_{ih} - u_{ib}$$

[3] Approximately 63% of eligible patients in the study declined participation in HaH, illustrating why clinical eligibility does not translate automatically into adoption or effective capacity expansion.

denote the patient's money-metric value of HaH relative to B&M, including an idiosyncratic component observed by the patient. Acceptance occurs when $\Delta u_i \geq 0$. Given the hospital's pre-offer information, the acceptance probability and expected patient surplus from an offer are

$$\rho_i = \mathbb{P}(\Delta u_i \geq 0), \qquad \mathcal{Q}_i = \mathbb{E}\left[\max\{\Delta u_i, 0\}\right].$$

The probability ρ_i determines whether an offer reallocates treatment, while $\mathcal{Q}_i$ measures the value of the additional treatment option.

Let $\Delta\mathcal{H}_i$ denote the monetized social value of health changes under HaH beyond those incorporated in the patient's own valuation, and let $\Delta\mathcal{C}_i$ denote the reduction in real treatment resources relative to B&M. Positive values favor home treatment. Acceptance-contingent consequences are understood as conditional expectations among accepting patients given the pre-offer information.

An accepted transfer also makes inpatient bed space available for other uses. Let $r_i^S(c)$ denote the social shadow value of this release: the value of additional capacity over the period in which the patient would otherwise occupy a B&M bed. It includes shorter waits, fewer delayed or forgone admissions, and lower crowding exposure for current and future patients. This value generally rises with utilization and expected demand (Joskow, 1980; Gaynor and Anderson, 1995). The finding that ICU strain increased mortality among hospitalized Covid-19 patients illustrates the consequences of scarce capacity for other patients (Gowrisankaran et al., 2026).

The shadow value reflects both the duration of the release and the scarcity of capacity during that period. A release during peak demand can be valuable even if brief, while a longer release preserves flexibility across subsequent arrivals. Thus, $r_i^S(c)$ values a temporary release conditional on acceptance; its expected contribution at the offer stage is $\rho_i r_i^S(c)$.

Social value of an HaH offer. A social planner chooses whether to offer HaH while respecting the patient's treatment choice. Let k_i denote the real cost of screening, coordinating, and extending an offer, borne by the hospital regardless of acceptance. Transfer and treatment costs incurred upon acceptance enter $\Delta\mathcal{C}_i$. The expected social gain from offering HaH is

$$G_i^S(c) = \mathcal{Q}_i + \rho_i \left[\Delta\mathcal{H}_i + \Delta\mathcal{C}_i + r_i^S(c)\right] - k_i. \tag{1}$$

The planner offers when $G_i^S(c) > 0$. Patient surplus already incorporates the patient's choice, while health, resource, and capacity gains are weighted by the probability of acceptance. Reimbursement transfers resources between payers and providers; its welfare consequences operate through behavioral incentives and any financing costs included in $\Delta\mathcal{C}_i$.

Hospital incentives. Let

$$\Delta\Pi_i = \Pi_{ih} - \Pi_{ib}$$

denote incremental hospital accounting profit from an accepted HaH placement. It reflects reimbursement and costs borne by the hospital, while $\Delta\mathcal{C}_i$ encompasses real resources used across providers and households.

Let $r_i^F(c)$ denote the hospital's private shadow value of the released B&M capacity. Freed bed space gives the hospital the option to accommodate another admission or retain operating slack. Its value combines the returns from these uses over the remaining period of the release. Benefits can extend across departments because inpatient patient-flow protocols influence emergency-department bottlenecks and delays (Armony et al., 2015). The private and social shadow values differ to the extent that the hospital internalizes these consequences for patients, staff, and the broader healthcare system.

Hospitals may also value patient welfare, health, or access to home care. Let m_i denote the expected nonpecuniary value of extending the offer. The hospital's expected gain is

$$G_i^F(c) = m_i + \rho_i \left[\Delta\Pi_i + r_i^F(c)\right] - k_i. \tag{2}$$

The hospital offers when $G_i^F(c) > 0$. For $\rho_i > 0$, this condition can be written as $r_i^F(c) > (k_i - m_i)/\rho_i - \Delta\Pi_i$: capacity value can justify an offer even when home treatment lowers accounting profit. Greater congestion raises the incentive to release a bed when it increases this shadow value, while higher acceptance makes the release more likely relative to the cost of extending the offer.

Private and social incentives. The difference between social and private offer values is

$$\begin{aligned}\Delta G_i(c) &= G_i^S(c) - G_i^F(c) \\ &= \underbrace{\mathcal{Q}_i + \rho_i \Delta\mathcal{H}_i - m_i}_{\text{demand/health wedge}} \quad + \quad \underbrace{\rho_i\left(\Delta\mathcal{C}_i - \Delta\Pi_i\right)}_{\text{resource/profitability wedge}} \quad + \quad \underbrace{\rho_i\left[r_i^S(c) - r_i^F(c)\right]}_{\text{capacity wedge}}.\end{aligned} \tag{3}$$

The first wedge compares patient and health benefits with the hospital's nonpecuniary valuation. The second compares social resource savings with accounting returns. Payment rules and the allocation of costs between the hospital and households can therefore make a transfer privately attractive even when its resource savings are small, or financially unattractive despite substantial social savings.

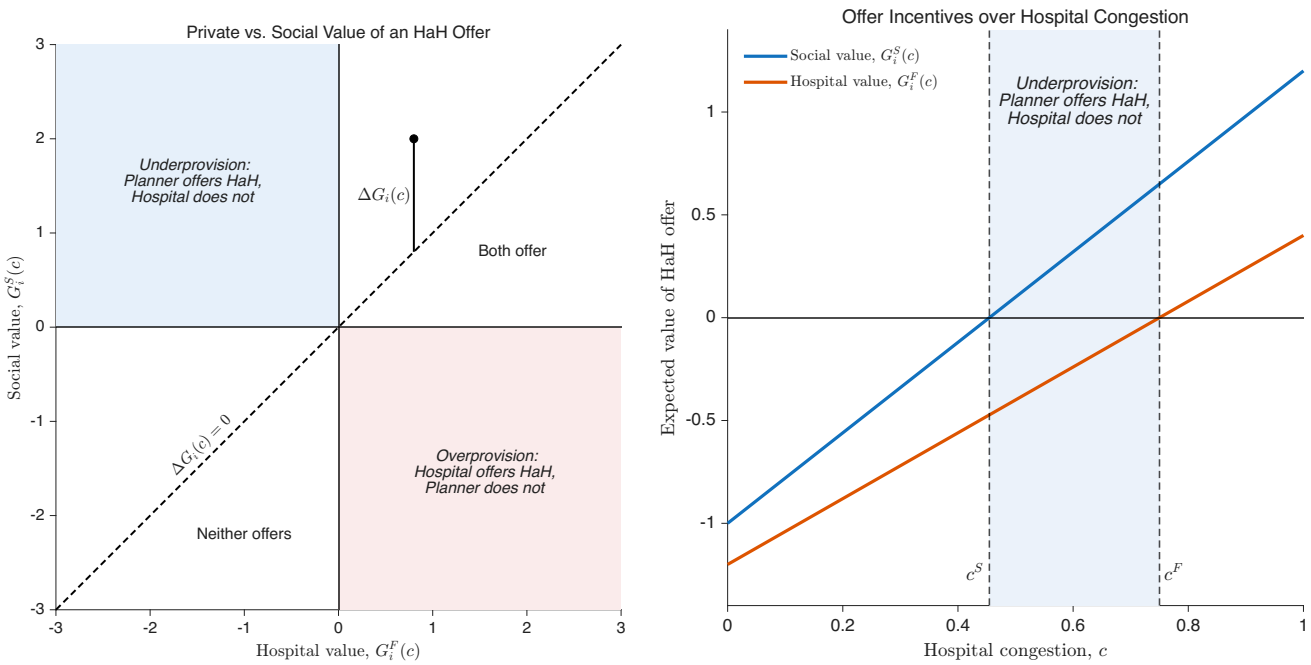
The capacity wedge compares the two shadow values of freed bed space. When the hospital captures only part of the benefit to other patients, its offer incentive falls short of the social value of the release. This gap can widen during congestion peaks as the consequences of delayed care

increase. The combined wedge varies across patients and operating states with demand, health gains, costs, and the timing of the capacity release.

HaH is underprovided for patient i when $G_i^S(c) > 0$ but $G_i^F(c) < 0$, and overprovided when $G_i^F(c) > 0$ but $G_i^S(c) < 0$. The hospital and planner can therefore disagree about both program scale and which patients should receive offers. Patient acceptance governs the capacity, health, and resource gains realized under either policy. Valuable candidates combine willingness to receive home care with substantial gains from the resulting transfer.

Figure 1 illustrates these allocation differences. Panel (a) compares private and social offer values: the upper-left region denotes underprovision and the lower-right region denotes overprovision. The 45-degree line represents agreement in valuations. Panel (b) illustrates a case in which social offer value crosses zero at a lower congestion level than private offer value. The interval between these thresholds identifies states in which the planner offers HaH but the hospital does not.

Figure 1: Private and Social Incentives for Hospital-at-Home Offers



(a) Private versus social offer value

(b) Congestion and offer thresholds

Empirical scope. The empirical analysis uses this framework to connect patient acceptance, hospital offer incentives, and the shadow value of freed capacity. The joint model uses variation in accounting returns and operating conditions, conditional on patient characteristics and acceptance beliefs, to recover the hospital's valuation of capacity relief in profit-equivalent units. A separate

expected consumer-surplus term captures the hospital's valuation of making HaH available to patients. Accounting data measure hospital costs, and the survey provides evidence on household preferences and non-medical spending. The health evidence in Section 4 motivates the baseline focus on treatment allocation, with clinical needs and escalation risk entering hospital offer incentives. Counterfactuals report hospital returns, potential bed-days released, and patient option value in normalized utility units. These outcomes quantify the principal allocation mechanisms, with the three wedges organizing their interpretation from private and social perspectives.

4 Data and Reduced-Form Evidence

4.1 Data, Assessment Sample, and Offer Documentation

The introduction of hospital-at-home care at a large southeastern health system provides a unique opportunity to study patient adoption and hospital capacity allocation from the program's inception to current maturation. I construct a novel proprietary dataset combining information from four sources. From the health system, I obtain (i) rich electronic health records (EHR), (ii) administrative data on hourly bed census, and (iii) accounting data on costs and revenues, spanning September 2021 through February 2026. The data cover two large urban medical centers and two satellite hospitals in less densely populated cities. I supplement these data with (iv) a novel survey I administer to patients who are eligible for HaH. This survey includes additional information on unobserved household expenses from receiving HaH, patient beliefs and preferences, and caregiver income and employment. Appendix D describes the survey and its supplementary role in characterizing patient preferences.

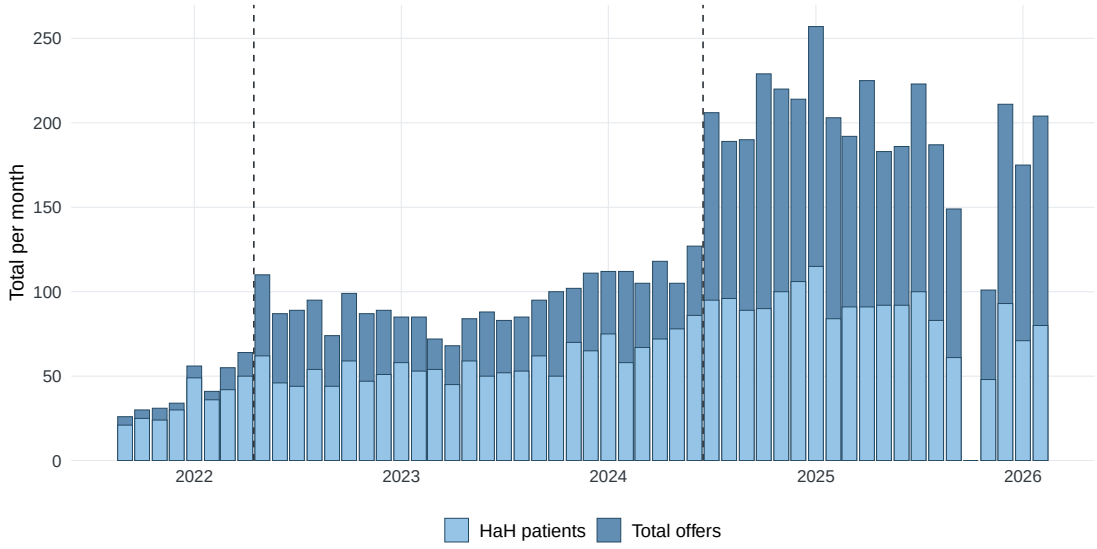
In addition to data on patient diagnoses and procedures, the EHR data provides information on patient demographics, primary and secondary payors, post-stay outcomes, and patient acquisition through the HaH screening process (see Figure A.1 for a detailed overview of the stages of screening). My analysis also employs patient-level revenues and costs. Additional proprietary data on negotiated payor contracts help characterize provider returns and patient cost-sharing. Baseline price difference across treatment locations has little variation, thus highlighting the importance of the survey to provide supplementary information on additional hospitalization-related household expenses. These comprehensive data allow me to model the complex process of patient admission to a HaH program, while accounting for the various trade-offs hospitals and patients face in choosing in-hospital or at-home care.

The selection stages are summarized in Table 1. Roughly 80% of condition-location-payor eligible encounters are screened, about 18% of screened encounters have a recorded offer, and approximately half of these observed offers are accepted. These recorded frequencies describe how

patients progress from eligibility to treatment, although incomplete documentation understates offers in earlier periods. The program expanded substantially during the study period. Figure 2 plots monthly observed offers and accepted placements. Accepted placements are observed throughout and therefore provide a consistent measure of treatment activity. Recorded offers also reflect changes in EHR procedures that improved documentation of patient declinations.[4]

To account for undocumented declined offers, I use overlapping reporting indicators to construct hospital-regime documentation probabilities, q_i, treating the most recent "modern regime" documentation as complete. These probabilities and the number of assessments inform a separate reconstruction of offer totals for the reduced-form evidence. Appendix C.1 describes this adjustment and its assumptions. The subsequent analysis examines how hospitals choose whom to offer HaH and how patients respond. It therefore focuses on encounters with an observed HaH assessment, taking the preceding screening decision as given. Section 7 reports the final estimation sample and the restrictions needed to construct its inputs.

Figure 2: Hospital-at-Home Offers and Admissions Over Time



[4]The October 2025 federal government shutdown interrupted HaH activity, followed by a shorter partial shutdown in January 2026. These operational interruptions are distinguished from the documentation changes marked in Figure 2.

Table 1: Hospital admissions and HaH selection, September 2021–February 2026

	All	Location & Payor eligible	Screened	Screened, not offered	Offered	Declined	Accepted
Encounters	76,481	41,595	34,563	28,234	6,329	2,938	3,391
Unique patients	49,830	26,613	22,881	19,535	5,219	2,597	2,899
Age (SD)	66.2 (17.1)	70.0 (16.3)	71.1 (15.8)	71.0 (16.0)	71.4 (14.8)	71.8 (14.3)	71.1 (15.1)
Male	0.468	0.461	0.464	0.468	0.447	0.432	0.461
White	0.653	0.682	0.681	0.672	0.724	0.708	0.737
Black	0.257	0.233	0.233	0.248	0.167	0.199	0.139
Latino/Hispanic	0.042	0.041	0.042	0.038	0.059	0.048	0.069
Commercial	0.135	0.072	0.054	0.052	0.063	0.055	0.069
Medicare	0.680	0.797	0.815	0.815	0.815	0.834	0.798
Medicaid	0.108	0.069	0.080	0.083	0.066	0.071	0.062
Dual Medicaid coverage	0.148	0.169	0.167	0.177	0.123	0.143	0.106
Multiple payors	0.364	0.484	0.482	0.487	0.460	0.452	0.468
Total LOS, days (SD)	6.4 (5.7)	6.2 (5.3)	6.4 (5.4)	6.5 (5.6)	6.0 (4.2)	5.4 (4.2)	6.5 (4.1)
DRG weight (SD)	1.5 (0.9)	1.4 (0.8)	1.4 (0.8)	1.4 (0.8)	1.2 (0.6)	1.3 (0.6)	1.2 (0.6)
ED entry	0.826	0.915	0.945	0.938	0.975	0.972	0.978
Nearest hospital, mins. (SD)	38.5 (35.4)	23.5 (14.0)	22.7 (12.6)	22.8 (12.8)	22.4 (11.8)	22.7 (11.5)	22.2 (12.0)
Net revenue, $000 (SD)	18.8 (26.0)	15.7 (18.3)	15.2 (16.8)	15.8 (17.8)	12.3 (10.9)	12.6 (10.7)	12.0 (11.0)
Neighborhood traits							
Household-size (SD)	2.6 (4.4)	2.5 (1.0)	2.5 (1.0)	2.5 (1.0)	2.5 (0.6)	2.5 (0.5)	2.5 (0.6)
College/graduate share	0.426	0.496	0.500	0.499	0.506	0.491	0.518
Log per-capita income (SD)	10.6 (0.5)	10.7 (0.5)	10.7 (0.5)	10.7 (0.5)	10.8 (0.4)	10.7 (0.4)	10.8 (0.4)
Share below 150% poverty line	0.197	0.169	0.168	0.170	0.159	0.165	0.154

Notes: Subset to NC residents and inpatient conditions with 3+ HaH discharges. Numbers reflect shares unless SD is indicated. Screened means evidence of human-review for HaH suitability or evidence of HaH treatment. Offered includes recorded offers or accepted HaH; declined means recorded offer without acceptance. ACS education status is bachelor plus graduate share, income is log(1 + per-capita income), and poverty is the source share below 150% of the poverty line. All monetary values described in this manuscript are expressed in February 2026 U.S. dollars ($).

4.2 Selection and Repeated Patient Decisions

Patients who receive HaH offers differ from other screened patients, and those who accept differ further from those who decline. Relative to screened patients without an observed offer, offered patients have lower clinical severity, are more likely to be White and less likely to be Black or Medicaid insured, and have lower reimbursement. Among patients with an observed offer, those who accept are somewhat younger and more likely to be male, White, and commercially insured than those who decline. These differences motivate examining hospital offers and patient acceptance as separate decision margins. The hospital may also use information about patient suitability or preferences that is unobserved by the econometrician, thus accounting for observed patient char-

acteristics alone may not fully explain which patients receive offers.

The repeated nature of inpatient encounters provides information on persistent willingness to receive HaH. Figure 3 shows that repeat patients account for approximately 50% of screened encounters, 30% of offered encounters, and 25% of accepted placements. Repeated choices by the same patient can therefore help distinguish time-invariant differences in willingness to accept HaH from variation across encounters.

Figure 3: Share of Encounters Accounted for by Repeat Patients

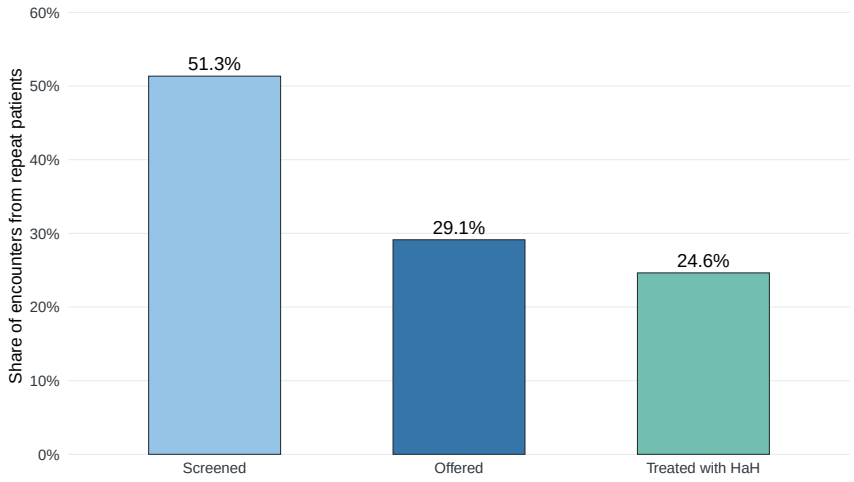
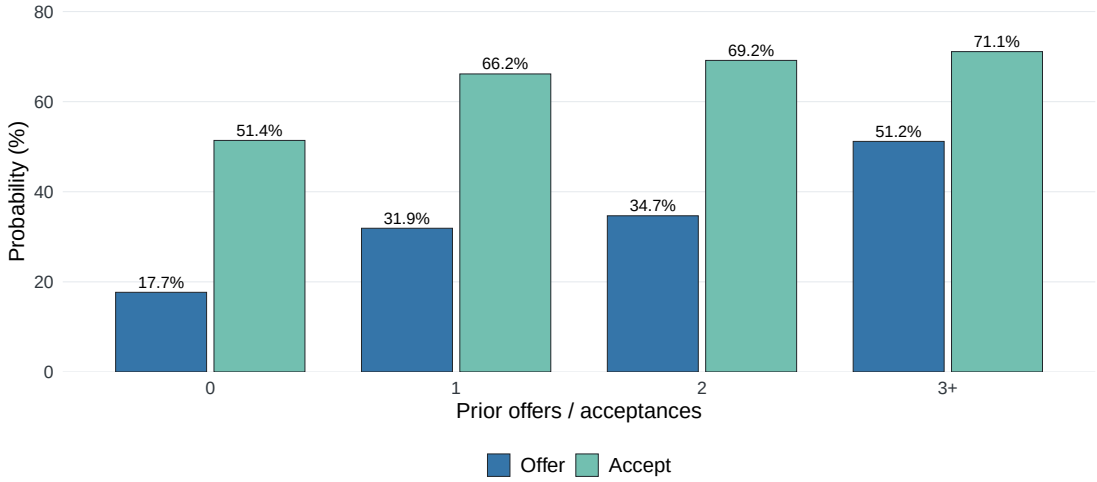


Figure 4 shows that acceptance conditional on an observed offer rises most between zero and one prior acceptance, in this first increment by 14.8 percentage points. It rises more modestly at two prior acceptances and then levels off, whereas offer rates continue to increase with prior HaH offers. Repeated declinations, especially without any prior acceptance, predict much lower subsequent acceptance. These patterns motivate an initial treatment-associated shift alongside time-invariant patient preferences and hospital targeting based on prior interactions. They do not by themselves distinguish stable tastes from the effects of earlier treatment or outreach, since selection into treatment and subsequent offers also shapes the observed rates (Heckman, 1981; Dube et al., 2010). The estimated model of HaH offers and patient acceptances that follows therefore combines a time-invariant patient effect with a rich measures of patient history and interaction with the HaH program.

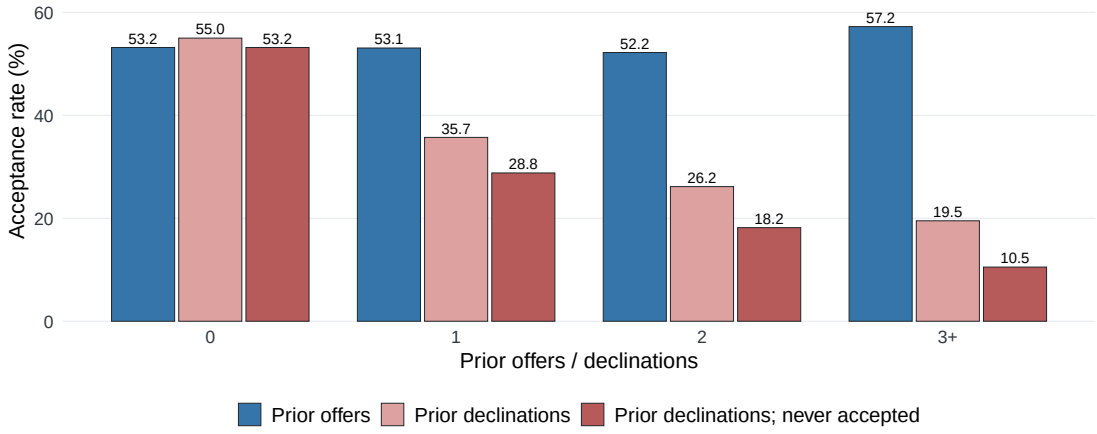
4.3 Capacity Pressure and Hospital Offers

Administrative census data capture the persistently high inpatient utilization in this market. Figure 5 plots the distribution of hourly capacity utilization separately by hospital. The two large med-

Figure 4: Probability of Outcome Across Offer and Acceptance History



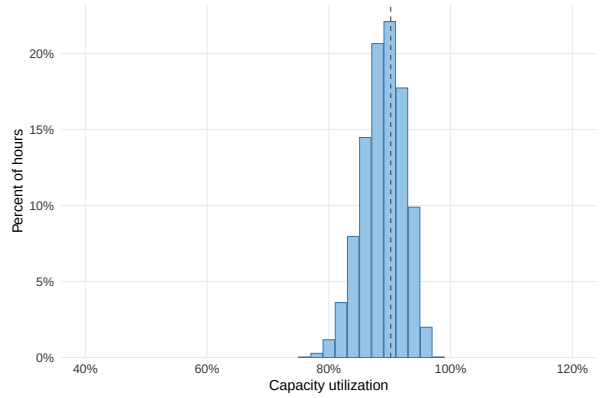
(a) Offer and Acceptance Rates by Prior Interactions



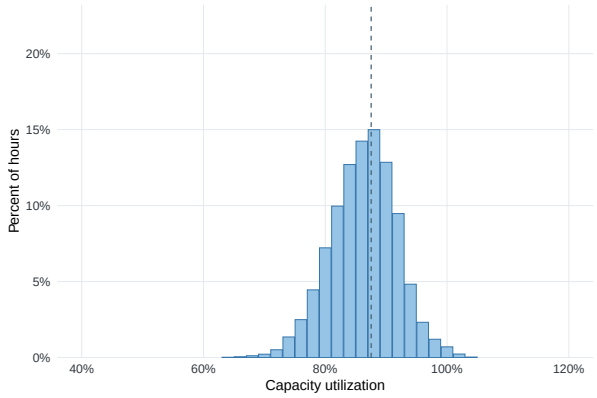
(b) Acceptance Rates by Prior Offers and Declinations

ical centers, Hospitals A and B, operate at high utilization for a large share of hours with mass concentrated around or above 90% utilization, while the two satellite facilities exhibit lower and more dispersed utilization. These levels are operationally meaningful because hospitals generally require slack bed capacity to move patients and resources throughout the system; utilization above roughly 85% is often associated with regular bed shortages and periodic capacity crises (Bagust et al., 1999; National Guideline Centre, 2018; Pratt and Wood, 2021). Thus, the same HaH technology is embedded in hospitals facing different degrees of capacity pressure, with the strongest congestion motive arising at the two urban facilities.[5]

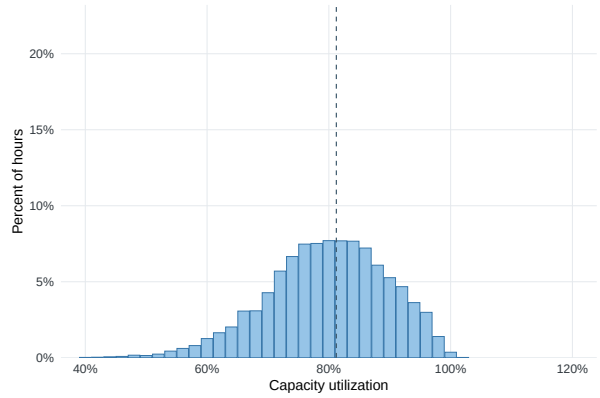
Figure 5: Average Brick-and-Mortar Capacity Utilization



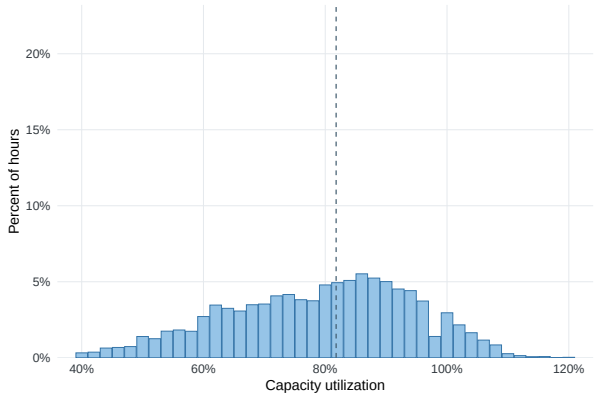
(a) Hospital A



(b) Hospital B



(c) Hospital C



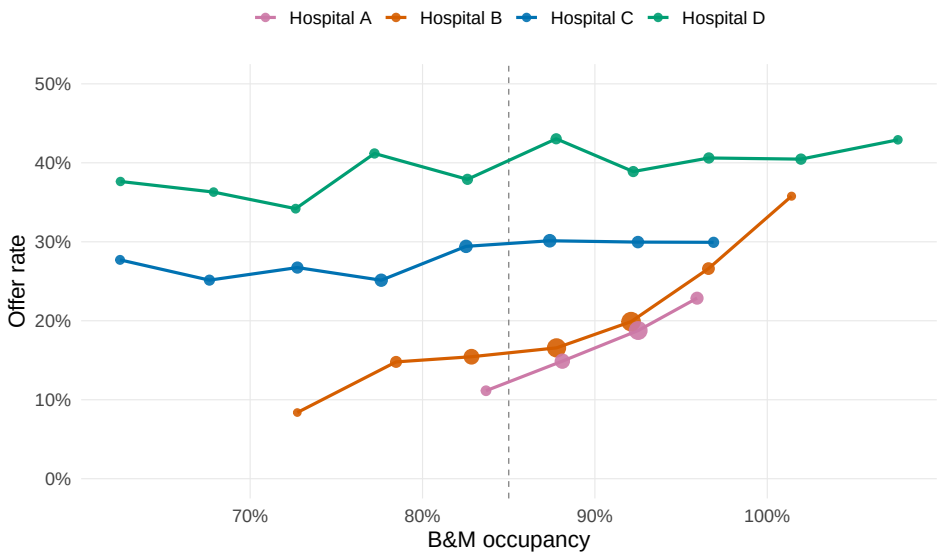
(d) Hospital D

Figure 6 relates empirical offer rates to B&M occupancy. Offer rates at the two large hospitals exhibit the most visible increase in occupancy, across the full distribution of occupancy measures

[5]The time-series analogues are reported in Figure A.2. These plots show substantial within-hospital variation in utilization over time, which provides within-hospital variation for estimating the conditional congestion response in screening and offer behavior in the analyses to come.

and the upper-tail of their support. The satellite facilities have higher offer rates over much of their support but flatter occupancy profiles. These level differences need not represent differences in capacity value alone: patient composition, operational differences, and persistent hospital policy also vary across facilities. The relevant comparison I shall examine is offer behavior changes with capacity pressure within a given hospital.

Figure 6: Documentation-Adjusted Offer Rates by B&M Occupancy



Notes: Bubble sizes correspond to the share of total hospital-hours spent in the corresponding B&M occupancy range.

Figure 7 provides evidence that HaH activity responds to capacity pressure. I estimate linear probability models for two sequential margins. The first outcome is whether an HaH-eligible patient is screened for home-care. The second is whether a reviewed patient receives an offer. In each specification, the slope on B&M occupancy varies by hospital. The figure reports the hospital-specific marginal effect of a 10 percentage-point increase in B&M occupancy. I estimate

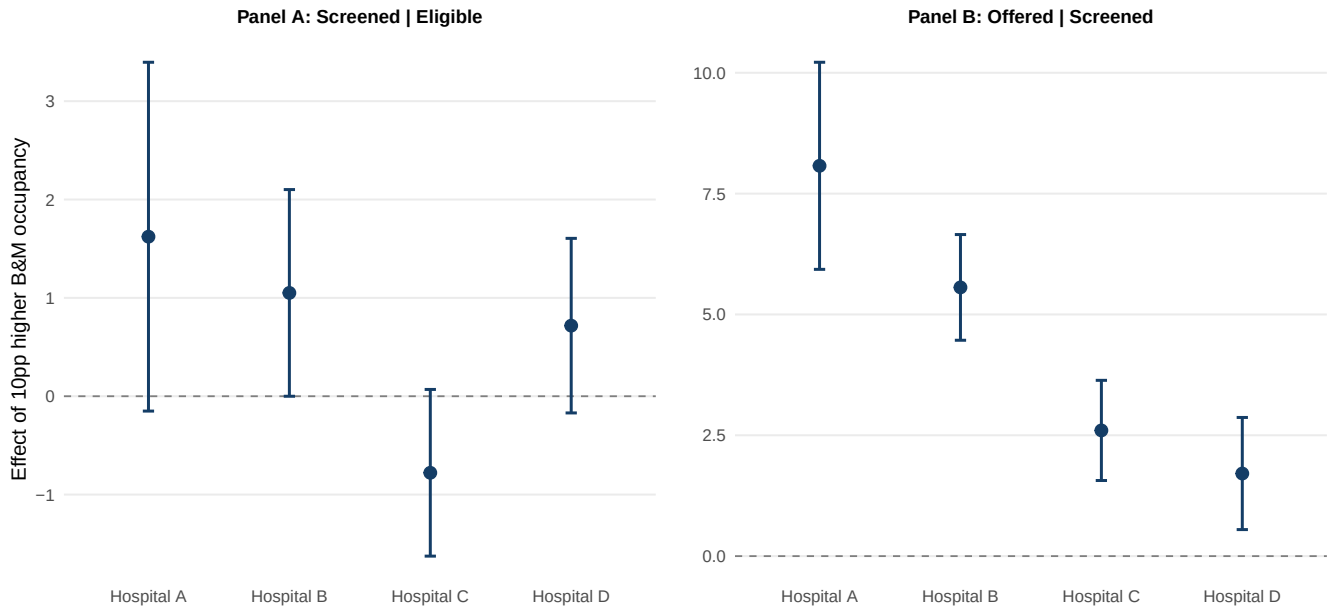
$$Screened_{ih} = \alpha^S + \beta_h^S Occ^{BM}_{h,t_i^A} + \theta_h^S Stock_{h,t_i^A} + X_i'\gamma^S + \delta_h + \tau^S_{ym(t_i^A)} + \eta^S_{dow(t_i^A)} + \varepsilon^S_{ih},$$

$$Offer_{ih} = \alpha^O + \beta_h^O Occ^{BM}_{h,t_i^O} + \theta_h^O Stock_{h,t_i^O} + X_i'\gamma^O + \delta_h + \tau^O_{ym(t_i^O)} + \eta^O_{dow(t_i^O)} + \varepsilon^O_{ih}.$$

Here, t_i^A is the admission time and t_i^O is the offer-decision time. $Occ^{BM}_{h,t}$ is B&M occupancy. $Stock_{h,t}$ contains the HaH census and the B&M emergency-department stock. The vector X_i contains patient demographics, payor, condition, and illness severity. The regressions include hospital, year-month, and day-of-week fixed effects. Standard errors are clustered at the hospital-day level.

Screening responses are weak, while offer responses are positive at all four hospitals. The two

Figure 7: Marginal Effects of Capacity Utilization by Decision Margin



(a) Marginal screening response

(b) Marginal offer response

large urban hospitals have the largest offer responses. These estimates do not directly identify the value of released B&M bed capacity, but they show that observed offer behavior changes with temporal capacity pressure. This is directly relevant to patient welfare, as substantial evidence supports the claim that meaningful capacity pressure also affects clinical quality and patient outcomes (Woodworth, 2020; Francetic et al., 2024; Gowrisankaran et al., 2026). The linear regressions summarize average conditional gradients.

Health outcomes and modeling scope. A fuzzy spatial regression discontinuity design exploits the program's geographic eligibility boundary to study health and utilization effects; Appendix B presents the design and summarizes the evidence developed in Capron et al. (2026). The estimates point to mixed effects, including lower mortality and fewer falls during the acute-care episode but more subsequent inpatient readmissions, with limited evidence of differences in several other health- and resource-related outcomes. These findings motivate a baseline model that abstracts from health outcomes directly while retaining escalation risk and acute-care needs as determinants of HaH offers.

5 Empirical Model of Hospital-at-Home Offers and Patient Acceptance

In this section, I specify and estimate a tractable joint model of demand for and supply of hospital-at-home care. Jointly modeling these decisions characterizes heterogeneity in patient willingness to receive inpatient-level care at home while recovering hospital offer incentives in a framework that accounts for endogenous offer selection and the dependence of realized capacity relief on patient acceptance. Because access to HaH occurs only after a patient presents to a hospital or its ED, I condition on the admitting hospital and abstract from hospital choice and competition across hospitals. The model instead focuses on the subsequent treatment-location margin: whether the hospital offers HaH and, conditional on an offer, whether the patient accepts care at home rather than remaining in the B&M setting.

The model conditions on an observed HaH assessment and does not separately estimate the preceding screening decision. An assessment produces two sequential decisions. The hospital first decides whether to offer at-home care to an eligible patient. If an offer is made, the patient decides whether to accept HaH or remain in B&M. Given most consequences of an offer are realized only after an accepted transfer, the hospital anticipates and internalizes the probability that the patient will accept. The model has three observed outcomes: an accepted HaH placement, a documented rejection, and no observed offer. The panel structure of the model allows willingness to accept to vary across patients through a time-invariant preference component, with repeat encounters exploited to distinguish this heterogeneity from encounter-specific choice variation. The hospital values the capacity released by an accepted transfer while balancing this benefit against the corresponding exposure to HaH staffing demands, location profitability, and patient care needs.

5.1 Setup and patient acceptance

Let i index HaH assessments and let $j(i)$ denote the patient associated with assessment i admitted to hospital $b(i)$. Let $O_i \in \{0,1\}$ indicate the hospital makes an actual HaH offer and let $A_i \in \{0,1\}$ indicate the patient accepts. Define

$$p_i = \mathbb{P}(O_i = 1 \mid \mathcal{I}_i^O), \qquad \rho_i = \mathbb{P}(A_i = 1 \mid O_i = 1, \mathcal{I}_i^O),$$

where $\mathcal{I}_i^O$ is the hospital's information at the time of the assessment.

The repeated-choice patterns in Section 4 motivate a specification that separates time-invariant preference heterogeneity from history dependence. Conditional on an offer, the patient chooses between HaH (h) and B&M ($b(i)$) care. Let ΔU_{ij} denote the utility of h relative to $b(i)$ at assessment i for patient $j(i)$ with time-invariant HaH preference ξ_j. I parameterize this indirect utility

difference as

$$\Delta U_{ij} = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \sigma_\xi \xi_j + \varepsilon_i, \tag{4}$$

where $\varepsilon_i \sim N(0,1)$ is a mean-zero normal utility-difference shock. The vector X_i contains race, payor, age, sex, distance to the admitting hospital, condition severity, neighborhood characteristics, and condition and hospital-by-quarter fixed effects. Let $\mathcal{H}_i^-$ denote the patient's complete HaH interaction history prior to assessment i, and let $h(\mathcal{H}_i^-)$ summarize this history through variables such as prior offers and acceptances. The vector P_i contains out-of-pocket price differences borne by the patient.[6] The patient-specific term $\xi_j \sim N(0,1)$ captures time-invariant unobserved willingness to receive acute care at home. The patient accepts HaH when $\Delta U_{ij} \geq 0$.

5.2 Hospital Offer Decision

While patient acceptance determines whether a release actually occurs, hospital offer selection first determines which transfers are attempted. The hospital may value consequences beyond contemporaneous accounting profit. The model therefore uses observed and latent offers to recover the private capacity value of scarce bed stock together with the hospital's valuations on accounting returns, HaH program strain, patient preferences, and clinical risk.[7]

The hospital observes the assessed patient, their clinical condition, and the current operating state before choosing whether to offer HaH. Let ρ_i denote the probability that the patient will accept based on the information available to the hospital at the assessment, and let Q_i denote the corresponding expected patient surplus from having the HaH option. The deterministic component of the hospital's value from offering rather than not offering is

$$\Gamma_i(\rho_i; \Theta_S) = Z_i'\kappa + \rho_i \Delta W_i, \tag{5}$$

where Z_i contains determinants of the offer decision whose effects do not require acceptance and ΔW_i is the incremental value of an accepted HaH placement. The hospital offers HaH when $\Gamma_i + \nu_i \geq 0$, where $\nu_i \sim N(0,1)$ is a mean-zero normal offer payoff shock. This rule determines the offer probability p_i. I specify the value of an accepted placement as

$$\Delta W_i = \beta_\Pi \Delta\Pi_i + \beta_H H_i + \beta_E E_i^B + \lambda_B' M_i^B + \lambda_H M_i^H. \tag{6}$$

[6] Age, condition severity weight, home-to-hospital distance, household size, neighborhood college share, log per-capita income, and the share below 150% of the poverty line are standardized using fixed reference means and standard deviations. Prior interaction histories are raw cumulative counts. Indicators are not standardized. Appendix C.2 provides a discussion of the relevant transformations.

[7] Consistent with health-economics models in which providers place weight on mission-driven payoffs as well as financial incentives (Ellis and McGuire, 1986; Jack, 2005; Brekke et al., 2011).

These terms measure expected accounting profits, escalatory care risks, hospitalization duration, released B&M capacity, and HaH program strain, respectively. The vector

$$\Theta_S = (\kappa, \beta_\Pi, \beta_H, \beta_E, \lambda_B, \lambda_H)$$

collects the supply-side parameters, with κ containing all direct offer coefficients, including the nuisance coefficients ω below.

The vector Z_i contains the offer-stage determinants that do not require patient acceptance. To distinguish the components of structural interest from nuisance controls, let

$$Z_i'\kappa = \kappa_O + \kappa_H f(\mathcal{H}_i^-) + \kappa_S S_i + \kappa_Q Q_i + \zeta_i'\omega. \tag{7}$$

The parameter κ_O is the common net offer term. A negative value is consistent with outreach, vetting, and administrative effort incurred before acceptance, but its level also absorbs baseline benefits and depends on the reference categories and regressor centering. It is therefore interpreted as a baseline offer friction rather than a separately measured financial cost. The variable $f(\mathcal{H}_i^-)$ is the strictly prior expected number of true offers made to the patient. It includes recorded offers and the posterior probabilities of undocumented declined offers. Its coefficient κ_H allows prior interactions to affect current offer value through repeated outreach, familiarity with the patient and caregivers, or persistent targeting.

S_i measures acute severity and clinical care burden at the assessment based on rich EHR flowsheet data. Care burden combines nursing-assessment (NWASS) and fall-risk (Morse) scores, while acute severity uses organ-failure-risk (SOFA) scores. These measures enter outside the acceptance multiplier because they affect the decision to attempt a transfer regardless of whether the patient ultimately accepts. The term $\kappa_Q Q_i$ allows the hospital to value the expected patient surplus from having the option to choose HaH, separately from the acceptance-weighted consequences of an actual transfer. This is consistent with a mission-driven preference for providing specific services to patients who value them more. The final vector ζ_i contains condition type indicators, hospital and department fixed effects, and indicators for missing clinical EHR inputs and the partial federal government shutdown described in Section 4.[8]

The components of ΔW_i summarize the predicted consequences of an accepted transfer. Let E_i^B and E_i^H denote expected discounted remaining days in B&M and HaH care, respectively. These measures are constructed from assessment-origin discharge models that condition on hospital, condition, treatment location, current length of stay, and day of week, using a daily discount factor of $\delta = 0.99$. Appendix C.3 details their construction. Expected remaining B&M duration enters

[8] Appendix C.2 details the centering, scaling, and other transformations applied to the offer-model inputs.

separately and affects the firm's payoff through a conditional duration effect, meaning hospitals' willingness to offer a patient can vary with the anticipated remaining stay conditional on accounting returns, clinical risk, and capacity consequences. This allows the capacity terms to fully capture how the value of releasing those bed-days varies with hospital utilization.

Let $\Delta\Pi_i^{\$} = \pi_i^H - \pi_i^B$ denote the conditional HaH-minus-B&M dollar profit difference, where $\pi_i^o = R_i^o - C_i^o$ is expected remaining revenue minus direct treatment costs under option $o \in \{H, B\}$. Reimbursement parity across treatment locations implies $\Delta\Pi_i^{\$} = C_i^B - C_i^H$. The offer index uses the centered difference in \$1,000 units, denoted by $\Delta\Pi_i$ and defined in equation (C.13). Figure A.3 illustrates the remaining cost profiles and fitted discharge hazards by condition and treatment location. Appendix C.4 explains how episode-level cost estimates yield these conditional predictions of remaining direct treatment costs.

Let H_i denote the probability that the patient would require return transfer from HaH back to B&M care due to escalation in acute care needs. Because this risk arises only under HaH treatment, it enters ΔW_i and is therefore scaled by the probability of acceptance. To construct HaH program strain, let N_i^H denote HaH census and L_i denote HaH staffing at the time of the assessment. Define HaH utilization as

$$u^H(N^H, L) = \frac{N^H}{L}. \tag{8}$$

The program-strain function is

$$m^H(N^H, L) = L\left[u^H(N^H, L)\right]^2 = \frac{(N^H)^2}{L}. \tag{9}$$

The expected strain generated by adding the assessed patient is therefore

$$\begin{aligned} M_i^H &= E_i^H\left[m^H(N_i^H + 1, L_i) - m^H(N_i^H, L_i)\right] \\ &= E_i^H\ \frac{2N_i^H + 1}{L_i}. \end{aligned} \tag{10}$$

This quadratic specification implies that the contemporaneous strain from adding a patient is greater when the HaH program is more highly utilized. Multiplication by L_i allows the level of the strain function to scale with staffing levels.

The B&M term M_i^B is of central interest to the paper's primary research question. It measures occupancy-weighted bed-days released by an accepted HaH transfer. Let N_i^B denote B&M census and K_i denote staffed B&M capacity. Similarly, define B&M utilization as

$$u^B(N^B, K) = \frac{N^B}{K}. \tag{11}$$

The capacity-pressure function is

$$m^B(N^B, K) = K\left[u^B(N^B, K)\right]^2 = \frac{(N^B)^2}{K}. \tag{12}$$

Holding staffed capacity fixed at its assessment value, the corresponding hospital-specific released-capacity component is

$$\begin{aligned} M^B_{ib} &= \mathbb{1}\{b(i) = b\}\, E^B_i \left[m^B(N^B_i, K_i) - m^B(N^B_i - 1, K_i)\right] \\ &= \mathbb{1}\{b(i) = b\}\, E^B_i\, \frac{2N^B_i - 1}{K_i}. \end{aligned} \tag{13}$$

This component increases with current utilization and expected remaining B&M duration. With E^B_i entering the payoff separately, the capacity coefficients capture how the value of releasing those bed-days varies with occupancy, conditional on other payoff inputs.

Let M^B_i collect the hospital-specific components $\{M^B_{ib}\}_{b\in\mathcal{B}}$, with corresponding coefficient vector λ_B. Consequently,

$$\lambda'_B M^B_i = \sum_{b\in\mathcal{B}} \lambda_{B,b} M^B_{ib} \tag{14}$$

is the measured B&M capacity contribution to the hospital's offer index, conditional on acceptance. Each hospital has its own coefficient on the smooth released-capacity component. Released capacity may be retained as operating slack or used to admit another patient. The model therefore identifies the composite private value of making additional capacity available. It does not separately identify the portion attributable to reduced operating pressure, improved patient throughput, or additional pecuniary profits from backfill (i.e., admitting patients waiting in the queue).

5.3 Dynamic Model Counterpart

Appendix E develops the dynamic counterpart to the static specification. Both use the same measured payoff components and distinguish offer-stage consequences from those contingent on acceptance. The static model already values expected discounted profits and capacity exposure over the remaining stay; the dynamic version additionally accounts for how today's transfer changes future operating states and subsequent offer decisions. The static specification permits joint maximum likelihood estimation, which is asymptotically efficient under correct specification and standard regularity conditions (Newey and McFadden, 1994). The dynamic implementation uses two-step GMM, whose efficiency depends on the moments and their weighting (Hansen, 1982); appropriately constructed two-step estimators can attain the efficiency of maximum likelihood (Aguirregabiria et al., 2021). The practical advantage of the static baseline is that it avoids the additional

transition estimation and continuation-value approximations required by the dynamic implementation. The dynamic estimates indicate that continuation values contribute little to the hospital's offer incentives, providing further support for the static specification. Unless otherwise explicitly stated, the main-text estimates and counterfactuals therefore refer to the static model.

6 Estimation and Identification

6.1 Offer Documentation

Estimation must account for incomplete documentation of HaH offers. As described in Section 4.1, the hospital-regime rejection-capture probabilities, q_i, are constructed from reporting patterns and fixed before estimation. Accepted placements are always observed; a rejected offer is recorded with probability q_i. These inputs link actual offers to the three observed EHR outcomes.

Let $D_i \in \{0,1\}$ indicate whether a rejected offer is documented. The observed outcome $Y_i \in \{a,d,n\}$ denotes an accepted placement ($O_i = 1$, $A_i = 1$), a documented rejection ($O_i = 1$, $A_i = 0$, $D_i = 1$), or no observed offer ($O_i = 0 \vee O_i = 1$, $A_i = 0$, $D_i = 0$), respectively. Conditional on p_i and ρ_i, the observed outcome probabilities are

$$\mathbb{P}(Y_i = a) = p_i \rho_i, \tag{15}$$

$$\mathbb{P}(Y_i = d) = p_i (1-\rho_i) q_i, \tag{16}$$

$$\mathbb{P}(Y_i = n) = 1 - p_i + p_i (1-\rho_i)(1-q_i). \tag{17}$$

The last outcome combines actual non-offers with latent rejected offers that were not documented. The probabilities in equations (15)–(17) sum to one for every assessment.

The maintained measurement assumption is that, conditional on hospital and documentation regime, q_i describes capture among rejected offers and does not independently shift offer behavior or acceptance utility after conditioning on the included controls. Appendix C.1 describes this adjustment. The empirical offer frequencies in Section 4.1 do not use fitted p_i or ρ_i; the posterior probabilities below are separate model objects used to update patient histories.

Let $\widehat{O}_i \equiv \mathbb{P}(O_i = 1 \mid Y_i)$ denote the posterior probability that a true offer occurred. Accepted placements and documented rejections reveal an offer, so $\widehat{O}_i = 1$ for these outcomes. When no offer is recorded, either no offer occurred or an offer was declined and not documented. Therefore, the posterior probability of a true offer when no offer is documented is

$$\widehat{O}_i = \mathbb{P}(O_i = 1 \mid Y_i = n) = \frac{p_i (1-\rho_i)(1-q_i)}{1 - p_i + p_i (1-\rho_i)(1-q_i)}. \tag{18}$$

These posterior probabilities update the patient's prior HaH interaction histories, which enter the demand and supply specifications below. The update occurs only after evaluating the current outcome and therefore affects future assessments.

6.2 Selection on Unobservables

Hospitals may condition offer decisions on information about patient willingness that is unavailable to the econometrician. For example, staff may observe home support, patient or caregiver reluctance, or aspects of clinical suitability that are related to the patient's willingness to accept HaH. As a result, patients who receive offers may be selected on unobserved acceptance utility, so estimating acceptance only among observed offers would confound patient preferences with offer selection. I account for this dependence by modeling the joint distribution of the patient and hospital encounter shocks:

$$\begin{pmatrix} \varepsilon_i \\ \nu_i \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \right], \qquad -1 < r < 1. \tag{19}$$

Conditional on the modeled information, these shock pairs are independent across assessments and independent of the time-invariant patient type ξ_i. The hospital observes ν_i when making the offer and therefore uses

$$\varepsilon_i \mid \nu_i = \nu \sim N\big(r\nu, 1 - r^2\big)$$

to form its beliefs about patient acceptance and surplus. Accordingly, ρ_i and Q_i in equations (5) and (7) become conditional on the realized hospital shock, denoted $\rho_i^H(\nu)$ and $Q_i^H(\nu)$. Let

$$\mathcal{O}_i = \{\nu : \Gamma_i(\nu) + \nu \geq 0\}, \qquad p_i = \int_{\mathcal{O}_i} \phi(\nu)\, d\nu, \tag{20}$$

where ϕ is the standard-normal density. Let $\Theta_D = (\beta_U, \theta, \alpha, \sigma_\xi)$ collect the demand parameters and let $\Theta = (\Theta_D, \Theta_S, r)$ collect all jointly estimated parameters. Beliefs about time-invariant preferences use observed assessment histories, with earlier private disturbances not retained in that history.

6.3 Joint Likelihood and Estimation

Forecast capacity inputs. In estimation, I replace contemporaneous census in equations (10) and (13) with forecasts of B&M and HaH census over the patient's remaining stay. Separate hospital-level B&M and program-wide HaH forecasts are estimated at multiple future horizons using only census and calendar information available at the assessment, with the forecasting models

updated monthly on an expanding historical window. Let $\widehat{N}_i^o(\tau)$ denote forecast census at horizon τ under setting $o \in \{B,H\}$ and let $S_i^o(\tau)$ denote remaining-stay survival. The exposure-weighted forecast census is

$$\overline{N}_i^o = \frac{\int_0^\infty \delta^\tau S_i^o(\tau)\widehat{N}_i^o(\tau)\,d\tau}{E_i^o}, \qquad o \in \{B,H\}, \tag{21}$$

with $\delta = 0.99$. Staffed B&M capacity and HaH staffing are held fixed at their assessment values. These forecasts capture expected pressure during the remaining stay without specifying a structural model of future admissions or discharges. This is the preferred specification because the value of a capacity release depends on congestion over the period in which the patient would otherwise occupy the bed, rather than only on the operating state at the assessment. Appendix A compares these estimates with an alternative that holds B&M and HaH census fixed at their assessment-time values over the remaining stay. Appendix C.3 provides the forecasting and exposure-construction details.

Conditional beliefs and offer-region integration. Let ξ_g index the support points for time-invariant patient preferences, and let w_{ig}^- denote the probability, immediately before assessment i, that patient $j(i)$ is represented by point g. Let

$$v_{ig} = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \sigma_\xi \xi_g, \qquad a_{ig}(\nu) = v_{ig} + r\nu.$$

Let $s_\varepsilon = \sqrt{1-r^2}$ denote the conditional standard deviation of the patient encounter shock. The node-specific conditional acceptance probability and the hospital's acceptance belief are

$$\rho_{ig}^H(\nu) = \Phi\left(\frac{a_{ig}(\nu)}{s_\varepsilon}\right), \tag{22}$$

$$\rho_i^H(\nu) = \sum_g w_{ig}^- \rho_{ig}^H(\nu), \tag{23}$$

where Φ and ϕ are the standard-normal distribution and density functions, respectively. Expected patient surplus from having the option to choose HaH, conditional on ν, is

$$Q_i^H(\nu) = \sum_g w_{ig}^- \left[a_{ig}(\nu)\Phi\left(\frac{a_{ig}(\nu)}{s_\varepsilon}\right) + s_\varepsilon \phi\left(\frac{a_{ig}(\nu)}{s_\varepsilon}\right)\right]. \tag{24}$$

The weights w_{ig}^- depend only on the patient's strictly prior observed history and are updated as described below. Because the hospital does not observe ξ_j, the offer region is common across preference nodes.

At each trial parameter vector, I locate all zero crossings of $\Gamma_i(\nu) + \nu$ and integrate over the

union of offer intervals. Define

$$J_{ig} = \int_{\mathcal{O}_i} \rho_{ig}^{H}(v)\phi(v)\,dv, \qquad J_i = \sum_g w_{ig}^{-} J_{ig}, \qquad \rho_i = \frac{J_i}{p_i}. \tag{25}$$

Here J_i is the joint probability of an offer and acceptance. Because v_i also informs acceptance and patient surplus, the offer region can contain multiple intervals; the calculation does not impose a unique cutoff.

Observed-outcome likelihood. Recall $Y_i \in \{a, d, n\}$ denotes an accepted offer, a documented rejection, or no observed offer, respectively. Conditional on preference node g, the probability of the observed outcome at assessment i is

$$\begin{aligned} \ell_{ig}(Y_i;\Theta) = J_{ig}^{\mathbb{1}\{Y_i=a\}} \left[q_i(p_i - J_{ig})\right]^{\mathbb{1}\{Y_i=d\}} \\ \times \left[1 - p_i + (1-q_i)(p_i - J_{ig})\right]^{\mathbb{1}\{Y_i=n\}}. \end{aligned} \tag{26}$$

Let $\mathcal{T}_j$ denote the chronologically ordered assessments for patient j that enter the likelihood, and let F_j^0 denote the distribution of the time-invariant patient taste immediately before the first assessment in $\mathcal{T}_j$. Conditional on the time-invariant patient effect, the patient-level likelihood contribution is the product of equation (26) across assessments. Integrating over the patient effect gives

$$\begin{aligned} L_j(\Theta) &= \int \prod_{i \in \mathcal{T}_j} \ell_i(Y_i;\Theta,\xi)\,dF_j^0(\xi) \\ &\approx \sum_{g=1}^{G} w_{jg}^0 \prod_{i \in \mathcal{T}_j} \ell_{ig}(Y_i;\Theta), \end{aligned} \tag{27}$$

where w_{jg}^0 denotes the node weight immediately before the first likelihood assessment for patient j. Because the data cover the HaH program from inception, there is no unobserved pre-program HaH interaction history to initialize. At a patient's first program assessment, $\mathcal{H}_i^- = \varnothing$ and the type weights equal the population quadrature weights w_g. Any history-only assessments preceding the patient's first likelihood observation subsequently update these weights, yielding w_{jg}^0. The joint log likelihood that is maximized is

$$\mathcal{L}(\Theta) = \sum_j \log L_j(\Theta). \tag{28}$$

Sequential integration over time-invariant heterogeneity. I approximate the distribution of time-invariant patient preferences using Gauss–Hermite quadrature, with transformed standard-normal nodes ξ_g and prior weights w_g. The estimator evaluates equation (27) recursively. Imme-

diately before assessment i, the predictive likelihood contribution and subsequent posterior update are

$$L_i(\Theta) = \sum_{g=1}^{G} w_{ig}^{-} \ell_{ig}(Y_i; \Theta), \tag{29}$$

$$w_{ig}^{+} = \frac{w_{ig}^{-} \ell_{ig}(Y_i; \Theta)}{\sum_{k=1}^{G} w_{ik}^{-} \ell_{ik}(Y_i; \Theta)}. \tag{30}$$

The updated weights become the prior weights for the patient's next assessment. At the beginning of the full observed patient history, the weights equal the Gaussian quadrature weights w_g. Earlier EHR assessments outside the likelihood sample may update these weights before the first assessment in $\mathcal{T}_j$, thereby initializing w_{jg}^{0}, but contribute zero directly to the objective.[9] Because the offer probability and predictive likelihood are evaluated using w_{ig}^{-} before the current outcome updates the weights, the current outcome does not enter its own acceptance utility or offer index. Appendix C provides additional details on computation and estimation. Standard errors use the observed information matrix, conditional on the generated inputs, and do not adjust for clustering or first-stage estimation uncertainty.

6.4 Identification

The fixed documentation adjustment and complete reporting in the modern regime distinguish the probabilities of actual non-offers and incompletely recorded rejections. Conditional on the fixed documentation probabilities, variation in acceptances, documented rejections, and no observed offers identifies the latent offer and acceptance probabilities. Repeated observations for the same patient identify time-invariant preference heterogeneity through serial dependence in acceptance and rejection outcomes.

Observed patient characteristics, prior interaction histories, and repeated decisions identify systematic and time-invariant heterogeneity in acceptance utility. The patient out-of-pocket price coefficient additionally requires variation in patient prices conditional on these characteristics. Because such variation is limited in the empirical application due to parity reimbursement and cost-sharing, patient utility is reported in its normalized choice scale rather than converted to dollars. The household beliefs and burden survey provides variation in the expected difference in non-medical spending between HaH and B&M care, but coverage over the full sample is scarce. Appendix D provides an in-depth discussion of the survey-augmented demand model and its results.

Separating type updating from direct history dependence relies on a sequential-independence

[9] If history-only assessments are interspersed with likelihood assessments, the objective sums $\log L_i(\Theta)$ over included assessments while retaining intervening updates.

assumption. Conditional on the patient's time-invariant type, current observed demand covariates, and the contemporaneous hospital shock,

$$\varepsilon_i \perp\!\!\!\perp \mathcal{H}_i^- \mid \xi_j, X_i, P_i, \nu_i. \tag{31}$$

Thus, prior HaH outcomes can predict future acceptance by changing the posterior distribution over ξ_j or through the direct utility shift $h(\mathcal{H}_i^-)'\theta$, but not through additional serial dependence in the encounter-specific patient shock.

Supply-side variables excluded from patient utility shift the offer probability and help identify selection into offers. These variables include forecast B&M capacity pressure, HaH staffing exposure, direct-variable savings, assessment department, and an indicator for the partial federal-shutdown. Identification requires these variables to be conditionally independent of unobserved acceptance utility after controlling for observed patient characteristics, clinical condition, hospital-by-quarter demand effects, and prior patient history. Under this exclusion restriction and the maintained joint-normal specification, variation in these offer shifters helps identify r through changes in acceptance among selected offers. The offer intercept is a net baseline term under the stated normalizations that captures the non-pecuniary marginal cost of offering the program to an eligible assessment.

The hospital payoff coefficients are identified in offer-index units because the variance of the normal offer disturbance is normalized to one. The direct-variable savings coefficient provides a dollar normalization. Ratios to β_Π express other offer-index components in profit-equivalent units when combined with a \$1,000 scaling conversion. The duration coefficient is identified from variation in expected remaining B&M days conditional on the other payoff inputs. The hospital-specific capacity coefficients use within-hospital variation in forecast occupancy and staffed capacity interacting with that duration, separately from the duration effect itself. The patient-surplus coefficient uses variation in expected patient surplus beyond acceptance-weighted placement consequences. The interpretation is conditional on the cost, duration, and census predictions, the functional form of the capacity terms, and the included clinical and operational controls. The resulting capacity contribution is composite, it does not separately identify the financial return from admitting another patient. Normalization and centering affect component levels, while changes in capacity value between two states are invariant to additive centering constants.

7 Results

7.1 Patient Acceptance

The panel structure uses repeated assessments of the same patient to distinguish time-invariant differences in willingness to accept HaH from changes associated with prior interactions, while updating beliefs about persistent preferences using strictly prior outcomes. Appendix Tables A.1 and A.2 compare time-invariant heterogeneity alone with specifications adding previously completed HaH treatment, prior acceptance and rejection counts, or both. The preferred specification in Table 2 includes all three history components. The estimated preference standard deviation (σ_ξ) indicates time-invariant unobserved heterogeneity across patients, while completing a first HaH treatment is positively associated with subsequent acceptance after accounting for these differences. There is no clear evidence of further increases with additional acceptances, suggesting an initial treatment-associated shift without cumulative reinforcement. Prior expected rejections remain negatively associated with acceptance beyond the information they provide through posterior updating. The main supply estimates remain broadly stable across these specifications, including when normal time-invariant heterogeneity is replaced by a discrete type distribution.

This distinction matters because expanding HaH requires hospitals to secure patient acceptance as well as identify clinically suitable transfers. Time-invariant preference differences suggest a role for targeting offers toward patients already willing to receive care at home. Prior decisions can help hospitals identify these patients and direct outreach more effectively. The first-treatment association raises a different possibility: willingness to accept may also change after experiencing HaH. If that association reflects an effect of treatment, facilitating a suitable patient's first use could increase acceptance at subsequent encounters, extending the benefit of outreach beyond the initial placement. Information about what treatment involves or support with an initial transition could therefore complement targeting based on existing preferences. Taken together, the estimates suggest that repeat acceptance reflects both time-invariant preferences and an additional shift associated with initial treatment. This provides a basis for targeting patients already inclined toward HaH, alongside suggestive evidence that facilitating suitable patients' first use could strengthen subsequent willingness to accept. The absence of clear gains from additional acceptances points toward an initial-exposure effect rather than cumulative reinforcement.

Conditional acceptance utility is lower for Black patients than for White patients and lower for Medicare patients than for commercially insured patients; the Medicaid difference is not statistically distinguishable from zero. Acceptance utility also declines with the neighborhood share of residents below 150 percent of the federal poverty line. Distance to the hospital, sex, and household size are not statistically distinguishable from zero in the joint model; neighborhood educa-

tional attainment is positively associated with acceptance at the 10 percent level. The demographic associations may reflect differences in household support, housing conditions, or preferences for home care, but the baseline coefficients do not separately identify these channels. Appendix D provides supplementary evidence from patient and caregiver responses.

The encounter-shock correlation is negative and precisely estimated. The estimate of -0.731 implies that a higher unobserved direct payoff from offering HaH is associated with lower unobserved patient willingness, after conditioning on observed covariates and time-invariant preference type. The coefficient is evidence against independent patient and hospital encounter shocks within the maintained specification.

7.2 Hospital Offers

Table 3 reports selected estimates from the hospital offer equation. The coefficients describe the relative importance of financial returns, patient acceptance and option value, capacity relief, and clinical concerns in the offer index. Their economic magnitudes depend on the units of the regressors and, for acceptance-contingent inputs, the probability of transfer.

Financial returns and patient surplus. The direct-variable profit coefficient is 0.179 per \$1,000. At an illustrative conditional acceptance belief $\rho_i^H(v) = 0.471$, corresponding to the model-implied average acceptance rate among offered patients, an additional \$1,000 of profit conditional on an accepted transfer raises the offer index by about 0.084 when the other components are held fixed. This however is a local financial incentive, so it does not imply that HaH has positive absolute accounting returns or lower total episode costs.

The patient-option-value coefficient is 0.861. A 0.10-unit increase in expected patient surplus $Q_i^H(v)$, measured in normalized utility units, raises this direct component of the offer index by 0.086 when the other components are held fixed. At the reference belief, that direct contribution is equivalent to about \$1,020 in additional profit conditional on an accepted transfer. Changes in expected acceptance also scale the financial, capacity, and clinical consequences in $\rho_i^H(v)\Delta W_i$. The estimates therefore indicate substantial targeting toward patients with greater expected surplus from having the HaH option, beyond the mechanical weighting of measured placement consequences. This is consistent with a mission-related benefit from providing patients with a valued treatment option.

Clinical concerns and offering frictions. Care burden, acute severity, and HaH escalation risk reduce offer value. Because escalation risk enters conditional on acceptance, a ten-percentage-point increase in risk changes the index by approximately $0.471(-0.820)(0.10) \approx -0.039$ at the

Table 2: Patient Acceptance Estimates

	Estimate	Standard error
Asian	0.129^{***}	(0.041)
Black/African American	-0.038^{*}	(0.020)
Latino/Hispanic	0.155^{***}	(0.029)
Medicaid	-0.019	(0.034)
Medicare	-0.075^{***}	(0.027)
Home-to-hospital distance	0.007	(0.007)
Age	0.056^{***}	(0.009)
Male	0.014	(0.012)
Neighborhood average household size	-0.000	(0.008)
Neighborhood college education share	0.018^{*}	(0.010)
Neighborhood log(per-capita income)	-0.015	(0.012)
Neighborhood share below 150% FPL	-0.019^{**}	(0.009)
Prior expected HaH declinations	-0.161^{***}	(0.031)
Ever previously treated with HaH	0.149^{***}	(0.035)
Count of prior HaH acceptances	-0.046	(0.030)
Condition severity weight	-0.034^{**}	(0.015)
Encounter-shock correlation, r_ν	-0.731^{***}	(0.031)
Time-invariant HaH preference SD, σ_ξ	0.228^{***}	(0.076)
Condition fixed effects	Yes	
Hospital-by-quarter fixed effects	Yes	
Assessments	31,944	
Patients	21,354	

Notes: Age, condition weight, home-to-hospital distance, household size, neighborhood household size, college share, log per-capita income, and neighborhood poverty share use fixed reference means and standard deviations. Prior acceptances and expected declinations are raw cumulative counts. Previously treated with HaH indicates a distinct prior HaH-treated encounter from this specific health system. Indicators are unstandardized. Omitted categories are White for race and Commercial for payor. The specification also includes condition and hospital-by-quarter fixed effects, and "other" race and payor categories. Standard errors are conditional on the generated inputs and use the observed information matrix. The preference-SD standard error uses the delta method. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

reference belief. Clinical burden and severity enter directly and are measured in standardized units.

The HaH-strain coefficient is negative and statistically significant. An increase in the uncentered forecast-weighted strain measure M_i^H from 26.7 to 41.7, spanning the middle 50% of assessment-level values, lowers the offer index by approximately $0.471(-0.004)(41.7-26.7) \approx -0.031$, holding the acceptance belief and other offer-index components fixed. This therefore provides evidence that the measured strain term discourages offers on a similar magnitude to pre-

dicted escalation risk. Both of these effects are modest when compared with the stronger incentives presented by financial returns and patient surplus. However, it does not establish that staffing constraints and incentives would remain constant under substantial program expansion outside the estimated support.

Strictly prior expected offers predict additional offers, consistent with repeated outreach, familiarity, and persistent targeting. The negative common offer term captures a baseline friction in initiating HaH offers, consistent with the staff time and coordination required to assess suitability, contact patients and caregivers, and arrange a potential transfer. These efforts begin before acceptance is known, so the hospital weighs them against the expected benefits of an accepted placement when deciding whom to offer HaH.

Capacity pressure. The hospital-specific linear B&M relief coefficients are positive and statistically significant at all four hospitals. The separate B&M exposure coefficient is negative, distinguishing the baseline association with remaining treatment duration from the value of capacity relief as forecast occupancy rises. Holding other inputs fixed, higher forecast occupancy increases the estimated capacity contribution at each hospital. The central finding in profit-equivalent terms is described below.

7.3 Profit-Equivalent Capacity Value

The profit coefficient permits the capacity-relief component to be expressed in profit-equivalent units. For assessment i, the model-implied gross private value of B&M capacity released by an accepted HaH placement is

$$\widehat{CV}_i^B = \frac{1{,}000}{\widehat{\beta}_\Pi} \widehat{\lambda}_{B,b(i)} E_i^B \frac{2\overline{N}_i^B - 1}{K_i}. \tag{32}$$

This measure values the focal patient's expected discounted B&M bed-days at the hospital's forecast occupancy and current staffed capacity. Because both capacity relief and accounting profits enter conditional on acceptance, their common acceptance multiplier cancels in the coefficient ratio. Multiplying $\widehat{CV}_i^B$ by ρ_i gives the expected capacity contribution per offer.

The point estimates imply positive capacity value at all four hospitals over observed operating conditions, with all four relief coefficients statistically distinguishable from zero. Dividing equation (32) by E_i^B gives the implied value per discounted bed-day released. Evaluating this capacity schedule at each hospital's median occupancy yields illustrative values of approximately \$1,570, \$1,150, \$1,180, and \$570 per discounted bed-day at hospitals A through D, respectively.[10] Hold-

[10]This calculation evaluates the capacity schedule at hospital-specific median operational occupancy among estimation-sample assessments: 91.8%, 89.3%, 82.4%, and 89.1% at hospitals A–D, respectively. Staffed capacity

Table 3: Hospital Offer Estimates

	Estimate	Standard error
Accounting profits (per $1,000)	0.179^{***}	(0.036)
HaH program strain	-0.004^{***}	(0.001)
HaH escalation risk	-0.820^{***}	(0.290)
Care-burden	-0.097^{***}	(0.007)
Acute-severity	-0.044^{***}	(0.006)
B&M stay-duration exposure	-0.353^{***}	(0.052)
Hospital A released-capacity	0.153^{***}	(0.026)
Hospital B released-capacity	0.115^{***}	(0.026)
Hospital C released-capacity	0.129^{***}	(0.027)
Hospital D released-capacity	0.058^{**}	(0.027)
Strictly prior expected offers	0.178^{***}	(0.015)
Patient option value	0.861^{***}	(0.052)
Baseline net offer term	-0.905^{***}	(0.043)
Supply-condition fixed effects	Yes	
Hospital fixed effects	Yes	
Department fixed effects	Yes	
Assessments	31,944	

Notes: All reported coefficients enter the offer equation additively. Acceptance-contingent variables are multiplied by $\rho_i^H(v)$. Direct-variable costs are centered within hospital-quarter-condition cells and expressed in $1,000 units. B&M relief, B&M stay-duration exposure, HaH strain, and escalation risk subtract fixed global reference means without rescaling. B&M relief and HaH strain use forecast census weighted by discounted remaining exposure, holding staffed capacity and HaH staffing fixed at assessment. Care burden and acute severity are standardized indices with zero-imputed underlying standardized scores. Prior expected offers are raw cumulative counts. The specification includes supply-condition, hospital, and department fixed effects as well as indicators for missing clinical measures. Standard errors are conditional on the generated inputs and use the observed information matrix. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

ing staffing and other inputs fixed, a ten-percentage-point increase in exposure-weighted forecast occupancy raises this value per discounted bed-day by approximately $171, $129, $144, and $65, respectively.

7.4 Model Fit

I assess the model's fit by comparing predicted and empirical outcome frequencies in aggregate and across hospital occupancy, calendar time, and predicted probabilities. These comparisons evaluate the model's ability to reproduce both overall placement rates and variation in hospital selection

is set to the corresponding hospital-specific sample median: 632, 500, 108, and 45 beds.

Table 4: Observed and Predicted Outcome Rates

Outcome	Empirical	Predicted	Difference
Accepted HaH placement	0.1028	0.1020	−0.0008
Documented rejection	0.0903	0.0885	−0.0018
No observed offer	0.8069	0.8095	0.0026
Latent offer, documentation-adjusted	0.2213	0.2167	−0.0046

Notes: Empirical rates are observed frequencies for the first three rows and a separately reconstructed, documentation-adjusted offer rate for the fourth. Recall, p_i denotes the predicted offer probability and J_i the predicted joint probability of an offer and acceptance, integrating over the hospital payoff shock and the strictly prior patient-taste distribution. The reported predicted rates average J_i, $q_i(p_i - J_i)$, $1 - J_i - q_i(p_i - J_i)$, and p_i, respectively. Differences are predicted minus empirical rates in probability units. The latent-offer reference is estimated rather than directly observed (Appendix C.1).

across assessments. All comparisons are in sample: parameters are estimated using the full estimation sample, while patient histories use strictly prior outcomes. Table A.3 reports outcome log loss and accepted-placement Brier scores for the baseline model and alternative specifications, both overall and for selected patient-history subgroups.

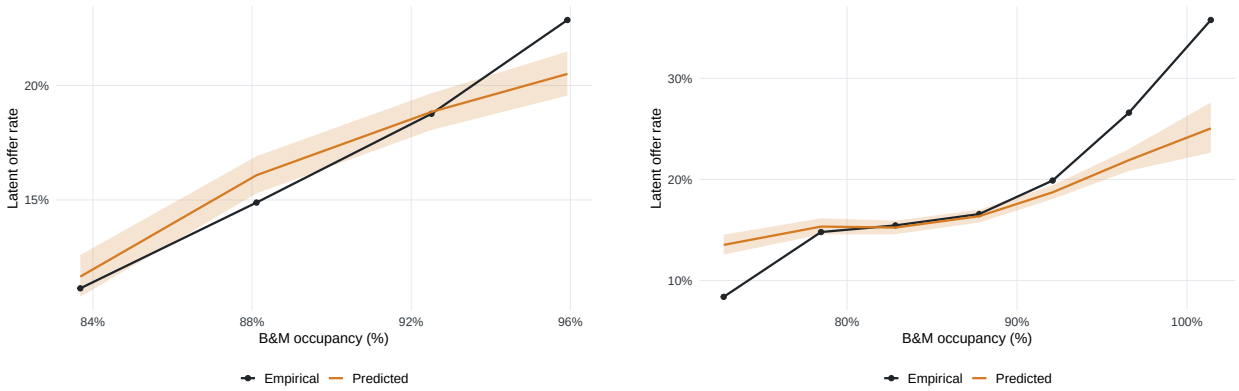
Table 4 shows close agreement between aggregate empirical and predicted frequencies. Each of the three observed outcomes is reproduced within 0.3 percentage points. Accepted placements are particularly closely matched, with a predicted rate of 10.2% compared with 10.3% observed. Predicted documented rejections and assessments with no observed offer are similarly close to their empirical counterparts. The predicted latent-offer rate of 21.7% also closely matches the separately constructed documentation-adjusted reference of 22.1%.

Figure 8 shows that the model captures the broad positive association between B&M occupancy and hospital offer rates. The fitted profiles reproduce differences in offer levels across hospitals and track the empirical profiles over much of the observed occupancy range. Agreement is especially close in several intermediate occupancy bins, although some empirical increases at the highest occupancy levels are steeper than the fitted profiles. These comparisons summarize fit across observed states. Figure 9 shows that the model also captures substantial variation in quarterly offer rates. Predictions follow the broad increases, declines, and subsequent fluctuations in the empirical series, while reproducing persistent differences across hospitals. Agreement is generally closer after the initial program period, with some early-regime offer rate extreme peaks and troughs less fully reproduced.

Figure 10 provides complementary evidence that predicted probabilities correspond closely to empirical outcome frequencies. Across accepted placements, documented rejections, and assessments with no observed offer, the calibration points generally follow a 45-degree line. The latent-offer panel likewise shows close agreement with the documentation-adjusted reference across the

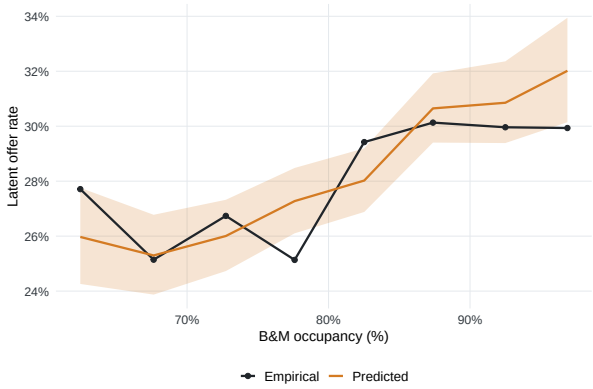
predicted-probability distribution. Small departures remain in selected bins, including underprediction in the highest accepted-placement decile. The accepted-placement panel evaluates the joint probability J_i of an offer and acceptance, while the latent-offer panel uses the separately reconstructed empirical reference. Across all fit comparisons, the estimated model closely reproduces aggregate outcome frequencies and critically captures the main patterns of variation across assessments, hospital occupancy, and calendar time. The calibration results further indicate that the model distinguishes assessments with substantially different probabilities of placement and other observed outcomes.

Figure 8: Empirical and Predicted Offer Rates by B&M Occupancy

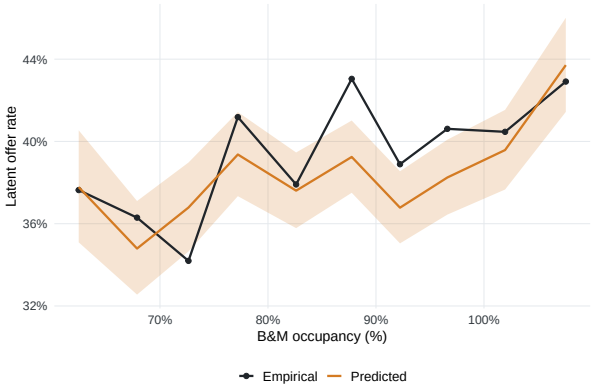


(a) Hospital A

(b) Hospital B



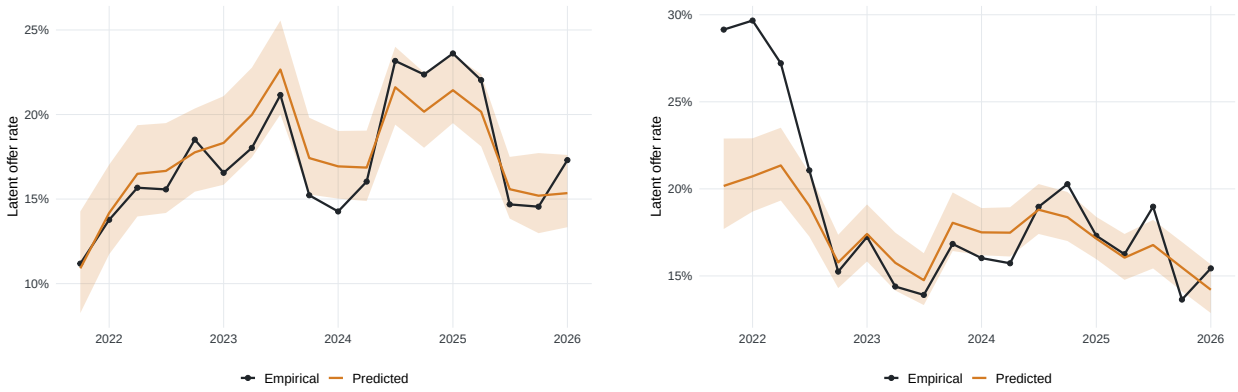
(c) Hospital C



(d) Hospital D

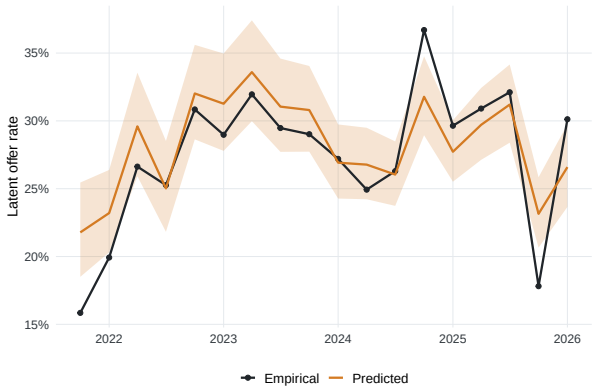
Notes: Points show assessment-weighted mean offer rates and mean contemporaneous occupancy within each hospital's occupancy bins. Empirical offer rates use the documentation-adjusted reference; modeled rates average predicted offer probabilities. Shading indicates pointwise 95% confidence bands for the modeled mean offer rates, calculated by the delta method using the full estimated parameter covariance matrix, including posterior and history updating. Generated inputs and documentation parameters are held fixed. All comparisons are in sample.

Figure 9: Empirical and Predicted Offer Rates by Quarter

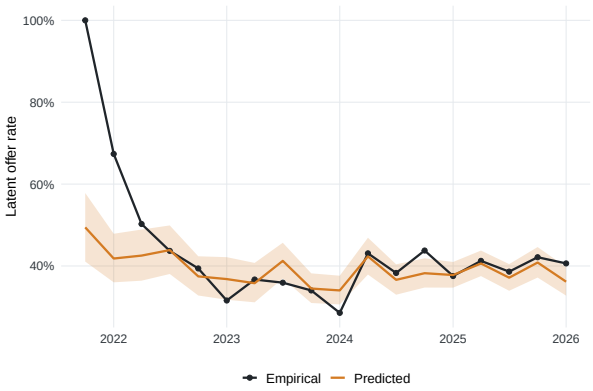


(a) Hospital A

(b) Hospital B



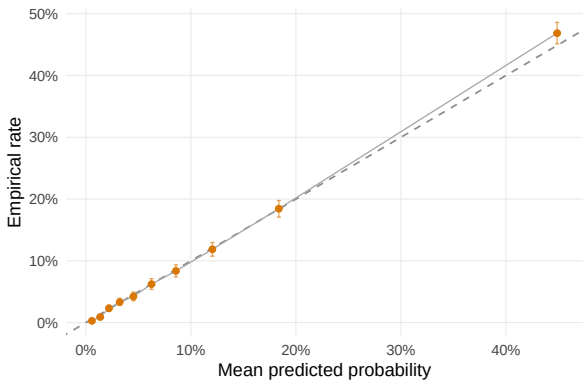
(c) Hospital C



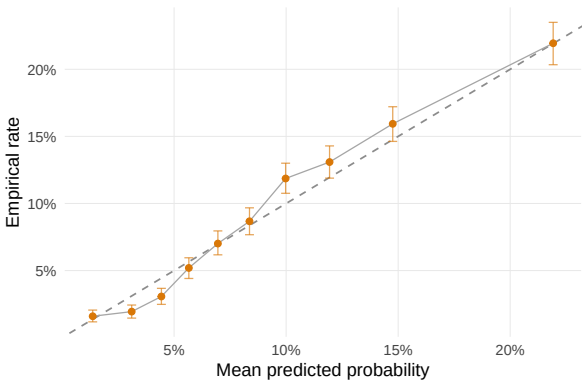
(d) Hospital D

Notes: Points show assessment-weighted mean offer rates within hospital-quarter cells. Empirical offer rates use the documentation-adjusted reference; modeled rates average predicted offer probabilities. Shading indicates pointwise 95% confidence bands for the modeled mean offer rates, calculated by the delta method using the full estimated parameter covariance matrix, including posterior and history updating. Generated inputs and documentation parameters are held fixed. Hospital D's first displayed quarter contains only two assessments, while the following quarter contains 101. The final quarter includes observations through February 2026. All comparisons are in sample.

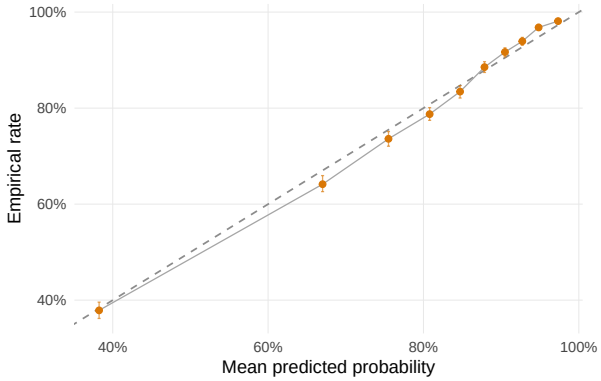
Figure 10: Calibration of Observed Outcomes and Documentation-Adjusted Offers



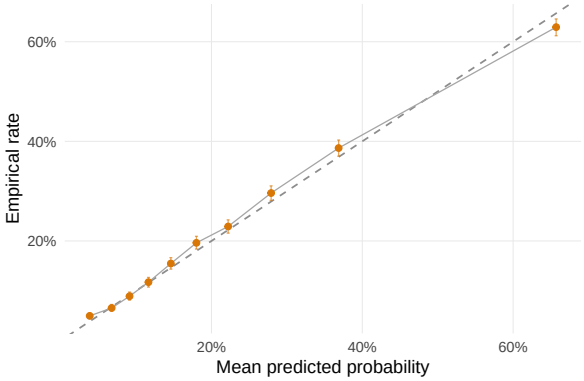
(a) Accepted placements



(b) Documented rejections



(c) No observed offers



(d) Latent offers

Notes: Each panel divides all modeled assessments into ten approximately equal-sized groups ranked by the corresponding model-implied predicted probability. Points compare the mean predicted probability with the empirical rate within each group; the dashed line indicates equality. Groups are formed separately for each panel. Vertical error bars indicate pointwise 95% confidence intervals for empirical rates, calculated using 2,000 patient-cluster bootstrap draws that retain each sampled patient's assessments together. Model estimates and predicted probabilities are held fixed. Panels (a)–(c) use observed outcome frequencies. Panel (d) uses the separately constructed documentation-adjusted offer reference, which is recomputed in each bootstrap draw with capture probabilities and capture hyperparameters held fixed. All comparisons are in sample.

8 Counterfactual Analysis

The counterfactual analysis examines how provider payment, geographic eligibility, capacity pressure, and information about patient demand affect offers and accepted transfers. I hold the estimated demand and supply parameters fixed, modify the affected patient or hospital inputs, and evaluate the offer policy in the new environment. Except in the information comparison described below, offer and joint offer–acceptance probabilities are computed from the conditional beliefs and offer-region integrals in equations (23) and (25).

Outcome accounting. Let $\mathcal{A}$ denote the set of assessment opportunities being evaluated. The mean true-offer rate is $|\mathcal{A}|^{-1}\sum_{i\in\mathcal{A}} p_i$, expected accepted placements are $\sum_{i\in\mathcal{A}} p_i\rho_i$, and acceptance among offers is $\sum_i p_i\rho_i/\sum_i p_i$. The latter weights acceptance by the probability that each patient receives an offer. Let T_i^B denote remaining B&M duration at assessment. Mean potential B&M bed-days released per assessment are

$$\frac{1}{|\mathcal{A}|}\sum_{i\in\mathcal{A}} p_i\rho_i\mathbb{E}(T_i^B \mid \mathcal{I}_i^O). \tag{33}$$

These are undiscounted days of focal B&M occupancy avoided under the model. They are not additional admissions or a prediction that beds remain empty. HaH exposure is calculated analogously using the remaining HaH duration.

Patient option value. Using the expected patient surplus $Q_i^H(v)$ defined in equation (24), expected option value created by access is

$$\int_{\mathcal{O}_i} Q_i^H(v) f_V(v)\, dv, \tag{34}$$

where $\mathcal{O}_i$ is the offer region in the counterfactual environment. No further acceptance multiplier is needed because $Q_i^H(v)$ already accounts for the option to decline. I report this normalized option value separately from hospital returns and bed-days, with its welfare interpretation conditional on the maintained demand specification and information structure. All reference means and standard deviations remain fixed at their estimation values. Counterfactual profit, capacity, duration, and strain inputs are transformed using those same references rather than re-centered after the shock. A change in supply can affect acceptance among offers by changing both the composition of offered patients and the hospital-payoff shock values at which offers are made. With demand inputs and pre-assessment preference weights held fixed, these changes reflect selection under the estimated correlated-shock distribution.

8.1 Provider Payment

There has been considerable discussion amongst policymakers surrounding the appropriate level of reimbursement for HaH care, with some insurers advocating for and negotiating towards lower than parity reimbursement levels. Therefore, the first experiment scales HaH reimbursement from 90% to 110% of the admission-specific payment received by the hospital, holding B&M reimbursement fixed. Under baseline payment parity, let R_i denote encounter i's observed total-episode reimbursement under either treatment location. A payment multiplier $m_P \in [0.90, 1.10]$ changes reimbursement for an accepted HaH placement to $m_P R_i$. The counterfactual profit input is therefore

$$\Delta\Pi_i(m_P) = \Delta\Pi_i + \frac{(m_P - 1)R_i}{1{,}000}, \tag{35}$$

where $m_P = 1$ reproduces the baseline and the estimation centering reference remains fixed. The adjustment applies to encounters eligible for HaH reimbursement at admission. This experiment changes provider returns without directly changing patient prices or other observed demand characteristics.

Figure 11 shows that higher HaH reimbursement increases potential B&M bed-days released, with larger gains from payment increases than losses from equally sized reductions. Across 31,944 assessment opportunities for 21,354 patients, baseline potential release is 14,289 bed-days. A 5% reimbursement increase adds 4,640 days, while a 10% increase adds 11,763 days; corresponding reductions decrease potential release by 2,582 and 4,355 days, respectively. This nonlinear response reflects changes in hospital offers and acceptance among selected patients.[11]

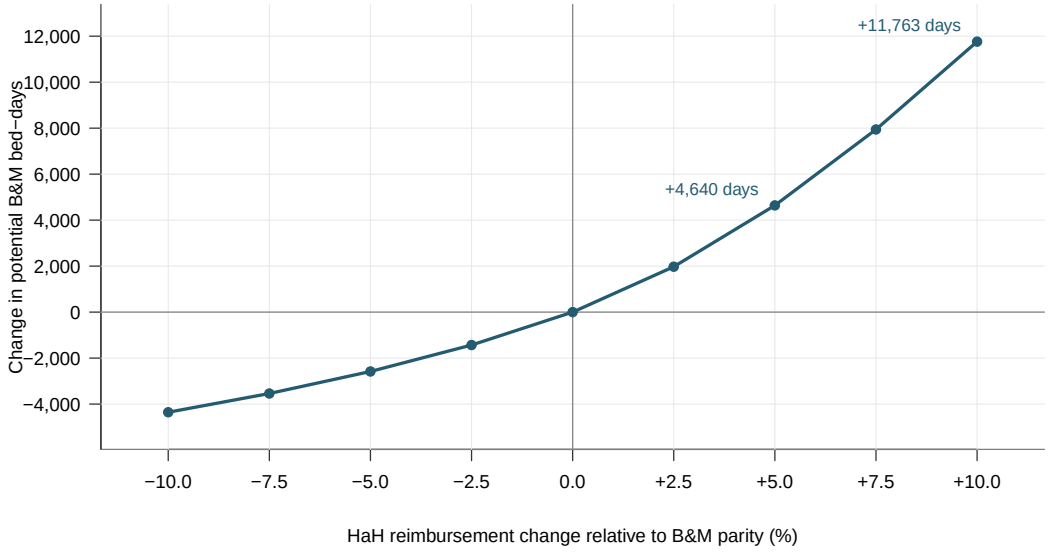
To value the additional throughput that released capacity could support, I apply hospital-specific contribution margins from the observed backfill population. For hospital b, let $\Delta BD_b(m_P)$ denote additional bed-days released, f_b the fraction filled, and c_b contribution dollars per occupied day. Potential additional hospital contribution is

$$\Delta V^{\$}(m_P) = \sum_{b \in \mathcal{B}} \Delta BD_b(m_P) f_b c_b. \tag{36}$$

Under persistent excess demand, I consider scenarios in which 50–100% of released bed-days are filled, allowing for delays, staffing constraints, and bed compatibility. Using historical hospital contribution margins to value this backfill, a 10% reimbursement increase generates \$7.8–\$21.1

[11] The simulations recompute offer and joint offer–acceptance probabilities under the estimated correlated-shock distribution, holding parameters, factual histories, pre-assessment preference weights, staffing, census, and forecast occupancy exposures fixed. These conditional assessment-level totals measure potential capacity release rather than realized system-wide throughput, since the exercise does not propagate policy-induced changes into subsequent histories, occupancy forecasts, or admissions.

Figure 11: Provider Reimbursement and Potential B&M Bed-Days Released



million in potential additional contribution across fill rates and accounting definitions. Additional HaH payments total approximately $11.4 million, including $4.1 million for baseline placements, while induced transfers increase remaining direct-variable treatment costs by approximately $15.5 million. The resulting $4.1 million reduction in focal contribution leaves a combined hospital cash gain of $3.7–$17.0 million.

Table 5 separately reports $13.7 million in incremental profit-equivalent B&M occupancy relief, calculated using equation (32) and the fixed forecast exposures. This is the gross structural value of the capacity-relief component, not additional accounting revenue or total provider surplus, and is not added to cash contribution. Its magnitude is substantial relative to the $7.8–$21.1 million potential backfill contribution, while the focal contribution loss offsets approximately 19–53% of those potential backfill gains.

8.2 Hospital Capacity Stress

The Covid-19 public health emergency catalyzed the system's adoption of HaH, motivating an assessment of how the program could strengthen system resilience during another emergency. This "Covid-2026" capacity-stress experiment examines the value of flexible home capacity when B&M demand rises, staffed capacity falls, and assessed patients shift toward infectious respiratory con-

Table 5: Reimbursement Increases, Released Capacity, and Hospital Contribution

HaH reimbursement increase	−10%	−5%	+5%	+10%
Additional B&M bed-days	−4,355	−2,582	4,640	11,763
Hospital contribution ($ millions)				
(1) Potential backfill contribution	[−7.45, −2.79]	[−4.44, −1.66]	[3.04, 8.25]	[7.75, 21.05]
(2) Focal contribution: (2.1)+(2.2)+(2.3)	3.89	2.18	−2.58	−4.09
(2.1) Induced transfer cost effect	6.29	3.69	−6.27	−15.48
(2.2) Reimbursement: baseline placements	−4.09	−2.05	2.05	4.09
(2.3) Reimbursement: induced placements	1.69	0.54	1.64	7.29
(3) Combined cash contribution: (1)+(2)	[−3.56, 1.10]	[−2.26, 0.52]	[0.45, 5.67]	[3.65, 16.95]
Structural profit equivalents ($ millions)				
B&M occupancy relief	−4.97	−2.96	5.42	13.65
Patient option value (firm valuation)	−3.74	−2.22	4.07	10.60
Patient option value				
Change relative to baseline (%)	−32.04	−18.98	34.85	90.79

Notes: All changes are relative to baseline reimbursement over the full assessment cohort, in February 2026 U.S. dollars ($). Reimbursement changes apply to eligible HaH episodes, with B&M payment unchanged. Estimated parameters, factual histories, pre-assessment preference weights, staffing, census, and forecast occupancy exposures are held fixed. Backfill ranges combine 50–100% fill with revenue including or excluding estimated settlements and direct or all reported variable costs. Combined cash contribution adds focal and potential backfill contribution within each accounting scenario. Structural occupancy-relief values use forecast-weighted capacity inputs and the model's $\delta = 0.99$ discount factor. Patient option value integrates expected patient surplus over the offer region, with no additional acceptance multiplier. Its percentage change is measured relative to aggregate baseline option value. The firm's valuation converts the change in option-value levels into profit-equivalent dollars using $1{,}000\widehat{\kappa}_Q/\widehat{\beta}_\Pi$ and does not measure patient willingness to pay. Both structural valuations are distinct from cash revenue and are reported separately, rather than added to combined cash contribution.

ditions. For hospital b, I increase the baseline B&M census forecast and reduce staffed capacity:

$$\widetilde{N}_i^B(t) = \widehat{N}_i^B(t) + \Delta N_{b(i)}, \qquad \Delta N_b \geq 0,$$
$$\widetilde{K}_i = m_{K,b(i)} K_i, \qquad 0 < m_{K,b(i)} \leq 1. \tag{37}$$

I reconstruct forecast-weighted census and the hospital-specific capacity inputs at the altered state, retaining the original centering references. The case-mix component reweights assessments within hospital-quarter cells toward infectious respiratory conditions, retaining each assessment's observed covariates and conditional cost, duration, and clinical predictions. This changes the composition of assessed patients and their average predicted remaining B&M and HaH exposure. Examining census, staffed capacity, and case mix separately before combining them distinguishes their

contributions to the response.

For each environment, I recompute offer regions and joint offer–acceptance probabilities under the estimated correlated-shock distribution. I report offers, accepted placements, potential B&M bed-days released, and profit-equivalent capacity value per discounted bed-day. The exercise evaluates conditional responses at stressed operating states; it does not simulate subsequent admissions or the evolution of census after additional HaH placements.

8.3 Information About Patient Demand

This information experiment compares pooled, baseline, and perfect knowledge of time-invariant patient preferences, holding patient utility, factual histories, and the estimated shock distribution fixed. All regimes retain the hospital's observation of its private payoff shock ν_i. Perfect information reveals the time-invariant preference type, but not the patient's encounter-specific choice shock.

Let w_{ig}^r denote the hospital's preference weights under information regime r, and let $Q_{ig}^H(\nu)$ denote expected patient surplus conditional on type g and hospital shock ν, as given by the node-specific expression in equation (24). The hospital's conditional acceptance belief and perceived patient surplus are

$$\rho_i^{H,r}(\nu) = \sum_g w_{ig}^r \rho_{ig}^H(\nu), \tag{38}$$

$$Q_i^{H,r}(\nu) = \sum_g w_{ig}^r Q_{ig}^H(\nu). \tag{39}$$

Both objects enter the offer decision: acceptance beliefs weight the consequences of a placement, while expected patient surplus enters directly.

The pooled benchmark uses population quadrature weights w_g; the baseline uses predictive weights w_{ig}^-. Under perfect information, the hospital's weights concentrate on the observed time-invariant type, yielding a type-specific offer region. Direct history terms in patient utility and hospital offer value remain at their factual values. Pooling therefore restricts the hospital's use of history to infer time-invariant preferences without erasing patients' experiences or removing the direct effects of prior interactions.

Hospital beliefs determine offers, while common evaluation weights preserve the underlying preference distribution across regimes. Let $\mathcal{O}_{ig}^r$ denote the offer region for type g under regime r. This region is common across types under pooled and baseline information, but can differ by type under perfect information. Offer probabilities, joint offer–acceptance probabilities, and expected

patient option value created by access are

$$
\begin{aligned}
p_i^r &= \sum_g w_{ig}^- \int_{\mathcal{O}_{ig}^r} f_\nu(\nu)\, d\nu, \\
J_i^r &= \sum_g w_{ig}^- \int_{\mathcal{O}_{ig}^r} \rho_{ig}^H(\nu) f_\nu(\nu)\, d\nu, \\
V_i^r &= \sum_g w_{ig}^- \int_{\mathcal{O}_{ig}^r} Q_{ig}^H(\nu) f_\nu(\nu)\, d\nu.
\end{aligned}
\tag{40}
$$

Replacing patient-specific preference weights with population weights reduces potential B&M capacity release by only 0.7 percent, indicating small incremental gains from posterior updating under the information otherwise retained in this comparison (Table 6). Perfect knowledge of time-invariant patient preferences produces substantially larger gains: offers increase by 4.4 percent, while accepted placements increase by 12.7 percent. Acceptance among offers rises from 47.1 to 50.8 percent, generating approximately 1,793 additional potential B&M bed-days and increasing patient option value by 23.5 percent. Better preference information therefore expands capacity relief through both broader access and improved targeting.

8.4 HaH Program Scale-Up

The program-scale experiment examines how additional HaH staffing capacity relaxes strain and changes offers and accepted placements. I increase effective HaH staffing from L_i to $\widetilde{L}_i = m_L L_i$, where $m_L \geq 1$. Holding forecast HaH census and remaining exposure fixed, the uncentered strain input becomes

$$
\widetilde{M}_i^H = E_i^H \frac{2\overline{N}_i^H + 1}{m_L L_i}. \tag{41}
$$

I apply the original centering reference and recompute offer regions and joint offer–acceptance probabilities under the estimated model.

Increasing effective HaH staffing by 50 percent raises expected placements by approximately 9 percent and generates 1,372 additional potential B&M bed-days (Table 6). Doubling staffing raises placements by approximately 14 percent and generates 2,129 additional days. The response is less than proportional to staffing expansion, indicating that relaxing measured program strain alone produces more limited gains under baseline operating conditions. These comparisons hold forecast HaH census, treatment costs, and duration predictions fixed and exclude additional staffing costs. They isolate the measured strain channel, assuming relationships estimated under observed operating conditions continue to apply at expanded staffing levels, without economies of scale, learning, operational reorganization, or endogenous growth in HaH census.

Table 6: Offers, Capacity Relief, and Patient Benefits Across Selected Counterfactuals

Panel A: Changes relative to baseline

Scenario	Offers	Accepted placements	Potential B&M bed-days released
Information: pooled	−18	−19	−96
Information: perfect	304	415	1,793
HaH staffing: +50%	418	293	1,372
HaH staffing: +100%	643	454	2,129

Panel B: Information, acceptance, and patient option value

Outcome	Pooled	Baseline	Perfect
Acceptance among offers (%)	46.9	47.1	50.8
Patient option value: change (%)	−1.0	0.0	23.5

Notes: All comparisons use the baseline model's assessment opportunities, holding estimated parameters and factual histories fixed. Baseline expected totals are 6,923 offers, 3,259 accepted placements, and 14,289 potential B&M bed-days released. Pooled information uses population preference weights; baseline information uses strictly prior predictive weights; perfect information reveals the time-invariant preference type. Direct history terms and encounter-shock information are retained, and outcomes use common factual predictive weights. Staffing increases relax the measured HaH strain term, holding forecast HaH census and other inputs fixed. Panel B reports the aggregate change in normalized expected patient option value relative to baseline.

9 Conclusion

Hospital-at-Home provides a flexible margin of inpatient capacity whose deployment depends on hospital selection and patient adoption. Hospitals determine which patients receive access, while patients determine whether that access translates into treatment at home and released B&M capacity. Jointly modeling these decisions connects patient willingness to the hospital's incentives to reallocate production when conventional capacity is constrained.

For the hospital, the value of an accepted transfer includes both the return to treating the patient at home and the opportunities created by released B&M capacity. The substantial private value of this release increases with occupancy and can motivate expansion even when induced transfers raise treatment costs. Reimbursement changes the financial return to a transfer, while the use of released beds determines how capacity relief translates into additional hospital contribution. Evaluating HaH therefore requires considering the consequences for the transferred patient alongside the capacity made available for other patients.

For patients, access to HaH provides a treatment alternative whose value differs across individuals. Hospital targeting determines who receives that option, and patient acceptance determines whether the associated capacity release occurs. Information about patient preferences can conse-

quently affect both access to a valued treatment option and the hospital's ability to translate offers into capacity relief. Expanding treatment resources alone does not determine how much effective capacity a program produces.

Improved information about time-invariant patient preferences improves the conversion of offers into accepted transfers, while additional HaH staffing relaxes program strain. The resulting capacity relief depends on how hospital offer incentives align with patients' willingness to receive care at home. Hospital financial returns, patient benefits, and social resource savings remain distinct: assessing efficiency also requires accounting for health outcomes, household inputs, and the consequences of congestion for other patients. More broadly, the paper provides a framework for flexible supply technologies whose deployment requires consumer adoption. Their value depends jointly on consumers' willingness to adopt, the returns to reallocating production, and the scarcity of conventional capacity.

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Appendices

A Supplementary Evidence and Model Specifications

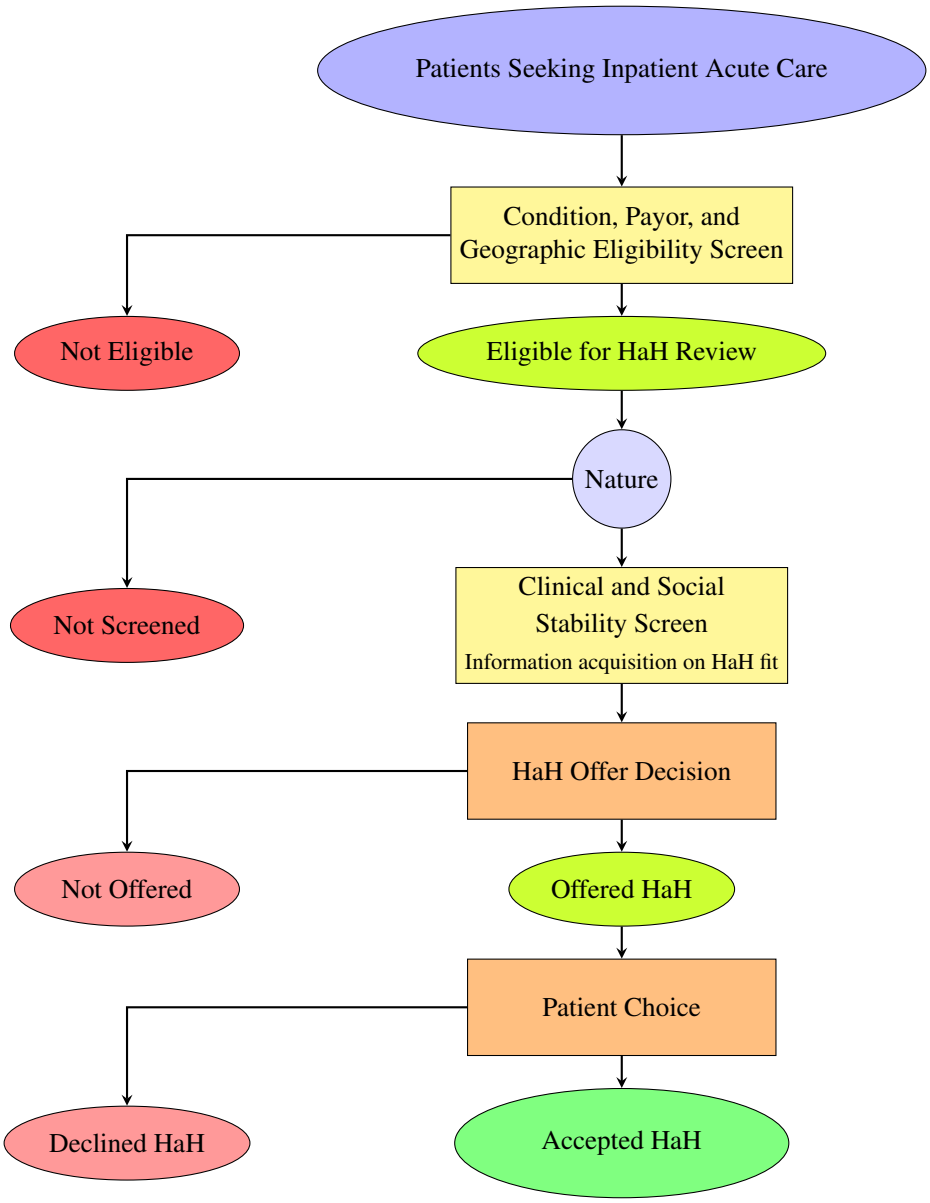
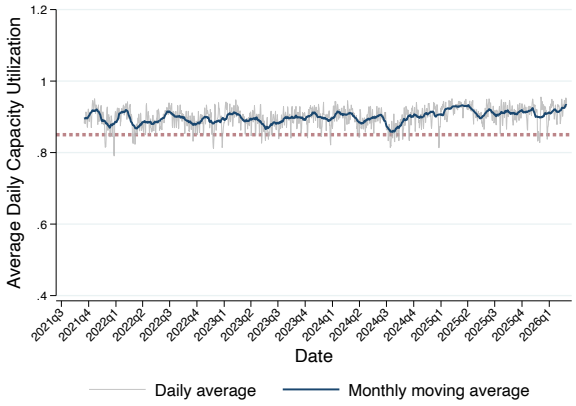


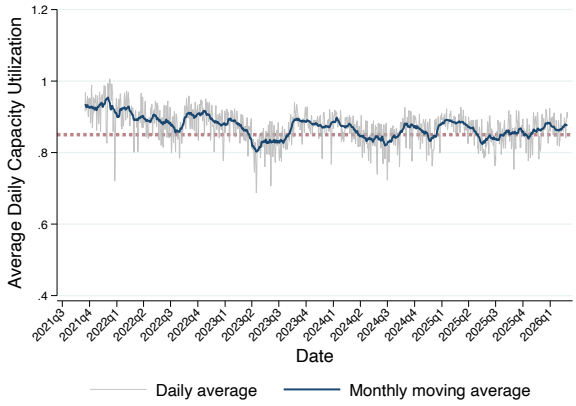
Figure A.1: Hospital-at-Home Patient Screening Process

Notes: Hospital-at-home programs allocate patients through sequential stages. First, hospitals apply condition, payor, and geographic eligibility criteria, although specific rules vary across systems and some programs are payor agnostic. Nature represents whether eligible patients are reviewed, allowing for "leakage" or operational oversight that are not due to active decisions. Clinical and social review may assess clinical stability, treatment and monitoring needs, home safety, caregiver support, and access to utilities and communication. Review findings inform the hospital's HaH offer decision. Offered patients choose whether to accept home-based care or remain in brick-and-mortar care. Dark red denotes eligibility or review exclusions, while light red denotes hospital or patient decisions.

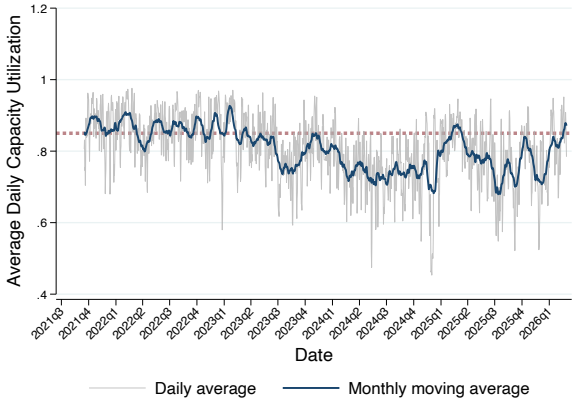
Figure A.2: Brick-and-Mortar Capacity Utilization Over Time



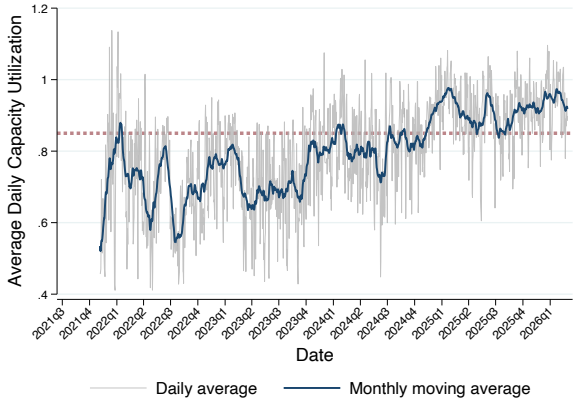
(a) Hospital A



(b) Hospital B

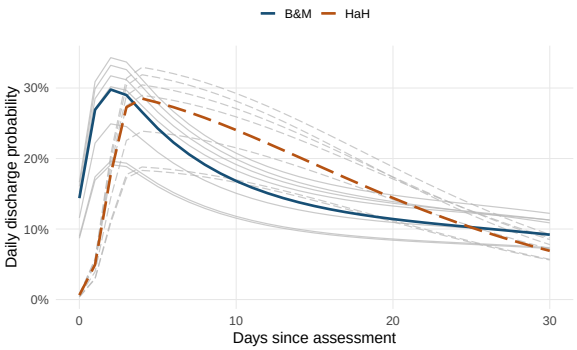


(c) Hospital C

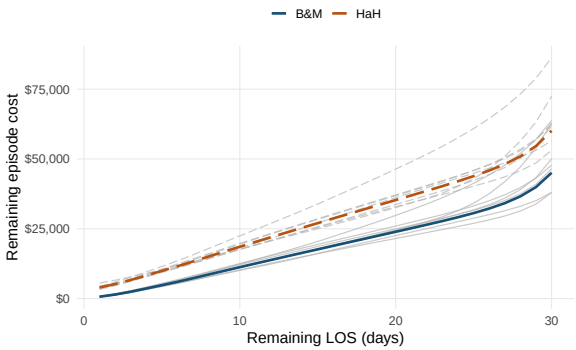


(d) Hospital D

Figure A.3: Predicted Discharge Hazards and Remaining Treatment Costs



(a) Discharge Hazards



(b) Treatment Costs

Time-invariant Preferences and Treatment History. Tables A.1 and A.2 compare alternative descriptions of demand history. Every specification retains time-invariant patient heterogeneity, joint-normal encounter shocks, forecast capacity inputs, and posterior updating. Column (1) excludes direct demand-history terms, column (2) adds previously completed treatment, and column (3) instead includes prior acceptance and expected rejection counts. The baseline in column (4) combines these terms. Column (5) uses the same history terms as the baseline model but replaces normal heterogeneity with a two discrete preference types. The baseline model has the lowest AIC among these specifications. Adding first treatment to the count-history model improves full-sample fit, with small reductions in in-sample terminal and first-post-treatment outcome log loss. Expected rejections retain a negative association with acceptance, while additional acceptances show no clear cumulative increase. Time-invariant heterogeneity remains, although its estimated magnitude falls and becomes less precise. The central supply coefficients remain broadly similar across specifications, including the two-type model. These results support the baseline's first-treatment flexibility.

Figure A.4 illustrates how prior choices inform the model's assessment of patient willingness. More prior acceptances are associated with higher posterior home preference, but substantial overlap indicates that acceptance counts alone do not fully summarize inferred preferences. This pattern illustrates the information available for patient targeting through posterior updating. Evidence on additional treatment-associated shifts comes from the joint estimates in Table 2.

Figure A.4: Posterior Home Preference by Prior HaH Acceptance History

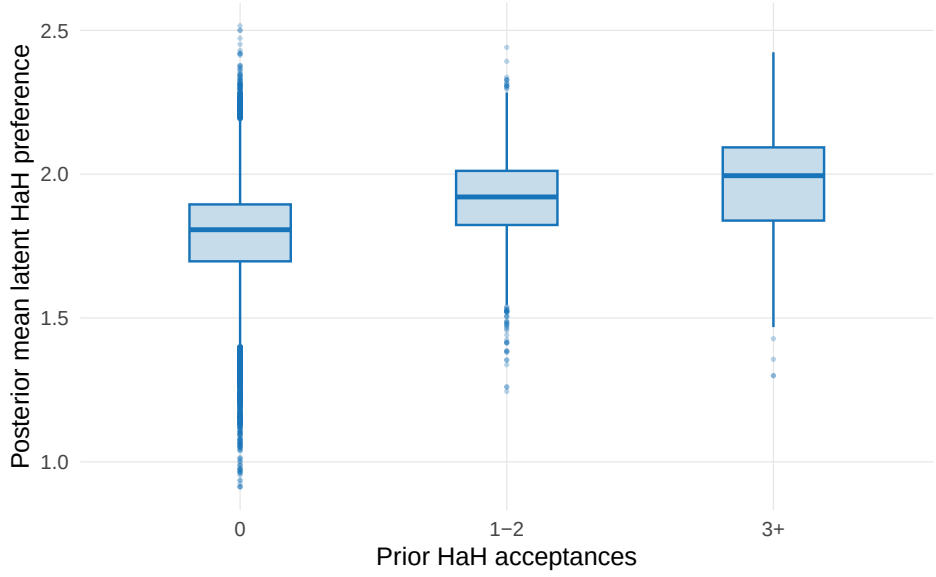


Table A.1: Demand History and Time-invariant Preferences

	Normally-distributed heterogeneity				Two types
	(1) No HaH history	(2) First treatment	(3) History counts	(4) Baseline specification	(5) Full history
Panel A: Patient acceptance					
Prior acceptance count	–	–	−0.013	−0.046	−0.048
			(0.032)	(0.030)	(0.031)
Expected prior rejection count	–	–	−0.132	−0.161	−0.159
			(0.034)	(0.031)	(0.033)
Previously completed treatment	–	0.070	–	0.149	0.150
		(0.034)		(0.035)	(0.038)
Encounter-shock correlation, r_ν	−0.749	−0.727	−0.763	−0.731	−0.731
	(0.030)	(0.033)	(0.029)	(0.031)	(0.031)
Time-invariant preference SD, σ_ξ	0.440	0.385	0.349	0.228	0.228
	(0.034)	(0.042)	(0.063)	(0.076)	–
Panel B: Hospital offers					
Profit (per $1,000)	0.205	0.199	0.191	0.179	0.179
	(0.038)	(0.037)	(0.036)	(0.036)	(0.036)
Patient option value	0.819	0.859	0.802	0.861	0.860
	(0.053)	(0.059)	(0.049)	(0.052)	(0.053)
B&M duration exposure	−0.389	−0.376	−0.378	−0.353	−0.353
	(0.054)	(0.054)	(0.053)	(0.052)	(0.052)
Hospital A occupancy relief	0.167	0.163	0.162	0.153	0.153
	(0.027)	(0.027)	(0.026)	(0.026)	(0.026)
Hospital B occupancy relief	0.125	0.123	0.120	0.115	0.115
	(0.027)	(0.027)	(0.026)	(0.026)	(0.026)
Hospital C occupancy relief	0.135	0.133	0.132	0.129	0.129
	(0.028)	(0.028)	(0.027)	(0.027)	(0.027)
Hospital D occupancy relief	0.060	0.063	0.056	0.058	0.058
	(0.029)	(0.028)	(0.028)	(0.027)	(0.028)
Fit					
Parameter count	169	170	171	172	173
Log likelihood	−16,217.4	−16,215.5	−16,207.9	−16,199.5	−16,199.4
AIC	32,772.8	32,771.0	32,757.8	32,743.0	32,744.8

Notes: All parameters are jointly re-estimated with common controls, fixed forecast and documentation inputs, posterior updating, and supply-history terms. History headings refer only to direct demand terms. Treatment requires a distinct prior HaH-treated encounter discharged before assessment. Counts are cumulative prior acceptances and expected declinations. Standard errors in parentheses use the observed Hessian conditional on generated inputs. Column (5) uses two ordered, mean-zero support points with estimated masses; its implied SD is reported without an SE. Normal and two-type heterogeneity are non-nested. Supply inputs retain the baseline units and structure.

Table A.2: Demand-History Restrictions and Sequential Prediction

Panel A: Nested likelihood-ratio tests

Expansion (columns)	Restrictions	LR statistic	p-value
First treatment: (2) vs. (1)	1	3.826	0.0505
History counts: (3) vs. (1)	2	18.980	0.0000756
Baseline: (4) vs. (3)	1	16.820	0.0000411

Panel B: In-sample outcome log-loss differences

Expansion minus restriction	Terminal assessments (21,354 rows)	First post-treatment (915 rows)
First treatment: (2) − (1)	−0.00006	−0.00532
History counts: (3) − (1)	−0.00029	+0.00052
Baseline: (4) − (3)	−0.00024	−0.01079

Notes: Column numbers refer to Table A.1. Panel A tests zero first-treatment coefficients or jointly zero acceptance and rejection count coefficients. The number of restrictions gives the χ^2 degrees of freedom. In Panel B, negative differences favor the expansion. Scores evaluate the three observed outcomes using strictly prior histories. Differences are calculated from reported log losses.

A.1 Alternative model specifications.

Column (3) of Table A.3 reports a static logit control-function alternative to the baseline model which uses the same forecast inputs, normal time-invariant patient heterogeneity, demand-history specification, and offer-documentation adjustment. This specification accounts for offer selection by including a term based on the offer probability in patient utility:

$$\Delta U_{ij}^{\Lambda} = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \sigma_\xi \xi_j + \varphi(1-p_i) + \varepsilon_i^{\Lambda}.$$

Here, $1-p_i$ is the generalized (score) residual from the logit offer equation, $O_i - p_i$, evaluated for an offered patient ($O_i = 1$). Including $\varphi(1-p_i)$ in acceptance utility serves as a control function for selection into offers, motivated by Petrin and Train (2010). Acceptance probabilities use the logistic distribution, and patient option value uses the corresponding logit inclusive value. The offer probability, acceptance belief, and option value are jointly determined through a fixed point: patient utility depends on the offer probability, while hospital offer incentives depend on anticipated acceptance and patient surplus. Parameters are jointly re-estimated under this specification. The coefficient φ captures selection through the utility adjustment, as opposed to the encounter-shock correlation r_ν under the baseline. This specification also supplies the first-stage offer policy and demand estimates for the dynamic comparison in Appendix E, preserving the estimated relationships between patient preferences, prior interactions, and offer selection. These first-stage objects are

Table A.3: Outcome Prediction Scores Across Static Specifications

Sample	Assessments	(1) Forecast census	(2) Contemporaneous census	(3) Control function (forecast census)
Panel A: Outcome log loss				
All assessments	31,944	0.50712	0.50722	0.50696
Repeat patients	16,402	0.50248	0.50269	0.50314
Terminal assessments	21,354	0.50170	0.50171	0.50111
First post-treatment	915	0.83069	0.83214	0.83004
Panel B: Accepted-placement Brier score				
All assessments	31,944	0.06937	0.06941	0.06918
Repeat patients	16,402	0.06819	0.06825	0.06809
Terminal assessments	21,354	0.06767	0.06768	0.06741
First post-treatment	915	0.16498	0.16559	0.16529

Notes: All specifications use the same estimation-sample assessments, with matched subgroup definitions. Column (1) represents the baseline model; Column (2) uses contemporaneous pressure exposures. Column (3) reports the logit control-function alternative with forecast occupancy exposure described in Appendix A.1. All three retain normal time-invariant patient heterogeneity and the full demand-history specification. Panel A reports the mean negative log predicted probability of the realized outcome: accepted placement, documented rejection, or no observed offer. Panel B reports the mean squared difference between the predicted joint probability of an offer and acceptance, J_i, and the accepted-placement indicator. Lower scores indicate better prediction within the same sample. Repeat patients includes all assessments of patients with multiple estimation-sample assessments. Terminal assessments includes each patient's last estimation-sample assessment, including singleton patients. First post-treatment includes the first qualifying assessment following prior HaH treatment, when that assessment is included in the estimation sample. All scores are in-sample sequential predictions: parameters are estimated using the full estimation sample, while patient histories use strictly prior outcomes.

held fixed while hospital payoff coefficients are re-estimated to incorporate continuation values. The logit structure permits a tractable conditional-choice-probability (CCP) inversion.

Table A.4 reports the joint-normal model using contemporaneous B&M and HaH census, rather than the forecast inputs used in the baseline. Patient-history and surplus estimates remain similar, while capacity coefficients are smaller. The log likelihood is $-16{,}203$, compared with $-16{,}200$ for the model in the main text.

Table A.4: Joint-Normal Estimates with Contemporaneous Census

	Estimate	Standard error
Panel A: Patient acceptance		
Asian	0.129^{***}	(0.042)
Black/African American	-0.036^{*}	(0.020)
Latino/Hispanic	0.158^{***}	(0.029)
Medicaid	-0.018	(0.034)
Medicare	-0.073^{***}	(0.027)
Home-to-hospital distance	0.007	(0.007)
Age	0.056^{***}	(0.009)
Male	0.015	(0.012)
Neighborhood average household size	-0.000	(0.008)
Neighborhood college education share	0.019^{*}	(0.010)
Neighborhood log(1 + per-capita income)	-0.015	(0.012)
Neighborhood share below 150% FPL	-0.019^{**}	(0.009)
Prior expected HaH declinations	-0.162^{***}	(0.031)
Ever previously treated with HaH	0.149^{***}	(0.035)
Count of prior HaH acceptances	-0.044	(0.031)
Condition severity weight	-0.034^{**}	(0.015)
Encounter-shock correlation, r_ν	-0.731^{***}	(0.030)
Time-invariant HaH preference SD, σ_ξ	0.226	(0.077)
Panel B: Hospital offers		
Accounting profits (per $1,000)	0.181^{***}	(0.036)
HaH program strain	-0.003^{***}	(0.001)
HaH escalation risk	-0.861^{***}	(0.296)
Care-burden	-0.097^{***}	(0.007)
Acute-severity	-0.044^{***}	(0.006)
B&M stay-duration exposure	-0.298^{***}	(0.040)
Hospital A released-capacity	0.119^{***}	(0.018)
Hospital B released-capacity	0.084^{***}	(0.019)
Hospital C released-capacity	0.086^{***}	(0.018)
Hospital D released-capacity	0.024	(0.019)
Strictly prior expected offers	0.179^{***}	(0.015)
Patient option value	0.858^{***}	(0.053)
Baseline net offer term	-0.917^{***}	(0.043)
Assessments	31,944	
Patients	21,354	
Log likelihood	$-16{,}202.628$	
AIC	32,749.26	

Notes: The contemporaneous occupancy alternative retains the baseline controls, fixed effects, input transformations, documentation probabilities, patient-history terms, and joint-normal encounter shocks, but replaces forecast census with contemporaneous census. Standard errors use the observed information matrix conditional on generated inputs and are not clustered. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Evidence on the Causal Health Effects of HaH

To estimate the causal effect of HaH on health outcomes, I exploit the program's geographic eligibility rule and implement a fuzzy spatial regression discontinuity design (RDD) (Lee and Lemieux, 2010; Keele and Titiunik, 2015). The complete design and analysis are developed in Capron et al. (2026). I summarize the econometric method, identifying variation, and estimates relevant to the demand and supply models here.

The program served patients residing within a fixed list of ZIP codes constructed to satisfy the 30-minute emergency-response requirement mandated by CMS. Using the operational ZIP-code list, rather than a reconstructed hospital isochrone, allows me to recover the eligibility boundary actually faced by patients. Panel (a) of Figure B.1 illustrates this boundary and the spatial distribution of the analytic sample and HaH receipt.

Let r_i denote patient i's signed distance in kilometers from the eligibility boundary, with $r_i < 0$ indicating residence inside the eligible area and $r_i \geq 0$ indicating residence outside. Geographic eligibility is therefore $E_i = \mathbb{1}\{r_i < 0\}$. Within bandwidth h around $r_i = 0$, the outcome equation follows the standard local-polynomial fuzzy-RD specification (Hahn et al., 2001; Calonico et al., 2014):[12]

$$Y_i = \alpha + \tau HaH_i + \gamma' X_i + \sum_{k=1}^{K} \left[\theta_k^- r_i^k E_i + \theta_k^+ r_i^k (1 - E_i) \right] + \varepsilon_i, \tag{B.1}$$

where $HaH_i = \mathbb{1}\{\text{treated with HaH}\}$ and X_i contains prespecified patient, admission, neighborhood, and hospital characteristics.

Because HaH receipt is endogenously determined, I instrument for HaH_i using geographic eligibility, E_i. Identification requires that potential outcomes and treatment receipt vary smoothly through the boundary absent the eligibility rule, that eligibility affects outcomes only through HaH receipt, and that eligibility weakly increases the probability of treatment. The discontinuity in the probability of receiving an HaH offer is shown in Panel (b) of Figure B.1. This offer discontinuity illustrates the access mechanism; the treatment first stage uses HaH receipt itself. The corresponding first-stage equation is

$$HaH_i = \pi_0 + \pi_1 E_i + \rho' X_i + \sum_{k=1}^{K} \left[\kappa_k^- r_i^k E_i + \kappa_k^+ r_i^k (1 - E_i) \right] + \upsilon_i. \tag{B.2}$$

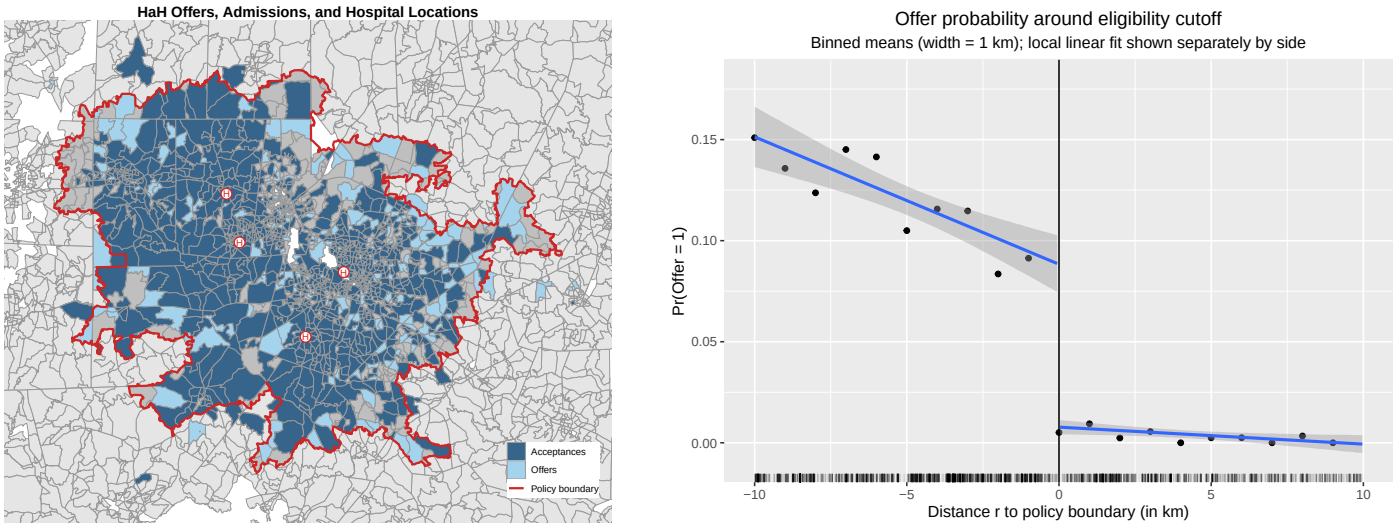
Letting ϕ_1 denote the reduced-form discontinuity in Y_i, the local average treatment effect for com-

[12] Flexible polynomials of order K are allowed to vary on either side of the cutoff. These polynomials are not required for identification but allow the relationship between distance and the outcome to differ across sides of the boundary.

pliers at the boundary is (Imbens and Angrist, 1994)

$$\tau = \frac{\phi_1}{\pi_1} = \frac{\lim_{r\uparrow 0}\mathbb{E}[Y_i \mid r_i = r] - \lim_{r\downarrow 0}\mathbb{E}[Y_i \mid r_i = r]}{\lim_{r\uparrow 0}\mathbb{E}[HaH_i \mid r_i = r] - \lim_{r\downarrow 0}\mathbb{E}[HaH_i \mid r_i = r]}. \tag{B.3}$$

Figure B.1: Program Treatment Boundary and Discontinuous Offer Probability



(a) Offers and Acceptances

(b) Discontinuity in HaH Offer Rate

The full set of results can be found in Table 1 of Capron et al. (2026). Residence inside the eligibility boundary increased HaH receipt by approximately 4 percentage points at the discontinuity, with first-stage F-statistics ranging from 45 to 159 across studied outcomes. The estimates point to mixed effects across margins: lower mortality and fewer falls during the acute-care episode, but more subsequent inpatient readmissions. Differences in longer-run mortality, length of stay, discharge destination, and several other outcomes were not statistically distinguishable from zero. HaH was also associated with fewer clinical orders, particularly imaging and consultations, without evidence that these services were simply shifted earlier in the inpatient stay prior to transfer home. Taken together, the results suggest broadly similar health outcomes across treatment settings, with evidence of both short-run benefits and downstream utilization costs. These local estimates motivate the baseline abstraction from direct health effects in Section 5, while escalation risk and acute-care needs remain determinants of HaH use. They do not establish equivalence across treatment settings or identify health effects for patients newly reached under expanded eligibility.

C Model Inputs and Estimation

C.1 Offer Documentation and Empirical Frequencies

Evidence from electronic health records indicates that flowsheet data miss some declined HaH offers, while accepted placements are observed throughout. I first use overlapping reporting fields to construct hospital-regime declination-capture probabilities. These probabilities are fixed inputs to both the joint demand–supply likelihood and a separate binomial model for reconstructing offers. Only the latter produces the bounded counts and rates used in the reduced-form evidence; those reconstructed outcomes are not used as inputs to the joint likelihood.

Capture of declinations across reporting regimes. New documentation procedures that made recording declined offers were rolled out throughout the program's tenure. Figure C.1 shows that older indicators remain in use after new fields appear. Their continued availability supports comparing capture across regimes. The adjustment assumes that capture by the older fields is transportable across adjacent regimes and that modern documentation is complete.

Let q_{br} be the probability that a declined offer is documented at hospital b in regime r, with $q_i = q_{b(i),r(i)}$. Modern capture is normalized to $q_{b,\text{modern}} = 1$. Within each hospital,

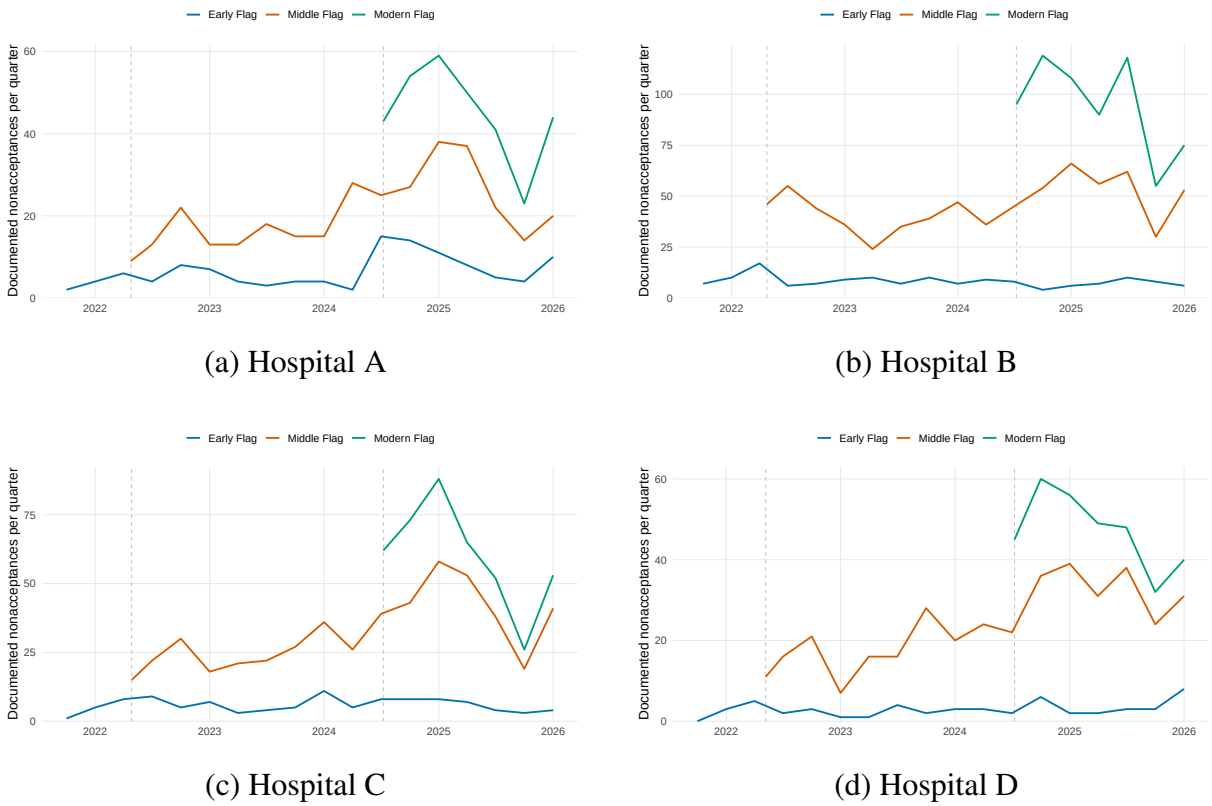
$$q_{b,\text{middle}} = \frac{\text{modern-regime declines with either older flag}}{\text{all modern-regime declines}},$$

$$q_{b,\text{early}} = q_{b,\text{middle}} \times \frac{\text{middle-regime declines with social-stability evidence}}{\text{all observed middle-regime declines}}.$$

Table C.1 reports the declination-capture probabilities used in joint maximum likelihood estimation. Each assessment is assigned $q_i = q_{b(i),r(i)}$ before estimation. These values are then held fixed at every likelihood evaluation and are not estimated with the demand and supply parameters. The joint model fits the observed acceptances, documented declines, and no-observed-offer outcomes using these capture probabilities. The table also reports implied overall offer capture, including acceptances, but only as a descriptive summary that it is not used as an additional likelihood input.

Reconstructing reduced-form offers. Directly scaling documented declines by $1/q_i$ can imply more offers than assessments for certain hospital-regime cells. I therefore fit a separate binomial model that uses the number of non-accepted assessments to limit possible hidden offers and pools information across months. Let c index hospital–month–regime cells, split at reporting transitions, with M_c total assessments, A_c acceptances, D_c documented declines, and $Z_c = M_c - A_c$ denoting

Figure C.1: Offer Documentation Across Reporting Regimes



(a) Hospital A

(b) Hospital B

(c) Hospital C

(d) Hospital D

Notes: Quarterly documented non-acceptances carrying each reporting flag. An assessment may carry multiple flags. Dashed vertical lines mark reporting transitions.

Table C.1: Reporting Regimes and Capture Probabilities

	Declination capture, q_{br}			Implied overall offer capture		
Hospital	Early	Middle	Modern	Early	Middle	Modern
A	0.163	0.675	1.000	0.521	0.867	1.000
B	0.108	0.572	1.000	0.421	0.750	1.000
C	0.156	0.665	1.000	0.553	0.829	1.000
D	0.088	0.665	1.000	0.423	0.833	1.000

Notes: Overall capture is $(A+D)/(A+D/q)$ within hospital-regime, where A and D count acceptances and documented declines.

total non-acceptances. If θ_c is the probability of a declined offer among non-accepted assessments,

$$D_c \mid Z_c, \theta_c \sim \text{Binomial}(Z_c, q_c\theta_c), \qquad \text{logit}(\theta_c) = \mu_{b(c),r(c)} + u_c, \quad u_c \sim N(0, \tau^2).$$

Hospital-regime means and a shared monthly variance are estimated by marginal maximum likelihood with normal quadrature, using the same fixed capture rates, $q_c = q_{b(c),r(c)}$. This empirical-Bayes fit is performed separately, outside the joint demand–supply estimator. The reconstructed offer count is

$$\widetilde{O}_c = A_c + D_c + (Z_c - D_c)\widehat{\mathbb{E}}\left[\frac{(1-q_c)\theta_c}{1-q_c\theta_c}\,\middle|\, D_c, Z_c, q_c\right].$$

The expectation averages the hidden-offer probability over the cell posterior at the fitted hyperparameters. Thus known offers count as one, while each no-observed-offer assessment receives this posterior probability. Counts lie between observed offers and assessments; modern counts equal observed offers. Group rates divide summed reconstructed counts by summed assessments. These are the reduced-form outcomes, replacing raw inverse-capture totals in the empirical evidence presented throughout. Table C.2 summarizes the reconstruction. Partial pooling stabilizes cell counts with little change in the aggregate total. The resulting latent offers are plotted in Figure C.2.

Table C.2: Descriptive Offer Reconstruction

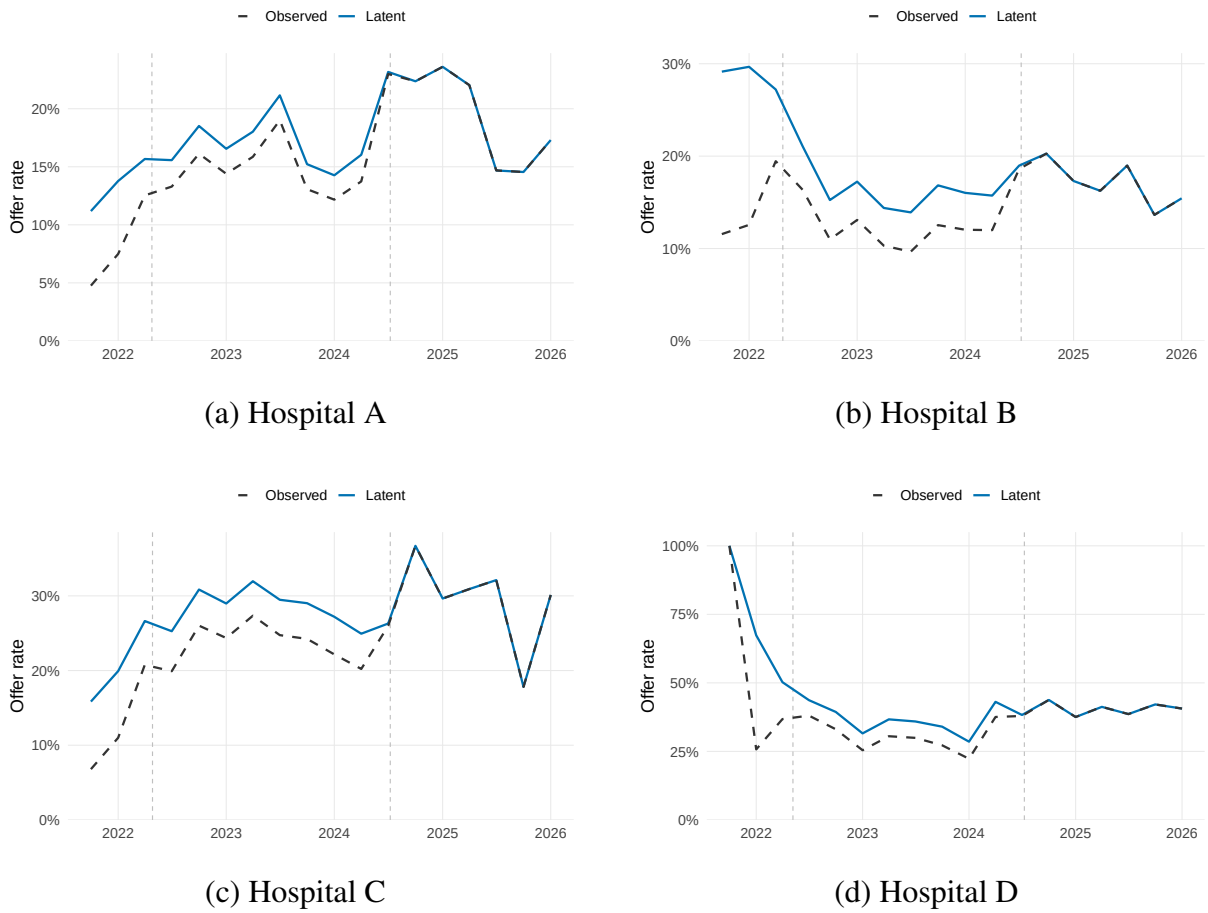
Assessments	31,944
Hospital–month–regime cells	211
Monthly heterogeneity SD, $\widehat{\tau}$	0.214
Observed offers	6,169
Raw inverse-capture total (benchmark)	7,071.0
Bounded reconstructed offers	7,070.7
Marginal log likelihood	−558.1
Complete-pooling log likelihood	−571.7
Early-cell held-out log-score difference	−0.052

Notes: Complete pooling sets $\tau = 0$ within hospital-regime. Log likelihoods include binomial constants. The held-out comparison leaves out one early-period cell at a time, with the negative difference indicating no predictive gain over complete pooling in those cells.

C.2 Input Transformations

Selected regressors are centered or standardized to express their effects relative to a common baseline and in interpretable units. The reference means and standard deviations used for these trans-

Figure C.2: Recovered Latent Offer Rates



(a) Hospital A

(b) Hospital B

(c) Hospital C

(d) Hospital D

Notes: Dashed vertical lines mark reporting transitions.

formations are fixed before estimation and retained in counterfactuals, so changes reflect altered inputs rather than a changing benchmark. Census, staffing, and duration enter their physical schedules in raw units; the transformations apply to the resulting regressors. The offer-index equations suppress this transformation in the notation used in the main text.

Standardized covariates. Demand covariates standardized by a reference mean and SD are age, condition weight, home-to-hospital distance, and census block group characteristics (household size, college share, log per-capita income, and share below 150% federal poverty line). Supply inputs use the latest valid NWASS, Morse, and SOFA records strictly before the actual assessment timing. Care burden averages standardized NWASS and Morse scores and then rescales the resulting index; acute severity uses rescaled SOFA. Missing component scores contribute zero on the standardized scale, with a separate indicator for each of the three measures. No additional clinical scores enter demand, and missing scores do not remove assessments. A standardized coefficient

refers to one reference SD, and dividing it by that SD gives the coefficient per original unit.

Centered payoff inputs. The profit regressor is the conditional HaH-minus-B&M dollar return, less its fixed hospital-quarter-condition reference, divided by \$1,000, with equation (C.13) providing the mapping. Hospital-specific B&M relief, B&M stay-duration exposure, HaH strain, and escalation risk subtract fixed global means without rescaling. Their coefficients retain the original units.

Acceptance, histories, and indicators. Payoffs that are conditional on acceptance use uncentered $\rho_i^H(\nu)$. Prior accepted placements, expected rejections, and expected true offers enter as raw cumulative counts. Categorical variables, fixed-effect indicators, flags for missing fields, and the federal shutdown indicator remain untransformed.

References and component values. For clarity here, write $M_i^{B,\mathrm{raw}}$ for the physical capacity vector and $\overline{M}^B$ for its fixed reference. Then

$$\rho_i^H(\nu)\lambda_B'(M_i^{B,\mathrm{raw}} - \overline{M}^B) = \rho_i^H(\nu)\lambda_B' M_i^{B,\mathrm{raw}} - \rho_i^H(\nu)\lambda_B'\overline{M}^B.$$

Centered component levels therefore depend on the reference. Section 5 suppresses this distinction, and the gross capacity-value measure in equation (32) uses the uncentered input. Differences in that measure across states retain their interpretation under a common reference.

C.3 Exposure Inputs and Numerical Estimation

Remaining treatment exposure. Let T_i^o denote counterfactual remaining duration in days under location $o \in \{B,H\}$, with its distribution predicted from information $\mathcal{I}_i^O$ at assessment. Superscripts B and H refer to B&M and HaH, as in Section 5.2. Let ℓ_i^{LOS} denote elapsed LOS, distinguishing it from the likelihood contribution $\ell_{ig}(Y_i;\Theta)$. The discharge models vary with hospital, condition, location, elapsed LOS, and day of week. Under the baseline zero transfer delay, the HaH path begins at the assessment.

For duration $T \geq 0$, the discounted exposure schedule is

$$A_\delta(T) = \sum_{t=0}^{\lfloor T \rfloor - 1} \delta^t + (T - \lfloor T \rfloor)\delta^{\lfloor T \rfloor}, \qquad \delta = 0.99, \tag{C.1}$$

where an empty sum is zero. The last term prorates a fractional final day. The exposure inputs

used in the supply model are

$$E_i^o = \mathbb{E}\Big[A_\delta(T_i^o) \mid \mathcal{I}_i^O\Big], \qquad o \in \{B,H\}. \tag{C.2}$$

Thus discounting is applied to each possible remaining duration before averaging over its predictive distribution.

To connect these inputs to discharge predictions, write $\mathcal{S}_i^o(u) = \mathbb{P}(T_i^o > u \mid \mathcal{I}_i^O)$ for remaining-stay survival. Then

$$E_i^o = \sum_{t=0}^{\infty} \delta^t \int_t^{t+1} \mathcal{S}_i^o(u)\,du. \tag{C.3}$$

For integer-valued durations this reduces to $\sum_{t\geq 0} \delta^t \mathcal{S}_i^o(t)$. If $\mu_i^o(t)$ is the exit probability for this duration during $(t,t+1]$ conditional on remaining in care at t, survival on the daily grid satisfies

$$\mathcal{S}_i^o(t) = \prod_{v=0}^{t-1}\{1-\mu_i^o(v)\}, \qquad \mathcal{S}_i^o(0) = 1.$$

Remaining B&M and HaH exposure therefore use their respective discharge paths. Section 5.2 multiplies these discounted days by capacity increments evaluated at exposure-weighted forecast census and the assessment's staffing state. The undiscounted bed-days in Section 8 instead use $\mathbb{E}[T_i^o \mid \mathcal{I}_i^O] = \int_0^\infty \mathcal{S}_i^o(u)\,du$.

Histories and sequential likelihood. Assessments are processed chronologically within patient. An acceptance increments prior accepted placements by one; a documented rejection increments prior rejections by one. A no-observed-offer outcome increments expected rejections by the posterior $\widehat{O}_i$ in equation (18). Expected true offers increase by $\widehat{O}_i$ for every outcome, with $\widehat{O}_i = 1$ for accepted placements and documented rejections. All increments occur after evaluating the current predictive likelihood, and all parameter-dependent histories are reconstructed at each trial vector $\Theta = (\Theta_D, \Theta_S, r)$. Using the likelihood notation of Section 6.3, the objective is

$$\mathcal{L}(\Theta) = \sum_j \sum_{i\in\mathcal{T}_j} \log L_i(\Theta), \qquad L_i(\Theta) = \sum_g w_{ig}^- \ell_{ig}(Y_i;\Theta). \tag{C.4}$$

The recursion updates node weights and histories at every retained assessment, including observations with zero direct objective weight. If all such observations precede the included history, this objective equals the integrated patient-level likelihood in equation (27). If they are interspersed, equation (C.4) specifies the selected predictive contributions with the intervening updates retained.

Gauss–Hermite quadrature uses 20 standard-normal nodes ξ_g and normalized prior weights w_g. The posterior mean of the time-invariant utility component before assessment i is $\sigma_\xi \sum_g w_{ig}^- \xi_g$; the

mean of the standardized latent type is $\sum_g w_{ig}^{-} \xi_g$.

Optimization and inference. I maximize the joint log likelihood using a quasi-Newton algorithm with analytic gradients, locating all zero crossings of $\Gamma_i(v) + v$ and integrating over the resulting offer intervals. Derivatives account for the conditional beliefs, offer-region boundaries, and sequential updates to patient histories and preference weights. Parameters are internally rescaled for numerical stability and returned to their original units for reporting. Standard errors use the observed information matrix and the delta method where needed.

C.4 Treatment Cost Model

The profit input in equation (6) uses predicted direct-variable costs remaining at assessment. Its construction combines a common LOS schedule, a HaH transfer increment, and a HaH-day cost difference. These are conditional accounting predictions rather than causal treatment effects.

Episode-level cost specification. For the episode associated with assessment i, let C_i^{ep} denote total direct treatment cost, L_i^{ep} total LOS, and h_i^{ep} HaH days. Let $D_i^H = \mathbb{1}\{h_i^{\text{ep}} > 0\}$ and let $c(i)$ denote the condition group. The specification is

$$C_i^{\text{ep}} = a_{c(i)} + g_{c(i)}(L_i^{\text{ep}}) + D_i^H s_{c(i)} + r_{c(i)} h_i^{\text{ep}} + Z_i^{C\prime}\gamma_C + \varepsilon_i^C, \tag{C.5}$$

$$g_c(L) = b_{c0}L + \sum_{k\in\{3,7,14,30\}} b_{ck}(L-k)_+, \tag{C.6}$$

where the transfer increment s_c captures the additional episode cost associated with entering HaH, while r_c measures the difference in cost per HaH day relative to the common schedule. Neither coefficient is constrained by its label to be positive. The controls Z_i^C include DRG-family, hospital, fiscal-year, age, sex, ED-entry, and condition-weight variables.

The model is estimated by penalized least squares on episode totals, selecting the ridge penalty by held-out root mean squared error across five calendar blocks. The common LOS schedule comes from the full screened sample used in the joint demand-supply estimation model. The HaH transfer and daily-rate increments come from a the sub-sample of documented-offer encounters. Therefore, I impose the assumption that their conditional predictive relationship extends beyond observed offers in order to form predictions for the full assessment sample.

Allocation to the remaining stay. Using the elapsed LOS and remaining durations defined in Appendix C.3, the common cost schedule for an integer remaining duration T is

$$G_c(\ell^{\text{LOS}}, T) = \sum_{t=0}^{T-1} \delta^t \left[g_c(\ell^{\text{LOS}} + t + 1) - g_c(\ell^{\text{LOS}} + t) \right]. \tag{C.7}$$

With zero transfer delay and A_δ from equation (C.1), conditional remaining costs are

$$C_i^B(T) = G_{c(i)}(\ell_i^{\text{LOS}}, T), \tag{C.8}$$

$$C_i^H(T) = G_{c(i)}(\ell_i^{\text{LOS}}, T) + s_{c(i)} + r_{c(i)} A_\delta(T). \tag{C.9}$$

The transfer increment is incurred at transfer, while the daily increment applies only to remaining HaH exposure. Fractional final-day cost increments are retained, using the same exposure convention as E_i^H. The supply model inputs average the two schedules over their respective predicted remaining-duration distributions:

$$\widehat{C}_i^B = \mathbb{E}\left[G_{c(i)}(\ell_i^{\text{LOS}}, T_i^B) \mid \mathcal{I}_i^O \right], \tag{C.10}$$

$$\widehat{C}_i^H = \mathbb{E}\left[G_{c(i)}(\ell_i^{\text{LOS}}, T_i^H) \mid \mathcal{I}_i^O \right] + s_{c(i)} + r_{c(i)} E_i^H. \tag{C.11}$$

Already-incurred costs and common episode controls cancel from the contrast. The common LOS schedule contributes when predicted duration differs by location, separating the duration channel from the HaH transfer and daily-rate increments.

Profit units and counterfactual use. Let $\Delta\Pi_i^{\$}$ denote the uncentered dollar profit difference between HaH and B&M conditional on an accepted placement. Under the baseline reimbursement-parity assumption, as supported by negotiated contracts and observed reimbursements, the profit difference becomes

$$\Delta\Pi_i^{\$} = \widehat{C}_i^B - \widehat{C}_i^H. \tag{C.12}$$

For the offer equation, the input is

$$\Delta\Pi_i = \frac{\Delta\Pi_i^{\$} - \overline{\Delta\Pi}^{\$}_{b(i),\text{qtr}(i),c(i)}}{1{,}000}, \tag{C.13}$$

where $\text{qtr}(i)$ denotes the quarter assigned to assessment i for reference-cell construction. Acceptance weights this input once through $\rho_i \Delta W_i$. A positive coefficient favors higher conditional returns, holding other components fixed, rather than implying positive absolute HaH profits. Payment counterfactuals add the dollar reimbursement change before applying the original normalization.

Hospital return outcomes use uncentered costs and revenues, keeping hospital accounting returns distinct from social resource savings.

D Survey Augmentation of Patient Choice Model

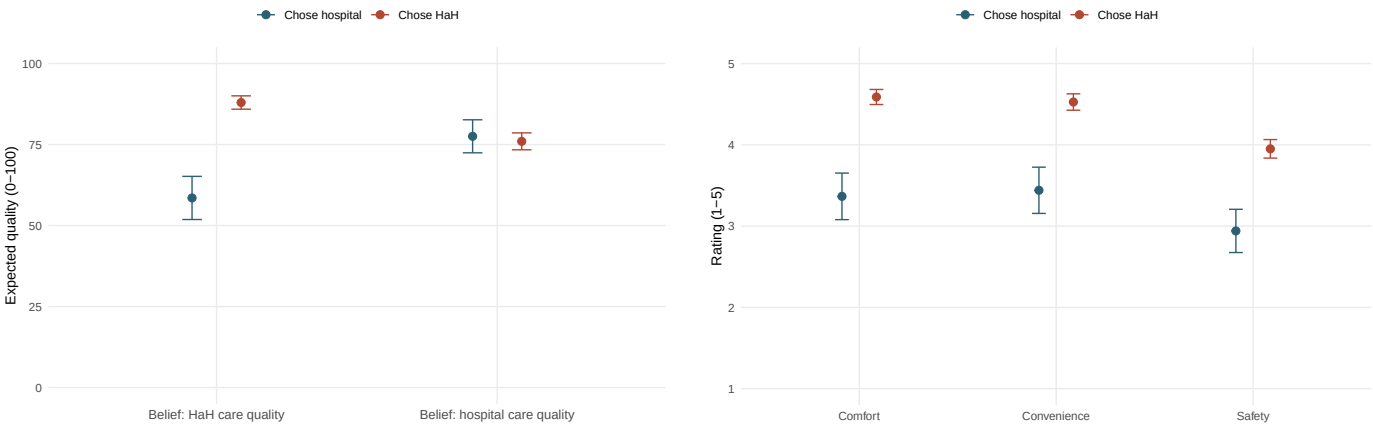
This appendix augments the joint demand–supply model in Section 5 with a novel survey measuring household beliefs over expected relative care quality and relative comfort, safety, and convenience under HaH, as it compares to B&M care. Survey answers enter the model as part of time-invariant patient preferences and are not part of the hospital's information set in estimation. Non-medical spending responses are relatively sparse and therefore enter only as a supplemental specification at the end of this appendix.

Table D.1: Survey Respondents and Reported Information

	Count (N)
Unique patients linked to a survey	453
Patient respondents	359
Caregiver respondents	94
Complete expected-quality difference	404
Any expected-experience information	453
Complete comfort, safety, and convenience ratings	450
Complete expected non-medical spending responses	297

Notes: Caregiver responses answer questions phrased to address the patient's expected beliefs and therefore are assumed to measure the associated household's care-choice preferences.

Figure D.1: Expected Care Quality and HaH Experience by Reported Choice

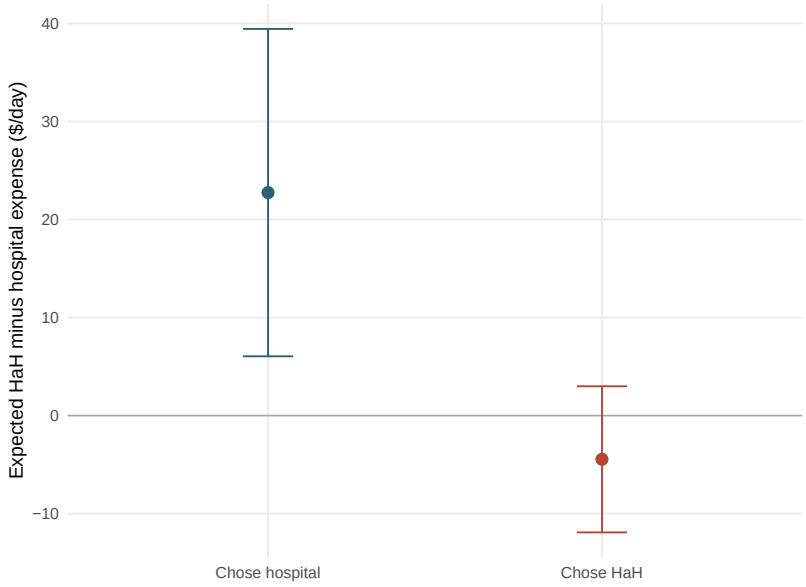


(a) Expected Quality

(b) Expected Experience

Notes: The left panel uses expected care-quality ratings on a 0-to-100 scale. The right panel uses a one-to-five Likert scale for expected comfort, convenience, and safety ratings. Points report respondent means and bars report 95 percent confidence intervals.

Figure D.2: Expected Non-Medical Spending Difference by Reported Choice



Notes: The outcome is expected daily household spending under HaH minus expected daily household spending under hospital care, summed across transportation, meals, child care, pet care, cleaning, and paid caregiver support. Points report means and bars report 95 percent confidence intervals.

D.1 Survey-Augmented Demand Model

Expected quality. Let $\Delta Q_j = \mathrm{Q}_j^H - \mathrm{Q}_j^B$ denote patient j's reported expected-quality difference between HaH and B&M, each rated from 0-to-100. I standardize this difference as

$$Q_j = \frac{\Delta Q_j - \bar{\Delta}_Q}{s_{\Delta Q}}, \qquad \bar{\Delta}_Q = 4.61, \quad s_{\Delta Q} = 34.64, \tag{D.1}$$

using the sample mean and standard deviation among the 404 respondents with both ratings. Higher Q_j indicates more favorable expected HaH quality relative to B&M. Conditional on first-assessment covariates X_j^0, I specify

$$Q_j \mid X_j^0 \sim N(\mu_j^Q, s_Q^2), \qquad s_Q = 0.958,$$

where μ_j^Q is a ridge-regression prediction. The prediction rule and residual standard deviation are held fixed during joint estimation. Observed scores are treated as exact measurements of a time-invariant quality-expectation trait, while missing scores are integrated out. Hospital beliefs use this conditional distribution, updated with strictly prior assessment outcomes, without observing the later survey response.

Expected experience. Let $F_j \sim N(0,1)$ denote a persistent expected-experience factor. For component $m \in \{\text{comfort}, \text{safety}, \text{convenience}\}$, let $Y_{jm}^S \in \{1,\ldots,5\}$ denote the ordered rating. The ordered-probit measurement model is

$$Y_{jm}^S = k \quad \Longleftrightarrow \quad \tau_{m,k-1} < \lambda_m^S F_j + e_{jm}^S \le \tau_{m,k}, \qquad e_{jm}^S \sim N(0,1), \tag{D.2}$$

where $\tau_{m,0} = -\infty$, $\tau_{m,5} = \infty$, and the four finite cutpoints are ordered. Conditional on F_j, component errors are independent, so an observed response contributes

$$h_{jm}(f) = \Phi\left(\tau_{m,Y_{jm}^S} - \lambda_m^S f\right) - \Phi\left(\tau_{m,Y_{jm}^S - 1} - \lambda_m^S f\right). \tag{D.3}$$

The loadings and cutpoints are estimated in a first-stage model and then held fixed. The loadings are 2.389 for comfort, 1.228 for safety, and 1.466 for convenience; higher F_j therefore indicates more favorable expected HaH experience. Missing components contribute a factor of one. Quality and experience are patient-specific traits that apply across the patient's assessments. The interpretation treats later answers as measurements of preferences present at the time of choice.

Augmented utility and hospital beliefs. Using the demand notation in Section 5.1, and letting superscript S denote the survey-augmented demand system, the survey adds two terms to time-invariant patient utility:

$$\Omega_j = \sigma_\xi \xi_j + \theta_Q Q_j + \theta_F F_j, \qquad \xi_j \sim N(0,1), \tag{D.4}$$

$$\Delta U_{ij}^S = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \Omega_j + \varepsilon_i. \tag{D.5}$$

The encounter shocks follow the baseline joint-normal distribution, with r_ν re-estimated. The reported survey models fix $\sigma_\xi = 0$, so the residual term drops out of equation (D.4). The baseline non-medical price difference is zero in this extension. Conditional on first-assessment covariates, the prior distribution of (Q_j, F_j, ξ_j) is

$$\mathsf{G}_j^0 = N(\mu_j^Q, s_Q^2) \times N(0,1) \times N(0,1). \tag{D.6}$$

This distinguishes the residual preference SD, σ_ξ, from the total conditional SD of time-invariant private utility,

$$s_\Omega = \left(\sigma_\xi^2 + \theta_Q^2 s_Q^2 + \theta_F^2\right)^{1/2}. \tag{D.7}$$

Let G_i^- denote the distribution of these traits updated with the patient's strictly prior observed assessment outcomes. For a trait vector $\mathbf{z} = (\tilde{q}, \tilde{f}, \xi)$, define

$$v_i^S(\mathbf{z}) = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \theta_Q \tilde{q} + \theta_F \tilde{f} + \sigma_\xi \xi, \qquad b_i^S(\mathbf{z}, \nu) = v_i^S(\mathbf{z}) + r_\nu \nu.$$

The type-specific acceptance probability conditional on the hospital disturbance is

$$\rho_i^S(\nu; \mathbf{z}) = \Phi\left(\frac{b_i^S(\mathbf{z}, \nu)}{s_\varepsilon}\right), \qquad s_\varepsilon = \sqrt{1 - r_\nu^2}. \tag{D.8}$$

The hospital integrates over its prior-history trait distribution to form

$$\rho_i^{H,S}(\nu) = \int \rho_i^S(\nu; \mathbf{z})\, d\mathsf{G}_i^-(\mathbf{z}),$$

$$Q_i^{H,S}(\nu) = \int \left[b_i^S(\mathbf{z}, \nu)\Phi\left(\frac{b_i^S(\mathbf{z}, \nu)}{s_\varepsilon}\right) + s_\varepsilon \phi\left(\frac{b_i^S(\mathbf{z}, \nu)}{s_\varepsilon}\right)\right] d\mathsf{G}_i^-(\mathbf{z}).$$

Here $Q_i^{H,S}$ is the conditional option surplus, distinct from the patient's surveyed quality trait Q_j.

With D_i denoting the non-surplus direct offer terms, the offer probability is

$$\begin{aligned} \Gamma_i^S(\nu) &= D_i + \kappa_Q Q_i^{H,S}(\nu) + \rho_i^{H,S}(\nu)\Delta W_i, \\ \mathcal{O}_i^S &= \{\nu : \Gamma_i^S(\nu) + \nu \geq 0\}, \qquad p_i^S = \int_{\mathcal{O}_i^S} f_\nu(\nu)\, d\nu. \end{aligned} \tag{D.9}$$

Here $\tilde{q}$ and $\tilde{f}$ denote draws of the time-invariant quality and experience traits Q_j and F_j, respectively. Survey answers inform the shared preference parameters through the measurement likelihood, but do not update G_i^- or enter the supply index directly. The historical offer–acceptance filter integrates the Gaussian composite Ω_j using 20 Gauss–Hermite nodes. Survey measurement integration is evaluated separately.

Augmented likelihood. Let $\widetilde{\Theta}$ collect the free baseline model parameters together with θ_Q and θ_F. Conditional on a trait vector $\mathbf{z} = (\tilde{q}, \tilde{f}, \xi)$, define the integrated joint placement probability

$$J_i^S(\mathbf{z}) = \int_{\mathcal{O}_i^S} \rho_i^S(\nu; \mathbf{z}) f_\nu(\nu)\, d\nu.$$

The contribution of observed assessment outcome $Y_i \in \{a, d, n\}$ is

$$\ell_i^S(Y_i; \mathbf{z}, \widetilde{\Theta}) = \begin{cases} J_i^S(\mathbf{z}), & Y_i = a, \\ q_i\{p_i^S - J_i^S(\mathbf{z})\}, & Y_i = d, \\ 1 - p_i^S + (1 - q_i)\{p_i^S - J_i^S(\mathbf{z})\}, & Y_i = n. \end{cases} \tag{D.10}$$

Its predictive probability is $L_i^S = \int \ell_i^S \, d\mathsf{G}_i^-$, and posterior weights update after the current outcome as in Section 6.3. Here q_i retains its baseline meaning as the documentation probability, distinct from the quality draw $\tilde{q}$. The same assessments from the baseline joint demand-supply contribute to the objective. Let $\mathcal{M}_j$ contain the observed experience components and define $H_j(\tilde{f}) = \prod_{m \in \mathcal{M}_j} h_{jm}(\tilde{f})$. For any trait distribution G, the joint density or probability of patient j's observed survey information is

$$m_j^S(\mathsf{G}) = \begin{cases} g_Q(Q_j^{\text{obs}}; \mathsf{G}) \displaystyle\int H_j(\tilde{f})\, d\mathsf{G}(\tilde{f}, \xi \mid Q_j = Q_j^{\text{obs}}), & Q_j \text{ observed}, \\ \displaystyle\int H_j(\tilde{f})\, d\mathsf{G}(\tilde{q}, \tilde{f}, \xi), & Q_j \text{ missing}, \end{cases} \tag{D.11}$$

where $g_Q(\cdot; \mathsf{G})$ is the marginal density of Q_j under G. An observed quality score enters as a density evaluated at its reported value. Missing quality is integrated out, and a patient with no observed survey components has $m_j^S = 1$.

Let G_j^Y denote the terminal trait distribution after the patient's complete observed assessment history, updated without survey answers, and let $\mathcal{R}$ denote patients with survey responses. The normalized survey-augmented log likelihood is

$$\mathcal{L}^S(\widetilde{\Theta}) = \sum_j \sum_{i \in \mathcal{T}_j} \log L_i^S(\widetilde{\Theta}) + \sum_{j \in \mathcal{R}} \log \left\{ \frac{m_j^S(\mathsf{G}_j^Y)}{m_j^S(\mathsf{G}_j^0)} \right\}. \tag{D.12}$$

Each patient's survey contributes once, irrespective of their number of assessments. The denominator is the marginal measurement density under the fixed first-stage prior and is constant with respect to the jointly estimated parameters. The normalized measurement contribution is zero when $\theta_Q = \theta_F = 0$, without a survey multiplier or population expansion weight. Non-respondents contribute through their outcome histories and the shared parameters. Predictions conditioned on survey answers are used for patient diagnostics separately from the hospital's historical beliefs.

All free demand and supply parameters, including the encounter-shock correlation, are jointly estimated conditional on the fixed residual SD. The augmented model assumes that the conditional quality distribution and experience measurement model also apply to non-respondents, and that later reports measure preferences present at the time of choice. Caregiver income, employment, and paid leave remain descriptive survey fields that are explored in the subsection that follows.

D.2 Conditioning on Reported Non-Medical Spending

The survey elicits household spending on transportation, meals, child care, pet care, cleaning, and paid caregiver support under each location. Let $\Delta C_i^{\$}$ denote the signed HaH-minus-B&M difference in dollars per day, and let $D_i^{\$}$ indicate a complete report matched to assessment i. Spending is used in daily units, without multiplication by length of stay, and applies only to the matched assessment.

The 297 matched reports comprise 237 acceptances and 60 documented declinations. This extension retains every assessment and adds a spending slope and reporting indicator to equation (D.5). Denoting this specification by superscript $\{S, \$\}$,

$$\Delta U_{ij}^{S,\$} = \Delta U_{ij}^S + \gamma_P D_i^{\$} \frac{\Delta C_i^{\$}}{100} + \delta_{\$} D_i^{\$}. \tag{D.13}$$

For $D_i^{\$} = 0$, the added terms are defined to be zero without imputing missing spending. The unrestricted slope γ_P measures the utility association with an additional \$100 per day of reported HaH spending relative to B&M. The reporting indicator allows the matched group to have a different utility level. This specification jointly re-estimates 175 parameters using equation (D.12), with acceptance probabilities formed from equation (D.13). Spending and report status enter the

offer decision through conditional acceptance and patient-surplus beliefs. Neither enters supply directly. Quality and experience answers remain outside the historical filter, and their normalized measurement contribution is unchanged.

This is a retrospective sensitivity analysis that conditions on report information unavailable in the historical EHR data that informs the offer decision. The reporting indicator absorbs an average group difference without fully correcting for survey selection. Spending answers lack exogenous price variation. Thus, γ_P is considered purely associational and is not used to convert utility into willingness to pay or monetary welfare.

D.3 Results

Table D.2 shows that more favorable expectations about HaH quality and experience are associated with greater willingness to accept. A one-SD increase in the reported quality difference raises acceptance utility by 0.076, while a one-SD increase in the experience factor raises it by 0.262. These estimates connect persistent differences in patients' choices to their stated expectations about care. Previously completed treatment remains positively associated with acceptance. The residual preference term $\sigma_\xi \xi_j$ captures a patient's lasting inclination toward or against HaH after accounting for observed covariates and, when included in the survey-augmented model, quality and experience expectations. It shifts utility across that patient's assessments, unlike the assessment-specific shock ε_i.

Total conditional time-invariant preference variation (s_Ω) includes quality and experience as well as the residual σ_ξ, and its recovered survey-augmented estimated SD is 0.272 compared with 0.228 in the baseline. Supply estimates retain the same substantive pattern as the baseline model. Hospitals favor transfers with greater financial returns and patient option value, while clinical burden and escalation risk discourage offers. The hospital-specific linear capacity terms are positive and statistically significant at all four hospitals. HaH strain is negative and statistically significant. Thus, incorporating measured patient expectations leaves the main interpretation of hospital incentives unchanged from the baseline model.

Table D.3 adds reported non-medical spending. Among matched reports, higher HaH spending relative to B&M has a negative coefficient, but it is not statistically distinguishable from zero ($p = 0.128$). The positive reporting-group coefficient indicates that respondents with complete spending reports differ substantially from other assessments, even after the included controls. Quality and experience remain positively associated with acceptance. Changes in the encounter-shock correlation and patient-option-value coefficient also show that conditioning on these reports affects the model's separation of patient choice and hospital targeting.

The joint log-likelihood gain of ~ 317 reflects both the spending slope and the reporting indi-

Table D.2: Baseline and Survey-Augmented Joint Estimates

	Baseline	Survey-augmented	
	Coefficient	Coefficient	SE
Patient acceptance			
Expected-quality difference, θ_Q	–	0.076^{***}	(0.018)
Expected-experience factor, θ_F	–	0.262^{***}	(0.042)
Encounter-shock correlation, r_ν	-0.731^{***}	-0.722^{***}	(0.031)
Home–hospital distance	0.007	0.000	(0.007)
Age	0.056^{***}	0.047^{***}	(0.009)
Male	0.014	0.023^{*}	(0.012)
Household size	0.000	0.001	(0.008)
College share	0.018^{*}	0.016	(0.010)
Log per-capita income	-0.015	-0.012	(0.011)
Share below 150% FPL	-0.019^{**}	-0.008	(0.009)
Prior expected rejections	-0.161^{***}	-0.147^{***}	(0.028)
Ever previously treated with HaH	0.149^{***}	0.139^{***}	(0.033)
Count of prior HaH acceptances	-0.046	-0.061^{**}	(0.027)
Condition acuity weight	-0.034^{**}	-0.035^{**}	(0.015)
Residual preference SD, σ_ξ	0.228^{***}	0.000	–
Total time-invariant preference SD, s_Ω		0.272^{***}	(0.041)
Hospital offers			
Profit, β_Π	0.179^{***}	0.187^{***}	(0.036)
HaH strain, λ_H	-0.004^{***}	-0.004^{***}	(0.001)
Escalation risk, β_H	-0.820^{***}	-0.851^{***}	(0.291)
Care burden	-0.097^{***}	-0.097^{***}	(0.007)
Acute severity	-0.044^{***}	-0.044^{***}	(0.006)
B&M stay-duration exposure	-0.353^{***}	-0.359^{***}	(0.052)
Hospital A occupancy release	0.153^{***}	0.153^{***}	(0.025)
Hospital B occupancy release	0.115^{***}	0.114^{***}	(0.026)
Hospital C occupancy release	0.129^{***}	0.129^{***}	(0.027)
Hospital D occupancy release	0.058^{**}	0.060^{**}	(0.027)
Patient option value, κ_Q	0.861^{***}	0.887^{***}	(0.053)
Likelihood assessments	31,944	31,944	
Unique patients	21,354	21,354	
Linked survey patients	–	453	
Outcome log likelihood	$-16{,}199.5$	$-16{,}194.0$	
Normalized joint log likelihood	$-16{,}199.5$	$-16{,}171.2$	
Accepted-placement Brier score	0.0694	0.0694	

Notes: Both models retain the baseline controls, fixed effects, documentation probabilities, and input transformations. Profit is measured in centered \$1,000 units. Capacity terms use forecast census, hospital-specific linear relief slopes, and a separate B&M duration exposure. The likelihood is maximized at the boundary $\sigma_\xi = 0$. The approximate one-sided 95% profile upper limit is 0.197 ($\sigma_\xi^2 = 0.039$); no regular Wald standard error is reported. Standard errors for the remaining parameters condition on $\sigma_\xi = 0$. For the survey-augmented model, standard errors appear in parentheses and use the observed Hessian, additionally conditioning on patient linkage and first-stage measurement inputs. The normalized survey measurement contribution enters the likelihood once per patient. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

cator. The full-sample accepted-placement Brier score falls from 0.0694 to 0.0676. Because every spending report follows a recorded offer, and the conditional model uses information unavailable to the hospital at the time, this gain does not establish better historical prediction and critically does not identify a population price response.

Table D.3: Survey-Augmented Estimates with Reported Spending

	Survey-augmented		With reported spending	
	Coefficient	SE	Coefficient	SE
Spending per \$100/day, γ_P	–		-0.109	(0.072)
Complete-report indicator, $\delta_{\$}$	–		0.919^{***}	(0.064)
Expected-quality difference, θ_Q	0.076^{***}	(0.018)	0.086^{***}	(0.015)
Expected-experience factor, θ_F	0.262^{***}	(0.042)	0.346^{***}	(0.044)
Encounter-shock correlation, r_ν	-0.722^{***}	(0.031)	-0.536^{***}	(0.037)
Profit, β_Π	0.187^{***}	(0.036)	0.177^{***}	(0.039)
Patient option value, κ_Q	0.887^{***}	(0.053)	1.390^{***}	(0.107)
Likelihood assessments	31,944		31,944	
Matched spending reports used	–		297	
Outcome log likelihood	$-16{,}194.0$		$-15{,}887.1$	
Normalized joint log likelihood	$-16{,}171.2$		$-15{,}854.5$	
Accepted-placement Brier-score: all assessments	0.069		0.068	
Accepted-placement Brier-score: terminal assessments	0.068		0.065	
Accepted-placement Brier-score: first post-treatment	0.165		0.161	

Notes: Both columns augment the baseline model with quality and experience. The second also conditions on reported non-medical spending and report status in conditional acceptance and surplus beliefs, jointly re-estimating all free demand and supply parameters. Both models recover a residual time-invariant SD at the boundary. Spending is measured per day. Standard errors in parentheses use the observed Hessian conditional on linkage and first-stage measurement inputs. Fit comparisons are in-sample and use different conditioning information. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Dynamic Offer Model

This appendix develops the dynamic counterpart to Section 5.2. The hospital values the same accounting, capacity, clinical, and offer-stage consequences, but now accounts explicitly for future assessment opportunities and operating states. The dynamic specification replaces the static model's valuation of remaining capacity exposure at the forecasted state with interval payoffs and continuation values. It retains Θ_S and the payoff functions defined in Section 5; the supply coefficients are estimated again under the dynamic specification.

E.1 The Hospital's Dynamic Offer Problem

Assessment timing and state. The risk set and patient information are those of the joint model. Let τ_i denote assessment time in hours. With the daily discount factor $\delta = 0.99$, the continuation discount and interval flow factor are

$$d_i = \delta^{(\tau_{i+1}-\tau_i)/24}, \qquad a_i = \int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s ds = \frac{1-d_i}{-\log \delta}. \tag{E.1}$$

Exactly tied assessments share an event time; their census changes are grouped before evaluating nonlinear capacity flows. Below, binary choice values refer to the focal assessment conditional on the other decisions at that event.

Write z_i for the full information state: the B&M census and staffed capacity at each facility, HaH census and staffing, current patient characteristics, calendar variables, and the prior-history and preference weights used in Section 5. Admitted patients occupying ED beds remain part of B&M census. Including patient histories distinguishes physical capacity changes from information acquired through an offer. The next-event law F_A governs the time to the next assessment, its patient characteristics, and the background operating state. It is estimated from consecutive assessment events, without constructing a separate daily admission model. Future assessment timing, patient mix, and staffing are uncertain; the hospital does not observe their future realizations.

Current payoffs. The direct offer terms $Z_i'\kappa$, remaining accounting return $\Delta\Pi_i$, and escalation risk H_i retain their definitions in equations (6)–(7). Accounting returns are booked once at the placement decision because $\Delta\Pi_i$ already values the expected remaining episode. Escalation remains an ex ante risk term; the baseline does not separately simulate return transfers. No explicit backfill-profit payoff is added.

For capacity flows, let $m^B(z)$ collect the sum of the common B&M scarcity functions across facilities and the individual hospital tail functions, using equation (12). Let $m^H(z)$ be the HaH

strain function in equation (9). These are the existing functions evaluated at the evolving state. Conditional on an accepted transfer, define the capacity differences over the first interval as

$$M_i^{B,0} = \mathbb{E}\left[\int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s \{m^B(z_i^B(s)) - m^B(z_i^H(s))\}\, ds \mid z_i, A_i = 1\right], \tag{E.2}$$

$$M_i^{H,0} = \mathbb{E}\left[\int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s \{m^H(z_i^H(s)) - m^H(z_i^B(s))\}\, ds \mid z_i, A_i = 1\right]. \tag{E.3}$$

Here B and H denote the focal patient's alternative treatment paths. If the state is constant over the interval, its discounted flow is multiplied by a_i. Survival and capacity reuse determine how long the two paths differ. The current offer-minus-no-offer payoff is

$$g_i(\Theta_S) = Z_i'\kappa + \rho_i\left[\beta_\Pi \Delta\Pi_i + \beta_H H_i + \beta_E E_i + \lambda_B' M_i^{B,0} + \lambda_H M_i^{H,0}\right]. \tag{E.4}$$

The acceptance probability enters once. The interval capacity terms replace, rather than supplement, the full remaining-exposure terms in the static index. Their subsequent flows enter continuation value. [13]

Dynamic problem. Let $\bar{\Gamma}_i(o, z_i; \Theta_S)$ be the expected current payoff from action o. Its capacity-flow component is the interval integral of $-\lambda_B' m^B(z) + \lambda_H m^H(z)$, so that $\bar{\Gamma}_i(1, z_i; \Theta_S) - \bar{\Gamma}_i(0, z_i; \Theta_S) = g_i(\Theta_S)$. These flow levels are retained in the value function; normalizing them away at every state would remove consequences of earlier transfers. The choice-specific value and Bellman equation are

$$v_i(o, z_i; \Theta_S) = \bar{\Gamma}_i(o, z_i; \Theta_S) + \mathbb{E}[d_i V_{i+1}(z_{i+1}; \Theta_S) \mid z_i, O_i = o], \tag{E.5}$$

$$V_i(z_i; \Theta_S) = \mathbb{E}_{\varepsilon_i}\left[\max_{o \in \{0,1\}} \{v_i(o, z_i; \Theta_S) + \varepsilon_i(o)\}\right]. \tag{E.6}$$

The first expectation integrates over future event timing, acceptance, survival, capacity reuse, staffing, and patient information. The discount factor remains inside this expectation because the next event time is uncertain. The second expectation integrates over current private offer shocks. Interval flows are not added again to the continuation term.

[13] Input units follow Appendix C.2: accounting returns are in \$1,000 units and capacity and risk measures retain their units. Reference offsets in centered placement features enter once in the assessment payoff, not again in every subsequent interval of that placement. The physical state-flow functions themselves are evaluated before this assessment-level centering.

Let $z_{i+1}^{a,i}$, $z_{i+1}^{d,i}$, and $z_{i+1}^{0,i}$ denote next states following acceptance, rejection, and no offer. Then

$$\begin{aligned}\Delta v_i(\Theta_S) = g_i(\Theta_S) &+ \rho_i \mathbb{E}\Big[d_i\{V_{i+1}(z_{i+1}^{a,i};\Theta_S) - V_{i+1}(z_{i+1}^{0,i};\Theta_S)\} \mid z_i, A_i = 1\Big] \\ &+ (1-\rho_i)\mathbb{E}\Big[d_i\{V_{i+1}(z_{i+1}^{d,i};\Theta_S) - V_{i+1}(z_{i+1}^{0,i};\Theta_S)\} \mid z_i, A_i = 0, O_i = 1\Big]. \end{aligned} \tag{E.7}$$

The counterfactual no-offer state is evaluated using the same underlying draws in each conditional comparison. Rejection and no offer have the same physical census path, but can imply different subsequent histories and beliefs. If these information effects are held fixed, the final line is zero and the continuation difference reduces to the acceptance-weighted HaH-minus-B&M comparison. Future value functions already incorporate acceptance by future patients; they do not multiply the focal ρ_i again.

Counterfactual state paths. An accepted transfer initially removes one patient from B&M census and adds one to HaH census. Let $s_i^B(h)$ and $s_i^H(h)$ be the remaining-stay survival probabilities at elapsed days h. The expected physical footprints are

$$w_i^B(h) = s_i^B(h)e_{b(i)}, \qquad w_i^H(h) = s_i^H(h)e_h + s_i^B(h)e_{b(i)}, \tag{E.8}$$

where the unit vectors add census to the originating hospital or HaH. At later assessment j, the physical states are constructed as

$$\mathcal{S}_j^{B,i} = \mathcal{S}_j^{-i} + w_i^B(h_{ij}), \qquad \mathcal{S}_j^{H,i} = \mathcal{S}_j^{-i} + w_i^H(h_{ij}), \qquad h_{ij} = \frac{\tau_j - \tau_i}{24}. \tag{E.9}$$

The common background $\mathcal{S}_j^{-i}$ removes the focal patient's realized footprint and any capacity use attributed to its release before either path is overlaid. This avoids adding the same replacement patient twice. Reuse affects the duration of the census difference, not a separate profit payoff. The B&M coefficient continues to value capacity on a composite basis. [14]

Both paths use common draws from $\widehat{F}_A$, with the fitted offer policy reevaluated at their respective states. Patient histories and preference weights update after simulated outcomes using the joint model's filtering rule. The maintained background arrival law does not respond to the focal offer beyond the specified capacity-reuse footprint. Changes in admission behavior or assessment intensity therefore require an extension of that law.

[14] Evaluating nonlinear capacity functions at these expected footprints is an approximation. It differs from averaging the functions over discrete survival and reuse realizations. The same convention must be used in current flows and continuation returns.

E.2 Conditional Choice Probabilities and Two-Step GMM

Conditional choice probabilities. I use a CCP estimator following Hotz and Miller (1993); Hotz et al. (1994); the equation-matching formulation is related to Pesendorfer and Schmidt-Dengler (2008). These methods represent continuation values using conditional choice probabilities and state transitions, avoiding repeated solution of the dynamic optimization problem during estimation (Aguirregabiria and Mira, 2010; Arcidiacono and Ellickson, 2011; Aguirregabiria et al., 2021).

The static logit control-function specification, described in Appendix A.1, supplies the fitted ex-ante true-offer policy $\widehat{p}(1 \mid z) = \widehat{p}(z)$ and acceptance belief $\widehat{\rho}(z)$. The former is a logit-based fitted probability, not a documented offer indicator or the posterior classification in equation (18). The documentation probabilities q_i remain empirical measurement inputs.

Conditional on z_i, assume independent standard Type-I extreme-value action shocks, with unit scale, and the maintained acceptance and transition laws. The Hotz–Miller inversion gives

$$\Delta\widehat{cv}_i = \log\left(\frac{\widehat{p}_i}{1-\widehat{p}_i}\right). \tag{E.10}$$

The first-stage policy must capture the state dependence relevant to the dynamic decision. Using the joint model's fitted policy does not equate its static payoff coefficients with the dynamic coefficients.

Fixed-policy returns. I hold the first-stage policy, demand parameters, and transition estimates fixed during supply estimation. At each simulated state, the fitted offer–acceptance fixed point is evaluated using these first-stage parameters. A trial Θ_S changes payoff weights, not the policy or acceptance map used to construct returns. This is a two-step estimator, distinct from joint maximization of the static likelihood.

For fixed policy $\widehat{p}$, the policy-value recursion includes the expected selected offer shock:

$$\begin{aligned} V^{\widehat{p}}(z;\Theta_S) &= \sum_o \widehat{p}(o \mid z)\left\{\bar{\Gamma}(o,z;\Theta_S) + \mathbb{E}[dV^{\widehat{p}}(z';\Theta_S) \mid z,o]\right\} + \eta(z), \\ \eta(z) &= \gamma_E - \sum_o \widehat{p}(o \mid z)\log\widehat{p}(o \mid z), \end{aligned} \tag{E.11}$$

where γ_E is Euler's constant under the standard extreme-value location normalization.

I simulate returns over 90 days, using common transition draws across branches and an approximate stationary tail. A future event j is discounted by $\delta^{(\tau_j-\tau_i)/24}$. Each return component is projected onto a flexible basis of the current state and patient characteristics. Because payoffs are

linear in Θ_S under the fixed policy,

$$\Delta v_i(\Theta_S) \approx \Delta\widehat{v}_i^0 + \widehat{\mathcal{D}}_i'\Theta_S, \qquad r_i(\Theta_S) = \Delta\widehat{cv}_i - \Delta\widehat{v}_i^0 - \widehat{\mathcal{D}}_i'\Theta_S. \tag{E.12}$$

The vector $\widehat{\mathcal{D}}_i$ collects current and continuation exposures to every coefficient in equation (E.4); $\Delta\widehat{v}_i^0$ includes the shock-value contribution. The tail approximation must preserve this affine decomposition.

Two-step GMM. Let $\mathcal{M}_i$ contain projected policy-value derivatives and any additional excluded state instruments used for overidentification. Define

$$g_n(\Theta_S) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{M}_i r_i(\Theta_S), \qquad \widetilde{\Theta}_S \in \arg\min_{\Theta_S} g_n(\Theta_S)' g_n(\Theta_S). \tag{E.13}$$

At $\widetilde{\Theta}_S$, I aggregate assessment-level moments by calendar date and estimate their long-run covariance $\widehat{\Omega}$ using a Bartlett-kernel HAC estimator. The second step is

$$\widehat{\Theta}_S \in \arg\min_{\Theta_S} g_n(\Theta_S)' \widehat{\Omega}^{-1} g_n(\Theta_S). \tag{E.14}$$

Objective coefficients are estimated without sign restrictions.

E.3 Identification and Inference

Identifying variation. The offer-shock normalization fixes the scale of Θ_S. Conditional on the generated demand and transition objects, remaining accounting returns identify β_Π through variation distinct from capacity consequences. The hospital-specific B&M terms use variation in census, capacity, remaining exposure, and their continuation paths. HaH census, staffing, and remaining HaH exposure distinguish λ_H from B&M relief; residual clinical variation identifies β_H and the clinical offer shifters. The coefficient on patient option value is identified separately from acceptance-weighted placement consequences only when their payoff exposures are sufficiently distinct. As in Section 5.2, it can reflect mission-related objectives and directly captures the firm's weight on patient surplus. The common offer term remains a net baseline friction under the stated centering. History and fixed-effect coefficients capture policy heterogeneity without separate welfare labels.

Local identification requires

$$\operatorname{rank}\left\{\mathbb{E}[\mathcal{M}_i \widehat{\mathcal{D}}_i']\right\} = \dim(\Theta_S). \tag{E.15}$$

This condition also requires valid moments, adequate offer-policy support, and a state and transition law that capture the hospital's relevant information. CCP inversion alone does not establish these assumptions. Diagnostics include derivative correlations, the minimum singular value and condition number of the moment Jacobian, and profile objectives. Particular attention is given to accounting versus capacity exposure, and acceptance targeting versus placement benefits.

Inference. For $\mathfrak{m}_i(\Theta_S) = \mathcal{M}_i r_i(\Theta_S)$, define $\mathfrak{m}_t(\Theta_S) = \sum_{i:\text{date}(\tau_i)=t} \mathfrak{m}_i(\Theta_S)$, retaining dates without assessments with zero moments. The HAC matrix allows dependence within dates and serial dependence across nearby dates. Standard errors use a calendar-date block multiplier bootstrap, assigning a common weight to assessments in each block while preserving chronology. Each replication refits the joint demand–offer model, estimated accounting, survival, reuse, escalation, and transition inputs, and the return projections, then reconstructs the HAC matrix and repeats both GMM steps. Empirical documentation probabilities remain fixed in this conditional inference; propagating their sampling uncertainty requires reconstructing them as well. The covariance of the replication estimates provides standard errors, including uncertainty in the reestimated generated inputs.

E.4 Dynamic Results and Comparison with the Static Model

Private capacity value. For comparison with the static model, the dynamic private capacity value is the offer-value difference between the full calculation and a no-release calculation that retains focal HaH treatment and HaH strain but removes the B&M capacity change and its continuation consequences. Focal acceptance, history updates, and payoff reference offsets are held common. Its dollar-equivalent value per accepted transfer is

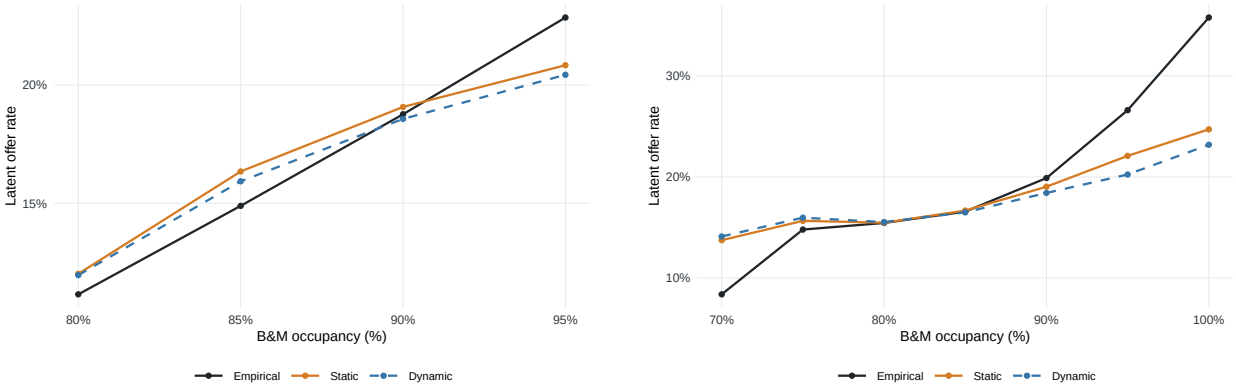
$$\widehat{CV}_i^{\text{Dyn}} = 1{,}000\,\frac{\Delta\widehat{v}_i^{\text{release}} - \Delta\widehat{v}_i^{\text{no-release}}}{\widehat{\rho}_i\widehat{\beta}_\Pi}. \tag{E.16}$$

Table E.1: Dynamic Hospital Offer Estimates

	Estimate	Standard error
Accounting profits (per $1,000)	0.653^{***}	(0.025)
HaH program strain	-0.0115^{***}	(0.0024)
HaH escalation risk	-3.429^{***}	(0.195)
Care-burden	-0.2270^{***}	(0.0022)
Acute-severity	-0.1090^{***}	(0.0016)
B&M stay-duration exposure	-0.830^{***}	(0.081)
Hospital A released-capacity	0.370^{***}	(0.045)
Hospital B released-capacity	0.299^{***}	(0.045)
Hospital C released-capacity	0.324^{***}	(0.049)
Hospital D released-capacity	0.113^{**}	(0.049)
Strictly prior expected offers	0.328^{***}	(0.003)
Patient option value	0.980^{***}	(0.016)
Baseline net offer term	-1.321^{***}	(0.016)
Assessments	31,944	
Unique patients	21,354	
Reestimated supply coefficients	72	
Moments used in GMM	77	
Conditional fit statistics		
Offer-CCP approximation RMSE (logit scale)	0.3009	
Occupancy-bin offer-rate weighted RMSE	0.0185	
Mean predicted offer rate	0.2144	
Mean documentation-adjusted empirical offer rate	0.2213	

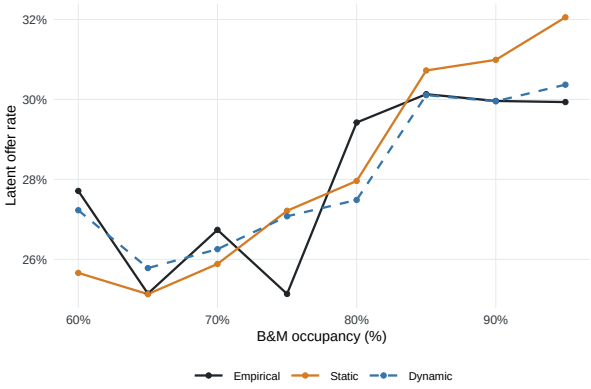
Notes: Estimates recovered using two-step conditional-choice-probability (CCP) GMM. All 72 supply coefficients are reestimated; demand parameters remain fixed at their static estimates described in Appendix A.1. Standard errors use calendar-HAC inference with bandwidth seven and condition upon demand parameters, generated inputs, and simulated continuation features. Estimates retain the static control-function model's logit normalization. Occupancy-bin fit compares predicted offers with the documentation-adjusted empirical reference, weighting cells by assessment counts. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure E.1: Empirical and Predicted Offer Rates by B&M Occupancy by Model Type

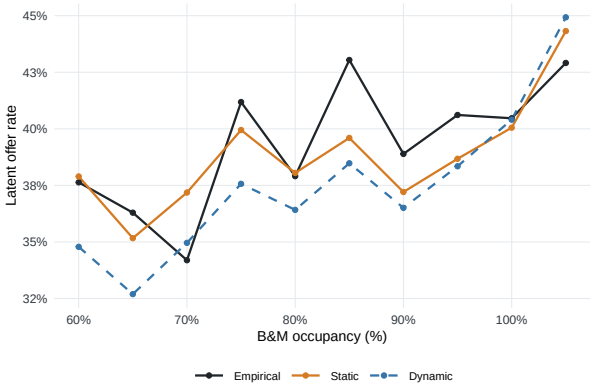


(a) Hospital A

(b) Hospital B



(c) Hospital C



(d) Hospital D

Notes: Points show mean offer rates within hospital-specific five-percentage-point occupancy bins, plotted at each bin's lower bound. Empirical rates are documentation-adjusted; static and dynamic rates average fitted offer probabilities. Bins require at least 50 assessments. All comparisons are in sample.